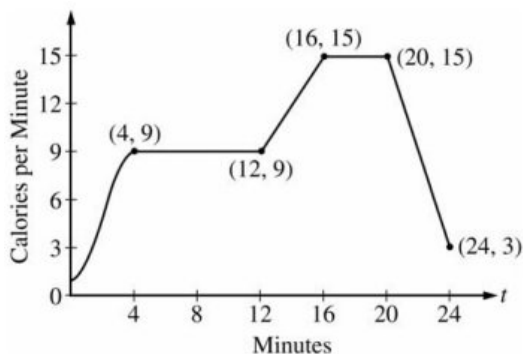


2-2-2b

1.



The rate, in calories per minute, at which a person using an exercise machine burns calories is modeled by the function f . In the figure above, $f(t) = -\frac{1}{4}t^3 + \frac{3}{2}t^2 + 1$ for $0 \leq t \leq 4$ and f is piecewise linear for $4 \leq t \leq 24$.

- Find $f'(22)$. Indicate units of measure.
- For the time interval $0 \leq t \leq 24$, at what time t is f increasing at its greatest rate? Show the reasoning that supports your answer.
- Find the total number of calories burned over the time interval $6 \leq t \leq 18$ minutes.
- The setting on the machine is now changed so that the person burns $f(t) + c$ calories per minute. For this setting, find c so that an average of 15 calories per minute is burned during the time interval $6 \leq t \leq 18$.

Part A

1 point is earned for $f'(22)$ and units

$$f'(22) = \frac{15 - 3}{20 - 24} = -3 \text{ calories/min/min}$$



0	1
---	---

The student response earns all of the following points:

1 point is earned for $f'(22)$ and units

$$f'(22) = \frac{15 - 3}{20 - 24} = -3 \text{ calories/min/min}$$

Part B

1 point is earned for f' on $(0, 4)$

2-2-2b

1 point is earned for showing f' has a max at $t = 2$ on $(0, 4)$

1 point is earned if shows $12 < t < 16, f'(t) < f'(2)$

1 point is earned for the answer

f is increasing on $[0, 4]$ and on $[12, 16]$.

On $(12, 16)$, $f'(t) = \frac{15 - 9}{16 - 12} = \frac{3}{2}$ since f has constant slope on this interval.

On $(0, 4)$, $f'(t) = -\frac{3}{4}t^2 + 3t$ and $f''(t) = -\frac{3}{2}t + 3 = 0$ when $t = 2$. This is where f' has a maximum on $[0, 4]$ since $f'' > 0$ on $(0, 2)$ and $f'' < 0$ on $(2, 4)$.

On $[0, 24]$, f is increasing at its greatest rate when $t = 2$ because $f'(2) = 3 > \frac{3}{2}$.



0	1	2	3	4
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The student response earns all of the following points:

1 point is earned for f' on $(0, 4)$

1 point is earned for showing f' has a max at $t = 2$ on $(0, 4)$

1 point is earned if shows $12 < t < 16, f'(t) < f'(2)$

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f is increasing on $[0, 4]$ and on $[12, 16]$.

On $(12, 16)$, $f'(t) = \frac{15 - 9}{16 - 12} = \frac{3}{2}$ since f has constant slope on this interval.

On $(0, 4)$, $f'(t) = -\frac{3}{4}t^2 + 3t$ and $f''(t) = -\frac{3}{2}t + 3 = 0$ when $t = 2$. This is where f' has a maximum on $[0, 4]$ since $f'' > 0$ on $(0, 2)$ and $f'' < 0$ on $(2, 4)$.

2-2-2b

On $[0, 24]$, f is increasing at its greatest rate when $t = 2$ because $f'(2) = 3 > \frac{3}{2}$.

Part C

1 point is earned for the method

1 point is earned for the answer

$$\int_6^{18} f(t) dt = 6(9) + \frac{1}{2}(4)(9 + 15) + 2(15) = 132 \text{ calories}$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the method

1 point is earned for the answer

$$\int_6^{18} f(t) dt = 6(9) + \frac{1}{2}(4)(9 + 15) + 2(15) = 132 \text{ calories}$$

Part D

1 point is earned for the setup

1 point is earned for the value of c

$$\text{We want } \frac{1}{12} \int_6^{18} (f(t) + c) dt = 15.$$

This means $132 + 12c = 15(12)$. So, $c = 4$.

OR

Currently, the average is $\frac{132}{12} = 11$ calories/min.

Adding c to $f(t)$ will shift the average by c . So $c = 4$ to get an average of 15 calories/min.

2-2-2b



0	1	2
---	---	---

The student response earns none of the following points:

1 point is earned for the setup

1 point is earned for the value of c

We want $\frac{1}{12} \int_6^{18} (f(t) + c) dt = 15$.

This means $132 + 12c = 15(12)$. So, $c = 4$.

OR

Currently, the average is $\frac{132}{12} = 11$ calories/min.

Adding c to $f(t)$ will shift the average by c . So $c = 4$ to get an average of 15 calories/min.

2-2-2b

2. Let $x = ky^2 + 2$, where $k > 0$.

(a) Show that for all $k > 0$, the point $\left(4, \sqrt{\frac{2}{k}}\right)$ is on the graph of $x = ky^2 + 2$.

(b) Show that for all $k > 0$, the tangent line to the graph of $x = ky^2 + 2$ at the point $\left(4, \sqrt{\frac{2}{k}}\right)$ passes through the origin.

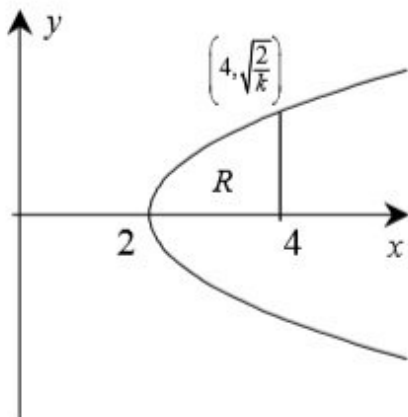
(c) Let R be the region in the first quadrant bounded by the x -axis, the graph of $x = ky^2 + 2$, and the line $x = 4$. Write an integral expression for the area of the region R and show that this area decreases as k increases.

Part A

1 point is earned for correctly substituting $x = 4$ and $y = \sqrt{\frac{2}{k}}$

$$4 = k\left(\frac{2}{k}\right) + 2$$

$$4 = 4$$



0

1

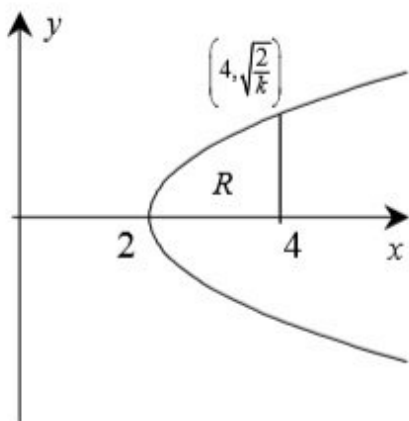
The student response earns all of the following points:

1 point is earned for correctly substituting $x = 4$ and $y = \sqrt{\frac{2}{k}}$

$$4 = k\left(\frac{2}{k}\right) + 2$$

$$4 = 4$$

2-2-2b

**Part B**

1 point is earned for the correct equation with $\frac{dy}{dx}$ or $\frac{dx}{dy}$

$$x = ky^2 + 2$$

$$1 = 2ky \frac{dy}{dx}$$

1 point is earned for the correct evaluation of $\frac{dy}{dx}$ or $\frac{dx}{dy}$ at $\left(4, \sqrt{\frac{2}{k}}\right)$

$$\frac{dy}{dx} \Big|_{y=\sqrt{2/k}} = \frac{1}{2\sqrt{2k}}$$

1 point is earned for the correct equation of tangent line or slope of line using $\left(4, \sqrt{2/k}\right)$ and $(0,0)$

the tangent line is

$$y - \sqrt{\frac{2}{k}} = \frac{1}{2\sqrt{2k}}(x - 4)$$

$$y = \frac{1}{2\sqrt{2k}}x \text{ which contains } (0,0)$$

or

$$\text{slope of line through } (0,0) \text{ and } \left(4, \sqrt{2/k}\right) \text{ is } \frac{\sqrt{2/k}}{4} = \frac{1}{2\sqrt{2k}}$$

which is the same as the slope of the tangent line

1 point is earned for correctly observing $(0,0)$ is a solution or observing the slopes are equal

2-2-2b



0	1	2	3	4
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The student response earns all of the following points:

1 point is earned for the correct equation with $\frac{dy}{dx}$ or $\frac{dx}{dy}$

$$x = ky^2 + 2$$

$$1 = 2ky \frac{dy}{dx}$$

1 point is earned for the correct evaluation of $\frac{dy}{dx}$ or $\frac{dx}{dy}$ at $\left(4, \sqrt{\frac{2}{k}}\right)$

$$\frac{dy}{dx} \Big|_{y=\sqrt{2/k}} = \frac{1}{2\sqrt{2k}}$$

1 point is earned for the correct equation of tangent line or slope of line using $\left(4, \sqrt{2/k}\right)$ and $(0,0)$

the tangent line is

$$y - \sqrt{\frac{2}{k}} = \frac{1}{2\sqrt{2k}}(x - 4)$$

$$y = \frac{1}{2\sqrt{2k}}x \text{ which contains } (0,0)$$

or

$$\text{slope of line through } (0,0) \text{ and } \left(4, \sqrt{2/k}\right) \text{ is } \frac{\sqrt{2/k}}{4} = \frac{1}{2\sqrt{2k}}$$

which is the same as the slope of the tangent line

1 point is earned for correctly observing $(0,0)$ is a solution or observing the slopes are equal

Part C

3 points are earned for correctly expressing the area as a definite integral

$$A = \int_0^{\sqrt{2/k}} (4 - (ky^2 + 2)) dy$$

or

$$A = \frac{1}{\sqrt{k}} \int_2^4 \sqrt{x - 2} dx$$

2-2-2b

<-1> error in limits

<-1> considers twice the region

<-2> error in integrand

1 point is earned for correctly showing that A decreases using algebra, geometry, or calculus

$$A = \frac{4\sqrt{2}}{3}k^{-0.5}$$

$$\frac{dA}{dk} = -\frac{2\sqrt{2}}{3}k^{-1.5} < 0 \text{ for all } k > 0$$

thus the area decreases as k increases



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

3 points are earned for correctly expressing the area as a definite integral

$$A = \int_0^{\sqrt{2/k}} (4 - (ky^2 + 2))dy$$

or

$$A = \frac{1}{\sqrt{k}} \int_2^4 \sqrt{x-2}dx$$

<-1> error in limits

<-1> considers twice the region

<-2> error in integrand

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$$A = \frac{4\sqrt{2}}{3}k^{-0.5}$$

$$\frac{dA}{dk} = -\frac{2\sqrt{2}}{3}k^{-1.5} < 0 \text{ for all } k > 0$$

thus the area decreases as k increases

2-2-2b

Let $x = ky^2 + 2$, where $k > 0$.

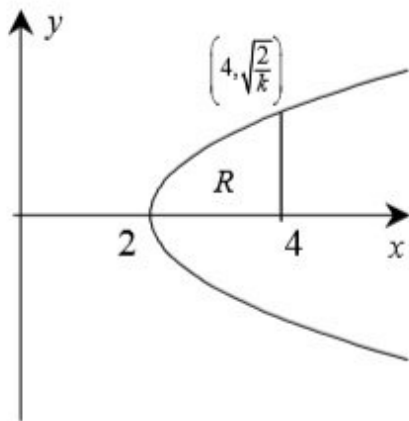
3. Show that for all $k > 0$, the point $\left(4, \sqrt{\frac{2}{k}}\right)$ is on the graph of $x = ky^2 + 2$.

Part A

1 point is earned for correctly substituting $x = 4$ and $y = \sqrt{\frac{2}{k}}$

$$4 = k\left(\frac{2}{k}\right) + 2$$

$$4 = 4$$



0

1

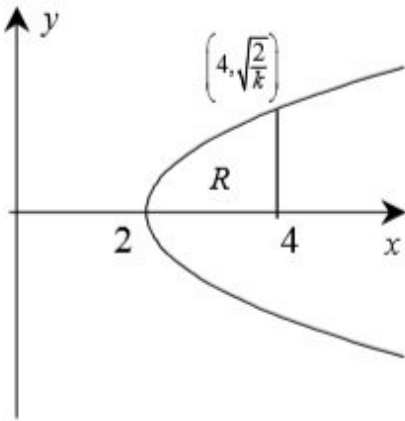
The student response earns one of the following points:

1 point is earned for correctly substituting $x = 4$ and $y = \sqrt{\frac{2}{k}}$

$$4 = k\left(\frac{2}{k}\right) + 2$$

$$4 = 4$$

2-2-2b



2-2-2b

4. Show that for all $k > 0$, the tangent line to the graph of $x = ky^2 + 2$ at the point $\left(4, \sqrt{\frac{2}{k}}\right)$ passes through the origin.

Part B

1 point is earned for the correct equation with $\frac{dy}{dx}$ or $\frac{dx}{dy}$

$$x = ky^2 + 2$$

$$1 = 2ky \frac{dy}{dx}$$

1 point is earned for the correct evaluation of $\frac{dy}{dx}$ or $\frac{dx}{dy}$ at $\left(4, \sqrt{\frac{2}{k}}\right)$

$$\frac{dy}{dx} \Big|_{y=\sqrt{\frac{2}{k}}} = \frac{1}{2\sqrt{2k}}$$

1 point is earned for the correct equation of tangent line or slope of line using $\left(4, \sqrt{\frac{2}{k}}\right)$ and $(0,0)$

the tangent line is

$$y - \sqrt{\frac{2}{k}} = \frac{1}{2\sqrt{2k}}(x - 4)$$

$$y = \frac{1}{2\sqrt{2k}}x$$

which contains $(0,0)$

slope of line through $(0,0)$ and $\left(4, \sqrt{\frac{2}{k}}\right)$ is $\frac{\sqrt{\frac{2}{k}}}{4} = \frac{1}{2\sqrt{2k}}$

which is the same as the slope of the tangent line

1 point is earned for correctly observing $(0,0)$ is a solution or observing the slopes are equal



0	1	2	3	4
---	---	---	---	---

The student response earns four of the following points:

2-2-2b

1 point is earned for the correct equation with $\frac{dy}{dx}$ or $\frac{dx}{dy}$

$$x = ky^2 + 2$$

$$1 = 2ky \frac{dy}{dx}$$

1 point is earned for the correct evaluation of $\frac{dy}{dx}$ or $\frac{dx}{dy}$ at $\left(4, \sqrt{\frac{2}{k}}\right)$

$$\frac{dy}{dx} \Big|_{y=\sqrt{\frac{2}{k}}} = \frac{1}{2\sqrt{2k}}$$

1 point is earned for the correct equation of tangent line or slope of line using $\left(4, \sqrt{\frac{2}{k}}\right)$ and $(0,0)$

the tangent line is

$$y - \sqrt{\frac{2}{k}} = \frac{1}{2\sqrt{2k}}(x - 4)$$

$$y = \frac{1}{2\sqrt{2k}}x$$

which contains $(0,0)$

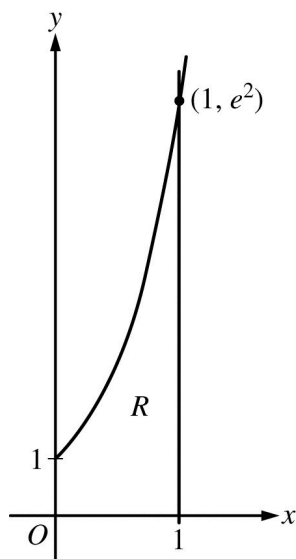
$$\text{slope of line through } (0,0) \text{ and } \left(4, \sqrt{\frac{2}{k}}\right) \text{ is } \frac{\sqrt{\frac{2}{k}}}{4} = \frac{1}{2\sqrt{2k}}$$

which is the same as the slope of the tangent line

1 point is earned for correctly observing $(0,0)$ is a solution or observing the slopes are equal

2-2-2b

5.



Let $f(x) = e^{2x}$. Let R be the region in the first quadrant bounded by the graph of $y = f(x)$ and the vertical line $x = 1$, as shown in the figure above.

- Write an equation for the line tangent to the graph of f at $x = 1$.
- Find the area of R .
- Region R forms the base of a solid whose cross sections perpendicular to the y -axis are squares. Write, but do not evaluate, an expression involving one or more integrals that gives the volume of the solid.

Part A

The response can earn up to 2 points:

1 point: For $f'(1)$

$$f(1) = e^2$$

$$f'(x) = 2e^{2x} \Rightarrow f'(1) = 2e^2$$

1 point: For the correct answer

An equation for the tangent line is $y = e^2 + 2e^2(x - 1)$.



0	1	2
---	---	---

The response earns all of the following points:

1 point: For $f'(1)$

2-2-2b

$$f(1) = e^2$$

$$f'(x) = 2e^{2x} \Rightarrow f'(1) = 2e^2$$

1 point: For the correct answer

An equation for the tangent line is $y = e^2 + 2e^2(x - 1)$.

Part B

The response can earn up to 3 points:

1 point: For the integral

1 point: For the antiderivative

1 point: For the correct answer

$$\text{Area} = \int_0^1 e^{2x} dx = \left[\frac{1}{2} e^{2x} \right]_{x=0}^{x=1} = \frac{1}{2}(e^2 - 1)$$



0	1	2	3
---	---	---	---

The response earns all of the following points:

1 point: For the integral

1 point: For the antiderivative

1 point: For the correct answer

$$\text{Area} = \int_0^1 e^{2x} dx = \left[\frac{1}{2} e^{2x} \right]_{x=0}^{x=1} = \frac{1}{2}(e^2 - 1)$$

Part C

The response can earn up to 4 points:

2 points: For the integrand

1 point: For correct limits

2-2-2b

1 point: For the correct answer

$$\text{Volume} = 1 + \int_1^{e^2} \left(1 - \frac{1}{2} \ln y\right)^2 dy$$



0	1	2	3	4
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The response earns all of the following points:

2 points: For the integrand

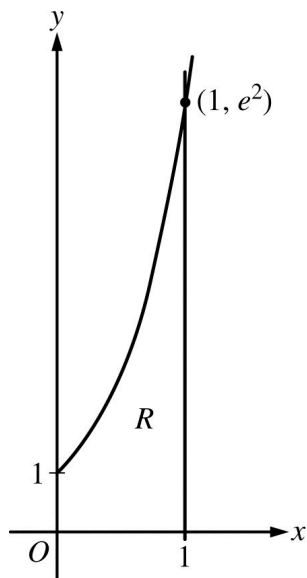
1 point: For correct limits

1 point: For the correct answer

$$\text{Volume} = 1 + \int_1^{e^2} \left(1 - \frac{1}{2} \ln y\right)^2 dy$$

2-2-2b

6.



Let $f(x)=e^{2x}$. Let R be the region in the first quadrant bounded by the graph of $y=f(x)$ and the vertical line $x=1$, as shown in the figure above.

Write an equation for the line tangent to the graph of f at $x=1$.

Part A

The question can earn up to 2 points:

1 point: for $f'(1)$

$$f(1) = e^2$$

$$f'(x) = 2e^{2x} \Rightarrow f'(1) = 2e^2$$

1 point: for the correct answer.

An equation for the tangent line is $y = e^2 + 2e^2(x - 1)$



0	1	2
---	---	---

The response earns both of the following points:

1 point: for $f'(1)$

$$f(1) = e^2$$

$$f'(x) = 2e^{2x} \Rightarrow f'(1) = 2e^2$$

1 point: for the correct answer.

An equation for the tangent line is $y = e^2 + 2e^2(x - 1)$

2-2-2b

2-2-2b

7. Let f be the function that is defined for all real numbers x and that has the following properties.

(i) $f'(x) = 24x - 18$

(ii) $f(1) = -6$

(iii) $f(2) = 0$

(a) Find each x such that the line tangent to the graph of f at $(x, f(x))$ is horizontal.

(b) Write an expression for $f(x)$.

(c) Find the average value of f on the interval $1 \leq x \leq 3$.

Part A

1 point is earned for antiderivative with or without C

1 point is earned for $f(1) = -6$ for student's $f(x)$

2 points are earned for setting $f'(x) = 0$ and solves

$$f'(x) = 12x^2 - 18x + C$$

$$f'(1) = -6 = 12 - 18 + C$$

$$\therefore C = 0$$

$$f'(x) = 6x(2x - 3) = 0$$

$$x = 0, \frac{3}{2}$$



0	1	2	3	4
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The student response earns all of the following points:

1 point is earned for antiderivative with or without C

1 point is earned for $f(1) = -6$ for student's $f(x)$

2 points are earned for setting $f'(x) = 0$ and solves

$$f'(x) = 12x^2 - 18x + C$$

$$f'(1) = -6 = 12 - 18 + C$$

$$\therefore C = 0$$

$$f'(x) = 6x(2x - 3) = 0$$

$$x = 0, \frac{3}{2}$$

2-2-2b

Part B

1 point is earned for antiderivative with or without C

1 point is earned for equating student's $f(2)$ and zero

1 point is earned for solving the resulting equation

$$f(x) = 4x^3 - 9x^2 + C$$

$$f(2) = 0 = 32 - 36 + C$$

$$\therefore C = 4$$

$$f(x) = 4x^3 - 9x^2 + 4$$



0	1	2	3
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The student response earns all of the following points:

1 point is earned for antiderivative with or without C

1 point is earned for equating student's $f(2)$ and zero

1 point is earned for solving the resulting equation

$$f(x) = 4x^3 - 9x^2 + C$$

$$f(2) = 0 = 32 - 36 + C$$

$$\therefore C = 4$$

$$f(x) = 4x^3 - 9x^2 + 4$$

Part C

1 point is earned for setting up integral with student's $f(x)$ from part (b)

1 point is earned for the antiderivative and evaluation

Note: 0/2 if integrand is not student's $f(x)$

2-2-2b

$$\begin{aligned} & \frac{1}{3-1} \int_1^3 4x^3 - 9x^2 + 4 \, dx \\ &= \frac{1}{2} [x^4 - 3x^3 + 4x]_1^3 \\ &= \frac{1}{2} [(81 - 81 + 12) - (1 - 3 + 4)] \\ &= \boxed{5} \end{aligned}$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for setting up integral with student's $f(x)$ from part (b)

1 point is earned for the antiderivative and evaluation

Note: 0/2 if integrand is not student's $f(x)$

$$\begin{aligned} & \frac{1}{3-1} \int_1^3 4x^3 - 9x^2 + 4 \, dx \\ &= \frac{1}{2} [x^4 - 3x^3 + 4x]_1^3 \\ &= \frac{1}{2} [(81 - 81 + 12) - (1 - 3 + 4)] \\ &= \boxed{5} \end{aligned}$$

2-2-2b

8. Let f be the function that is defined for all real numbers x and that has the following properties.

(i) $f''(x) = 24x - 18$

(ii) $f'(1) = -6$

(iii) $f(2) = 0$

Find each x such that the line tangent to the graph of f at $(x, f(x))$ is horizontal.

Part A

1 point is earned for antiderivative with or without c

1 point is earned for $f'(1) = -6$ for student's $f'(x)$

2 points are earned for setting $f'(x) = 0$ and solves

$$f'(x) = 12x^2 - 18x + C$$

$$f'(1) = -6 = 12 - 18 + C$$

$$\therefore C = 0$$

$$f'(x) = 6x(2x - 3) = 0$$

$$x = 0, \frac{3}{2}$$



0	1	2	3	4
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The student response earns four of the following points:

1 point is earned for antiderivative with or without c

1 point is earned for $f'(1) = -6$ for student's $f'(x)$

2 points are earned for setting $f'(x) = 0$ and solves

$$f'(x) = 12x^2 - 18x + C$$

$$f'(1) = -6 = 12 - 18 + C$$

$$\therefore C = 0$$

$$f'(x) = 6x(2x - 3) = 0$$

$$x = 0, \frac{3}{2}$$

2-2-2b

9. The function f is defined by $f(x) = \frac{x}{x+2}$. What points (x, y) on the graph of f have the property that the line tangent to f at (x, y) has slope $\frac{1}{2}$?
- (A) $(0, 0)$ only
- (B) $(\frac{1}{2}, \frac{1}{5})$ only
- (C) $(0, 0)$ and $(-4, 2)$ ✓
- (D) $(0, 0)$ and $(4, \frac{2}{3})$
- (E) There are no such points.

2-2-2b

10. The twice-differentiable function f is defined for all real numbers and satisfies the following conditions:

$$f(0) = 2, f'(0) = -4, \text{ and } f''(0) = 3.$$

- (a) The function g is given by $g(x) = e^{ax} + f(x)$ for all real numbers, where a is a constant. Find $g'(0)$ and $g''(0)$ in terms of a . Show the work that leads to your answer.
- (b) The function h is given by $h(x) = \cos(kx)f(x)$ for all real numbers, where k is a constant. Find $h'(x)$ and write an equation for the line tangent to the graph of h at $x = 0$.

Part A

1 point is earned for: $g'(x)$

1 point is earned for: $g'(0)$

1 point is earned for: $g''(x)$

1 point is earned for: $g''(0)$

$$g'(x) = ae^{ax} + f'(x)$$

$$g'(0) = a - 4$$

$$g''(x) = a^2e^{ax} + f''(x)$$

$$g''(0) = a^2 + 3$$



0	1	2	3	4
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The student response earns all of the following points:

1 point is earned for: $g'(x)$

1 point is earned for: $g'(0)$

1 point is earned for: $g''(x)$

1 point is earned for: $g''(0)$

$$g'(x) = ae^{ax} + f'(x)$$

$$g'(0) = a - 4$$

$$g''(x) = a^2e^{ax} + f''(x)$$

$$g''(0) = a^2 + 3$$

Part B

2-2-2b

2 points are earned for: $h'(x)$

1 point is earned for: $h'(0)$

1 point is earned for: $h(0)$

1 point is earned for the equation of tangent line

$$h'(x) = f'(x) \cos(kx) - k \sin(kx) f(x)$$

$$h'(0) = f'(0) \cos(0) - k \sin(0) f(0) = f'(0) = -4$$

$$h(0) = \cos(0) f(0) = 2$$

The equation of the tangent line is $y = -4x + 2$.



0	1	2	3	4	5
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The student response earns all of the following points:

2 points are earned for: $h'(x)$

1 point is earned for: $h'(0)$

1 point is earned for: $h(0)$

1 point is earned for the equation of tangent line

$$h'(x) = f'(x) \cos(kx) - k \sin(kx) f(x)$$

$$h'(0) = f'(0) \cos(0) - k \sin(0) f(0) = f'(0) = -4$$

$$h(0) = \cos(0) f(0) = 2$$

The equation of the tangent line is $y = -4x + 2$.

2-2-2b

11. The twice-differentiable function f is defined for all real numbers and satisfies the following conditions: $f(0) = 2$, $f'(0) = -4$, and $f''(0) = 3$.

The function h is given by $h(x) = \cos(kx)f(x)$ for all real numbers, where k is a constant. Find $h'(x)$ and write an equation for the line tangent to the graph of h at $x = 0$.

Part B

2 points are earned for: $h'(x)$

1 point is earned for: $h'(0)$

1 point is earned for: $h(0)$

1 point is earned for the equation of tangent line

$$h'(x) = f'(x) \cos(kx) - k \sin(kx) f(x)$$

$$h'(0) = f'(0) \cos(0) - k \sin(0) f(0) = f'(0) = -4$$

$$h(0) = \cos(0) f(0) = 2$$

The equation of the tangent line is $y = -4x + 2$.



0	1	2	3	4	5
---	---	---	---	---	---

The student response earns all of the following points:

2 points are earned for: $h'(x)$

1 point is earned for: $h'(0)$

1 point is earned for: $h(0)$

1 point is earned for the equation of tangent line

$$h'(x) = f'(x) \cos(kx) - k \sin(kx) f(x)$$

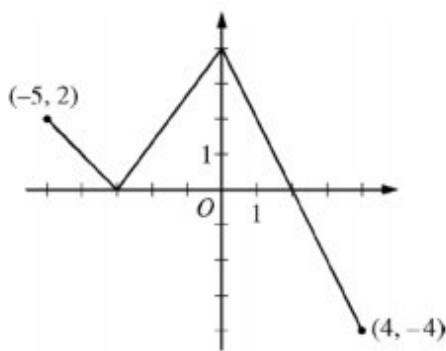
$$h'(0) = f'(0) \cos(0) - k \sin(0) f(0) = f'(0) = -4$$

$$h(0) = \cos(0) f(0) = 2$$

The equation of the tangent line is $y = -4x + 2$.

2-2-2b

12.

Graph of f

The function f is defined on the closed interval $[-5, 4]$. The graph of f consists of three line segments and is shown in the figure above. Let g be the function defined by $g(x) = \int_{-3}^x f(t) dt$.

The function p is defined by $p(x) = f(x^2 - x)$. Find the slope of the line tangent to the graph of p at the point where $x = -1$.

Part D

2 points are earned for $p'(x)$

1 point is earned for the answer

$$p'(x) = f(x^2 - x)(2x - 1) \quad p'(-1) = f(2)(-3) = (-2)(-3) = 6$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

2 points are earned for $p'(x)$

1 point is earned for the answer

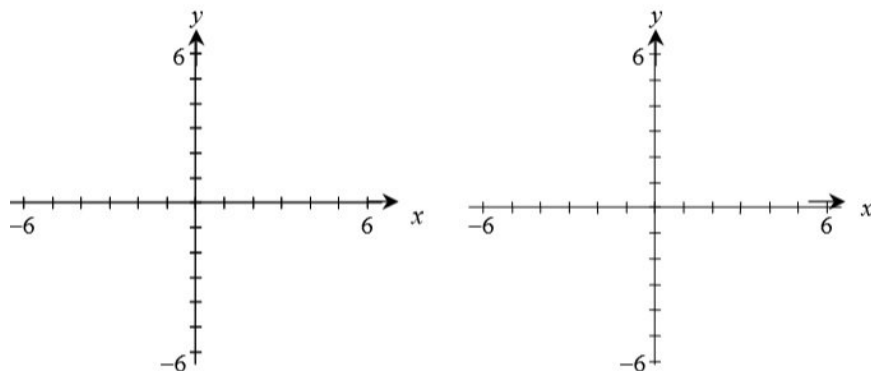
$$p'(x) = f(x^2 - x)(2x - 1)$$

$$p'(-1) = f(2)(-3) = (-2)(-3) = 6$$

2-2-2b

13.  This problem deals with functions defined by $f(x) = x + b \sin x$, where b is a positive constant and $-2\pi \leq x \leq 2\pi$.

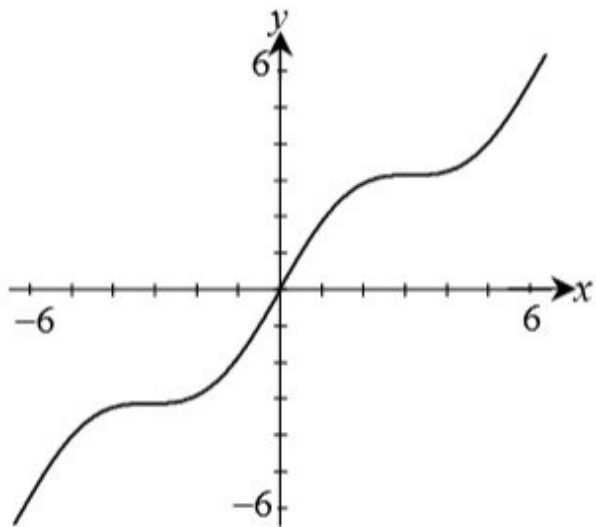
(a) Sketch the graphs of two of these functions, $y = x + \sin x$ and $y = x + 3 \sin x$.



- (b) Find the x -coordinates of all points, $-2\pi \leq x \leq 2\pi$, where the line $y = x + b$ is tangent to the graph of $f(x) = x + b \sin x$.
- (c) Are the points of tangency described in part (b) relative maximum points of f ? Why?
- (d) For all values of $b > 0$, show that all inflection points of the graph of f lie on the line $y = x$.

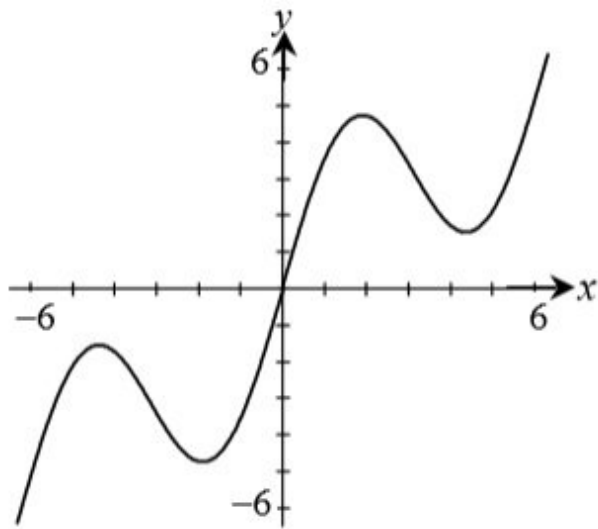
Part A

1 point is earned for the correct graph of $y = x + \sin x$ (increasing, through origin, cd - cu - cd - cu) shape of interest not extent



1 point is earned for the correct graph of $y = x + 3 \sin x$ (inc - dec - inc - dec - inc, through origin, cd - cu - cd - cu)

2-2-2b

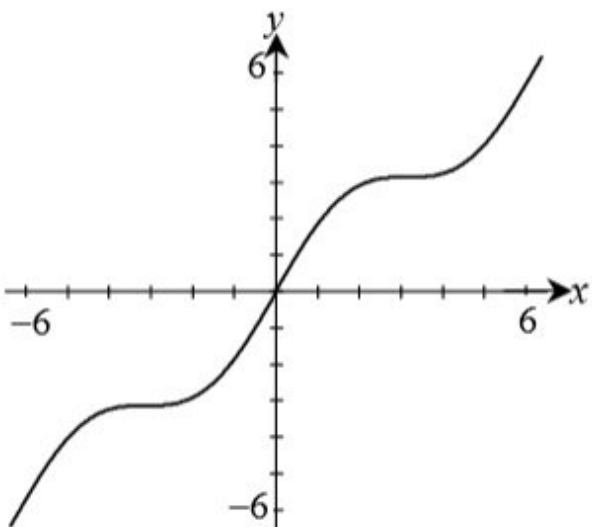


✓

0	1	2
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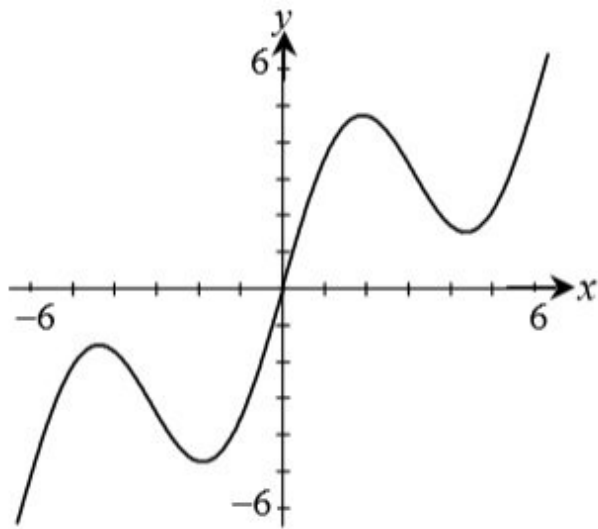
The student response earns all of the following points:

1 point is earned for the correct graph of $y = x + \sin x$ (increasing, through origin, cd - cu - cd - cu) shape of interest not extent



1 point is earned for the correct graph of $y = x + 3 \sin x$ (inc - dec - inc - dec - inc, through origin, cd - cu - cd - cu)

2-2-2b

**Part B**

1 point is earned for the correct derivatives of $x + b$ and $x + b \sin x$

$$y' = 1 = 1 + b \cos x$$

$$b \cos x = 0$$

$$\cos x = 0$$

$$y = x + b = x + b \sin x$$

$$b = b \sin x$$

$$1 = \sin x$$

1 point is earned for correctly setting the derivatives equal or checking candidates

1 point is earned for correctly setting y -values equal or checking candidates

1 point is earned for the correct answer from merged information

$$x = -\frac{3\pi}{2} \text{ or } \frac{\pi}{2}$$

Max: 0 - 1 - 1 - 0 if specific value(s) for b

Max: 0 - 0 - 1 - 0 if no derivative or bald answer



0	1	2	3	4
---	---	---	---	---

2-2-2b

The student response earns all of the following points:

1 point is earned for the correct derivatives of $x + b$ and $x + b \sin x$

$$y' = 1 = 1 + b \cos x$$

$$b \cos x = 0$$

$$\cos x = 0$$

$$y = x + b = x + b \sin x$$

$$b = b \sin x$$

$$1 = \sin x$$

1 point is earned for correctly setting the derivatives equal or checking candidates

1 point is earned for correctly setting y -values equal or checking candidates

1 point is earned for the correct answer from merged information

$$x = -\frac{3\pi}{2} \text{ or } \frac{\pi}{2}$$

Max: 0 - 1 - 1 - 0 if specific value(s) for b

Max: 0 - 0 - 1 - 0 if no derivative or bald answer

Part C

1 point is earned for the correct answer with reason

No, because $f'(x) = 1$ (or $f'(x) \neq 0$) at x -coordinates of points of tangency



0	1
---	---

The student response earns all of the following points:

1 point is earned for the correct answer with reason

No, because $f'(x) = 1$ (or $f'(x) \neq 0$) at x -coordinates of points of tangency

Part D

1 point is earned for correctly setting second derivative equal to 0

$$f''(x) = -b \sin x = 0$$

2-2-2b

$$\sin x = 0$$

1 point is earned for correctly showing $f(x) = x$

$$f(x) = x + b \cdot 0 = x$$

at x -coordinates of any inflection points

Note: 0/2 if specific value(s) for b



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for correctly setting second derivative equal to 0

$$f''(x) = -b \sin x = 0$$

$$\sin x = 0$$

1 point is earned for correctly showing $f(x) = x$

$$f(x) = x + b \cdot 0 = x$$

at x -coordinates of any inflection points

Note: 0/2 if specific value(s) for b

2-2-2b

14. This problem deals with functions defined by $f(x) = x + b \sin x$, where b is a positive constant and $-2\pi \leq x \leq 2\pi$.

Find the x -coordinates of all points, $-2\pi \leq x \leq 2\pi$, where the line $y = x + b$ is tangent to the graph of $f(x) = x + b \sin x$.

Part B

1 point is earned for the correct derivatives of $x + b$ and $x + b \sin x$

$$y' = 1 = 1 + b \cos x$$

$$b \cos x = 0$$

$$\cos x = 0$$

$$y = x + b = x + b \sin x$$

$$b = b \sin x$$

$$1 = \sin x$$

1 point is earned for correctly setting the derivatives equal or checking candidates

1 point is earned for correctly setting y -values equal or checking candidates

1 point is earned for the correct answer from merged information

$$x = -\frac{3\pi}{2} \text{ or } \frac{\pi}{2}$$

Max: 0 - 1 - 1 - 0 if specific value(s) for b

Max: 0 - 0 - 1 - 0 if no derivative or bald answer



0	1	2	3	4
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The student response earns four of the following points:

1 point is earned for the correct derivatives of $x + b$ and $x + b \sin x$

$$y' = 1 = 1 + b \cos x$$

$$b \cos x = 0$$

$$\cos x = 0$$

$$y = x + b = x + b \sin x$$

$$b = b \sin x$$

$$1 = \sin x$$

1 point is earned for correctly setting the derivatives equal or checking candidates

2-2-2b

1 point is earned for correctly setting y -values equal or checking candidates

1 point is earned for the correct answer from merged information

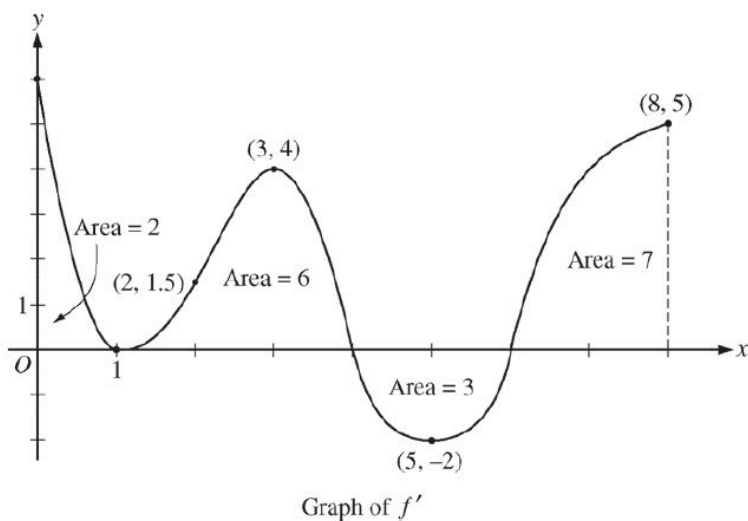
$$x = -\frac{3\pi}{2} \text{ or } \frac{\pi}{2}$$

Max: 0 - 1 - 1 - 0 if specific value(s) for b

Max: 0 - 0 - 1 - 0 if no derivative or bald answer

2-2-2b

15.



The figure above shows the graph of f' , the derivative of a twice-differentiable function f , on the closed interval $0 \leq x \leq 8$. The graph of f' has horizontal tangent lines at $x=1$, $x=3$, and $x=5$. The areas of the regions between the graph of f' and the x -axis are labeled in the figure. The function f is defined for all real numbers and satisfies $f(8)=4$.

The function g is defined by $g(x) = (f(x))^3$. If $f(3) = -5/2$, find the slope of the line tangent to the graph of g at $x = 3$.

Part D

2 points are earned for $g'(x)$

1 point is earned for the answer

$$g'(x) = 3[f(x)]^2 \cdot f'(x)$$

$$g'(3) = 3 \left[f(3) \right]^2 \cdot f'(3) = 3 \left(-\frac{5}{2} \right)^2 \cdot 4 = 75$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

2 points are earned for $g'(x)$

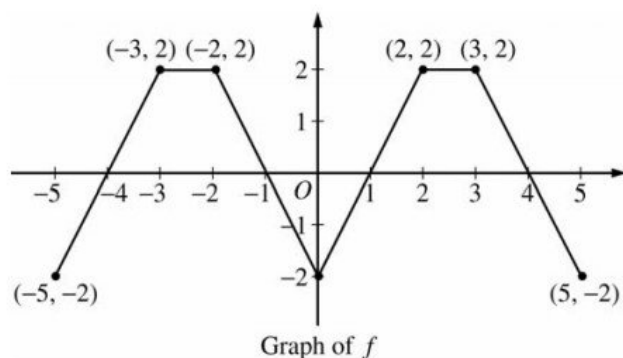
1 point is earned for the answer

$$g'(x) = 3[f(x)]^2 \cdot f'(x)$$

$$g'(3) = 3 \left[f(3) \right]^2 \cdot f'(3) = 3 \left(-\frac{5}{2} \right)^2 \cdot 4 = 75$$

2-2-2b

16.



The graph of the function f shown above consists of six line segments. Let g be the function given by

$$g(x) = \int_0^x f(t) dt.$$

(a) Find $g(4)$, $g'(4)$, and $g''(4)$.

(b) Does g have a relative minimum, a relative maximum, or neither at $x = 1$? Justify your answer.

(c) Suppose that f is defined for all real numbers x and is periodic with a period of length 5. The graph above shows two periods of f . Given that $g(5) = 2$, find $g(10)$ and write an equation for the line tangent to the graph of g at $x = 108$.

Part A

1 point is earned for: $g(4)$

1 point is earned for: $g'(4)$

1 point is earned for: $g''(4)$

$$g(4) = \int_0^4 f(t) dt = 3$$

$$g'(4) = f(4) = 0$$

$$g''(4) = f'(4) = -2$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for: $g(4)$

1 point is earned for: $g'(4)$

1 point is earned for: $g''(4)$

2-2-2b

$$g(4) = \int_0^4 f(t) dt = 3$$

$$g'(4) = f(4) = 0$$

$$g''(4) = f'(4) = -2$$

Part B

1 point is earned for the answer

1 point is earned for the reason

g has a relative minimum at $x = 1$

because $g' = f$ changes from negative to positive at $x = 1$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the answer

1 point is earned for the reason

g has a relative minimum at $x = 1$

because $g' = f$ changes from negative to positive at $x = 1$.

Part C

1 point is earned for: $g(10)$

1 point is earned for: $g(108)$

1 point is earned for: $g'(108)$

1 point is earned for the equation of tangent line

$g(0) = 0$ and the function values of g increase by 2 for every increase of 5 in x .

$$g(10) = 2g(5) = 4$$

2-2-2b

$$\begin{aligned}
 g(108) &= \int_0^{105} f(t)dt + \int_{105}^{108} f(t)dt \\
 &= 21g(5) + g(3) = 44
 \end{aligned}$$

$$g'(108) = f(108) = f(3) = 2$$

An equation for the line tangent to the graph of g at $x = 108$ is $y - 44 = 2(x - 108)$.



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

1 point is earned for: $g(10)$

1 point is earned for: $g(108)$

1 point is earned for: $g'(108)$

1 point is earned for the equation of tangent line

$g(0) = 0$ and the function values of g increase by 2 for every increase of 5 in x .

$$g(10) = 2g(5) = 4$$

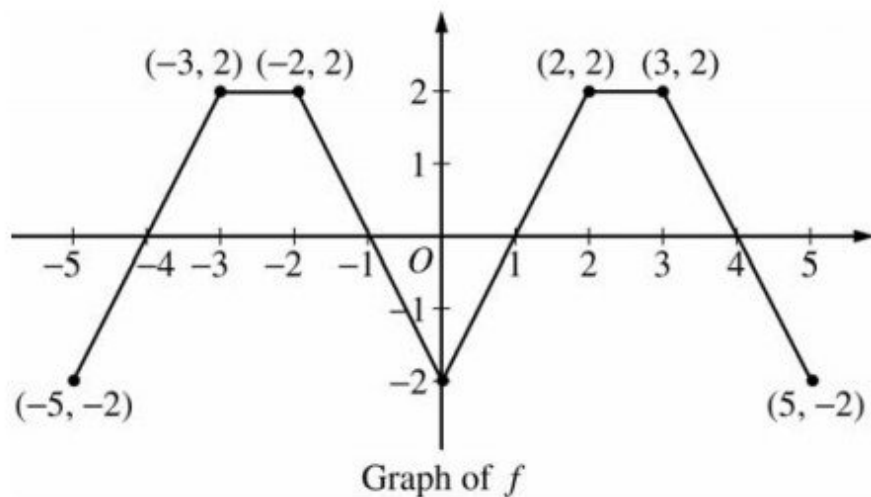
$$\begin{aligned}
 g(108) &= \int_0^{105} f(t)dt + \int_{105}^{108} f(t)dt \\
 &= 21g(5) + g(3) = 44
 \end{aligned}$$

$$g'(108) = f(108) = f(3) = 2$$

An equation for the line tangent to the graph of g at $x = 108$ is $y - 44 = 2(x - 108)$.

2-2-2b

17.



The graph of the function f shown above consists of six line segments. Let g be the function given by $g(x) = \int_0^x f(t) dt$.

Suppose that f is defined for all real numbers x and is periodic with a period of length 5. The graph above shows two periods of f . Given that $g(5) = 2$, find $g(10)$ and write an equation for the line tangent to the graph of g at $x = 108$.

Part C

1 point is earned for: $g(10)$

1 point is earned for: $g(108)$

1 point is earned for: $g'(108)$

1 point is earned for the equation of tangent line

$g(0) = 0$ and the function values of f increase by 2 for every increase of 5 in x .

$$g(10) = 2g(5) = 4$$

$$\begin{aligned} g(108) &= \int_0^{105} f(t) dt + \int_{105}^{108} f(t) dt \\ &= 21g(5) + g(3) = 44 \end{aligned}$$

$$g'(108) = f(108) = f(3) = 2$$

An equation for the line tangent to the graph of g at $x = 108$ is $y - 44 = 2(x - 108)$.



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

2-2-2b

1 point is earned for: $g(10)$

1 point is earned for: $g(108)$

1 point is earned for: $g'(108)$

1 point is earned for the equation of tangent line

$g(0) = 0$ and the function values of f increase by 2 for every increase of 5 in x .

$$g(10) = 2g(5) = 4$$

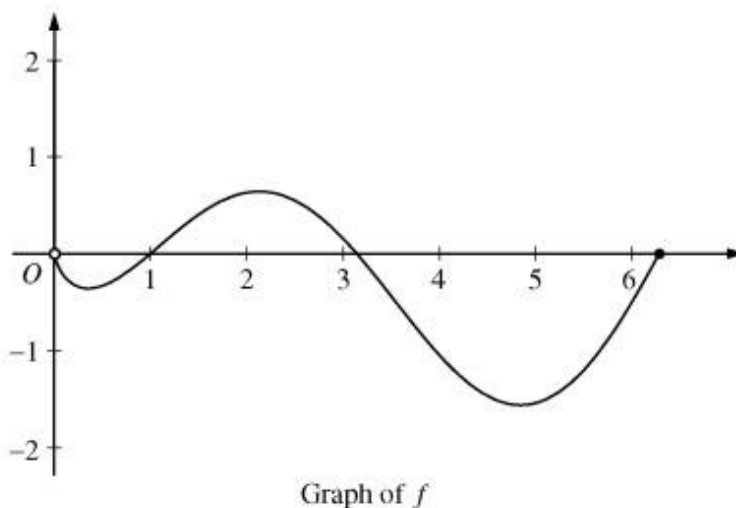
$$\begin{aligned} g(108) &= \int_0^{105} f(t)dt + \int_{105}^{108} f(t)dt \\ &= 21g(5) + g(3) = 44 \end{aligned}$$

$$g'(108) = f(108) = f(3) = 2$$

An equation for the line tangent to the graph of g at $x = 108$ is $y - 44 = 2(x - 108)$.

2-2-2b

18.



Let f be the function given by $f(x) = (\ln x)(\sin x)$. The figure above shows the graph of f for $0 < x \leq 2\pi$.

The function g is defined by $g(x) = \int_1^x f(t) dt$ for $0 < x \leq 2\pi$.

For $0 < x < 2\pi$ is there a value of x at which the graph of g is tangent to the x -axis? Explain why or why not.

Part D

1 point is earned for the answer of "yes" with $x = 1$

1 point is earned for the explanation

Yes, the graph g is tangent to the x -axis at $x = 1$ $g(1) = 0$ and $g'(1) = 0$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the answer of "yes" with $x = 1$

1 point is earned for the explanation

Yes, the graph g is tangent to the x -axis at $x = 1$ $g(1) = 0$ and $g'(1) = 0$

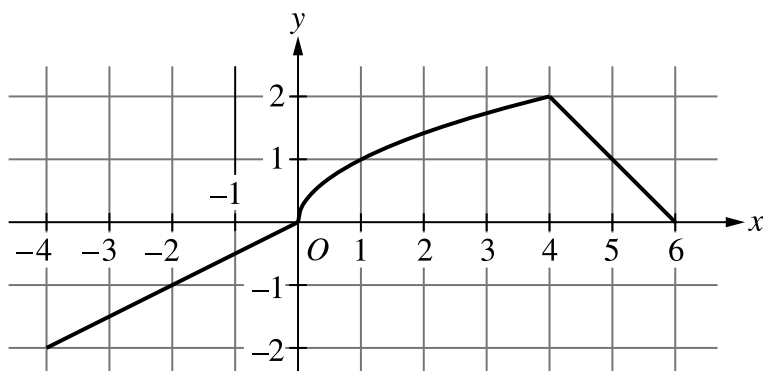
2-2-2b

19. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of g

The graph of the function g is shown on the interval $[-4, 6]$. The graph of g consists of two line segments and the portion of the graph of $y = \sqrt{x}$ for $0 \leq x \leq 4$. Let f be the function defined by

$$f(x) = g(x) + \int_{-4}^x g(t) \, dt.$$

(a) Find the value of $f(4)$.

(b) Show that f has a critical point at $x = -1$. Use the Second Derivative Test to determine whether f has a local minimum or a local maximum at $x = -1$. Show the work that leads to your answer.

(c) Find the value of $\lim_{x \rightarrow 4^-} \frac{g(x) - g(4)}{x - 4}$.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response does not need to present $\int_{-4}^0 g(t) \, dt$ to earn the first point. The value of ± 4 may be determined directly from the graph.

The inclusion of $g(4)$ is considered in the third point.

The second point is earned by finding the antiderivative $\frac{2}{3}t^{3/2}$ or by determining the area under this portion of the parabola is $\frac{2}{3} \cdot 8$.

2-2-2b

A response that finds an antiderivative of the form $k \cdot t^{3/2}$ where $k \neq 0$ and $k \neq \frac{2}{3}$ does not earn the second point but does earn the third point for the consistent answer of $-2 + k \cdot 8$.

The final answer does not need to be simplified. Any error in simplification does not earn the third point.



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ $\pm$ value of $\int_{-4}^0 g(t) dt$
- ☐ Antiderivative of $\sqrt{t}$
- ☐ Answer

Solution:

$$\begin{aligned}
 f(4) &= g(4) + \int_{-4}^4 g(t) dt = 2 + \int_{-4}^0 g(t) dt + \int_0^4 g(t) dt \\
 &= 2 + (-4) + \int_0^4 \sqrt{t} dt = -2 + \left[\frac{2}{3} t^{3/2} \right]_{t=0}^{t=4} \\
 &= -2 + \frac{2}{3} \cdot 8 = \frac{10}{3}
 \end{aligned}$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is only earned for the statement $f'(x) = g'(x) + g(x)$ or If there is any equating of unequal quantities, or miscommunication, for example, the response does not earn the first point.

If the first point is not earned, the second point can only be earned for a response that presents

If the first point is earned, a response can earn the second point by stating that

To earn the third point, a response must correctly find or and go on to state that

The fourth point can be earned without any of the first 3 points, provided the response declares a nonzero value or sign of and makes a consistent Second Derivative Test argument based on that value or sign.



0	1	2	3	4
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The student response accurately includes **all four** of the criteria below.

2-2-2b

- ☐ $f'(x) = g'(x) + g(x)$ OR $f'(-1) = g'(-1) + g(-1)$
- ☐ Shows $f'(-1) = 0$
- ☐ $f''(-1)$
- ☐ Answer with reason

Solution:

$$f'(x) = g'(x) + g(x)$$

$$f'(-1) = g'(-1) + g(-1) = \frac{1}{2} + \left(-\frac{1}{2}\right) = 0$$

Because $f'(-1) = 0$, f has a critical point at $x = -1$.

$$f''(x) = g''(x) + g'(x)$$

$$f''(-1) = g''(-1) + g'(-1) = 0 + \frac{1}{2}$$

Because $f'(-1) = 0$ and $f''(-1) > 0$, the Second Derivative Test indicates that f has a local minimum at $x = -1$.

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The response does not need to state that $\sqrt{x}$ is differentiable.

A minimal response to earn both points is $\frac{1}{2\sqrt{4}}$.

Special case (using L'Hospital's Rule): For $0 \leq x \leq 4$, $g(x) = \sqrt{x}$.

$$\lim_{x \rightarrow 4^-} (\sqrt{x} - g(4)) = 2 - 2 = 0 \text{ and } \lim_{x \rightarrow 4^-} (x - 4) = 0.$$

Because $\sqrt{x}$ is continuous on the interval $[0, 4]$ and differentiable on the interval $(0, 4)$,

$\lim_{x \rightarrow 4^-} [g(x) - g(4)] = 2 - 2 = 0$, and $\lim_{x \rightarrow 4^-} [x - 4] = 4 - 4 = 0$, L'Hospital's Rule may be applied:

$$\lim_{x \rightarrow 4^-} \frac{g(x) - g(4)}{x - 4} = \lim_{x \rightarrow 4^-} \frac{g'(x)}{1} = \lim_{x \rightarrow 4^-} \frac{\frac{d}{dx} \sqrt{x}}{1} = \lim_{x \rightarrow 4^-} \frac{\frac{1}{2\sqrt{x}}}{1} = \frac{1}{2\sqrt{4}} = \frac{1}{4}.$$

- A response earns the first point by evaluating the limits of the numerator and denominator.
- A response that equates the limit to $\frac{0}{0}$ does not earn the first point but is eligible for the second point. A response may say that the limit is of indeterminate form $\frac{0}{0}$.

2-2-2b

- The second point is earned by correctly applying L'Hospital's Rule (by presenting a limit of a ratio of derivatives) and finding the correct answer.



0	1	2
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The student response accurately includes **both** of the criteria below.

- ☐ Uses $g(x) = \sqrt{x}$
- ☐ Answer

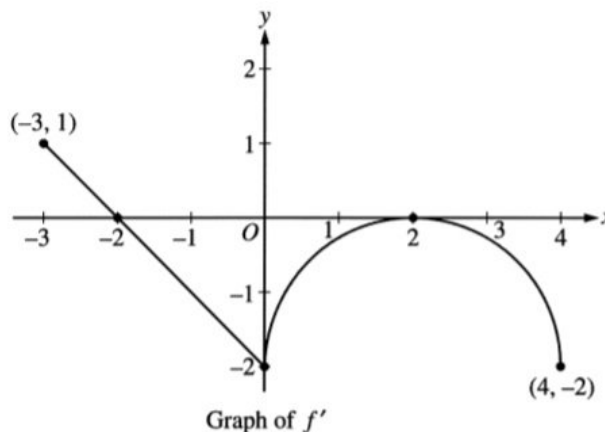
Solution:

For $0 \leq x \leq 4$, $g(x) = \sqrt{x}$.

Because $\sqrt{x}$ has a left-hand derivative at $x = 4$, $\lim_{x \rightarrow 4^-} \frac{g(x) - g(4)}{x - 4} = \frac{d}{dx} \sqrt{x} \Big|_{x=4} = \frac{1}{2\sqrt{x}} \Big|_{x=4} = \frac{1}{4}$.

2-2-2b

20.



Let f be a function defined on the closed interval $-3 \leq x \leq 4$ with $f(0) = 3$. The graph of f' , the derivative of f , consists of one line segment and a semicircle, as shown above.

- On what intervals, if any, is f increasing? Justify your answer.
- Find the x -coordinate of each point of inflection of the graph of f on the open interval $-3 < x < 4$. Justify your answer.
- Find an equation for the line tangent to the graph of f at the point $(0, 3)$.
- Find $f(-3)$ and $f(4)$. Show the work that leads to your answers.

Part A

1 point is earned for the correct interval

1 point is earned for the correct reason

The function f is increasing on $[-3, -2]$ since $f' > 0$ for $-3 \leq x < -2$.



0	1	2
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The student response earns all of the following points:

1 point is earned for the correct interval

1 point is earned for the correct reason

The function f is increasing on $[-3, -2]$ since $f' > 0$ for $-3 \leq x < -2$.

Part B

1 point is earned for the correct answers $x = 0$ and $x = 2$ only

1 point is earned for the justification

2-2-2b

$x = 0$ and $x = 2$

f' changes from decreasing to increasing at $x = 0$ and from increasing to decreasing at $x = 2$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the correct answers $x = 0$ and $x = 2$ only

1 point is earned for the justification

$x = 0$ and $x = 2$

f' changes from decreasing to increasing at $x = 0$ and from increasing to decreasing at $x = 2$

Part C

1 point is earned for the correct equation

$f(0) = -2$

Tangent line is $y = -2x + 3$.



0	1
---	---

The student response earns all of the following points:

1 point is earned for the correct equation

$f(0) = -2$

Tangent line is $y = -2x + 3$.

Part D

1 point is earned for the difference of areas of triangles, $\pm \left(\frac{1}{2} - 2 \right)$

$$\begin{aligned}
 f(0) - f(-3) &= \int_{-3}^0 f'(t) dt \\
 &= \frac{1}{2}(1)(1) - \frac{1}{2}(2)(2) = -\frac{3}{2}
 \end{aligned}$$

2-2-2b

1 point is earned for the correct answer for $f(-3)$ using FTC

$$f(-3) = f(0) + \frac{3}{2} = \frac{9}{2}$$

1 point is earned for the area of rectangle - area of semicircle, $\pm \left(8 - \frac{1}{2}(2)^2\pi\right)$

$$\begin{aligned} f(4) - f(0) &= \int_0^4 f'(t) dt \\ &= -\left(8 - \frac{1}{2}(2)^2\pi\right) = -8 + 2\pi \end{aligned}$$

1 point is earned for the correct answer of $f(4)$ using FTC

$$f(4) = f(0) - 8 + 2\pi = -5 + 2\pi$$



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

1 point is earned for the difference of areas of triangles, $\pm \left(\frac{1}{2} - 2\right)$

$$\begin{aligned} f(0) - f(-3) &= \int_{-3}^0 f'(t) dt \\ &= \frac{1}{2}(1)(1) - \frac{1}{2}(2)(2) = -\frac{3}{2} \end{aligned}$$

1 point is earned for the correct answer for $f(-3)$ using FTC

$$f(-3) = f(0) + \frac{3}{2} = \frac{9}{2}$$

1 point is earned for the area of rectangle - area of semicircle, $\pm \left(8 - \frac{1}{2}(2)^2\pi\right)$

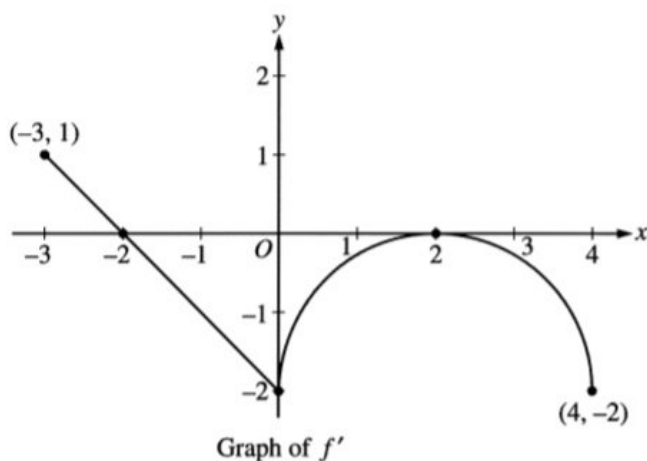
$$\begin{aligned} f(4) - f(0) &= \int_0^4 f'(t) dt \\ &= -\left(8 - \frac{1}{2}(2)^2\pi\right) = -8 + 2\pi \end{aligned}$$

2-2-2b

1 point is earned for the correct answer of $f(4)$ using FTC

$$f(4) = f(0) - 8 + 2\pi = -5 + 2\pi$$

21.



Let f be a function defined on the closed interval $-3 \leq x \leq 4$ with $f(0) = 3$. The graph of f' , the derivative of f , consists of one line segment and a semicircle, as shown above.

Find an equation for the line tangent to the graph of f at the point $(0, 3)$.

Part C

1 point is earned for the correct equation

$$f'(0) = -2$$

Tangent line is $y = -2x + 3$.



0	1
---	---

The student response earns one of the following points:

1 point is earned for the correct equation

$$f'(0) = -2$$

Tangent line is $y = -2x + 3$.

2-2-2b

22.  Grass clippings are placed in a bin, where they decompose. For $0 \leq t \leq 30$, the amount of grass clippings remaining in the bin is modeled by $A(t) = 6.687(0.931)^t$, where $A(t)$ is measured in pounds and t is measured in days.
- (a) Find the average rate of change of $A(t)$ over the interval $0 \leq t \leq 30$. Indicate units of measure.
- (b) Find the value of $A'(15)$. Using correct units, interpret the meaning of the value in the context of the problem.
- (c) Find the time t for which the amount of grass clippings in the bin is equal to the average amount of grass clippings in the bin over the interval $0 \leq t \leq 30$.
- (d) For $t > 30$, $L(t)$, the linear approximation to A at $t = 30$, is a better model for the amount of grass clippings remaining in the bin. Use $L(t)$ to predict the time at which there will be 0.5 pound of grass clippings remaining in the bin. Show the work that leads to your answer.

Part A

1 point is earned for the answer with units

$$\frac{A(30) - A(0)}{30 - 0} = -0.197 \text{ (or } -0.196) \text{ lbs/day}$$



0	1
---	---

The student response earns all of the following points:

1 point is earned for the answer with units

$$\frac{A(30) - A(0)}{30 - 0} = -0.197 \text{ (or } -0.196) \text{ lbs/day}$$

Part B

1 point is earned for $A'(15)$

1 point is earned for the interpretation

$$A'(15) = -0.164 \text{ (or } -0.163)$$

The amount of grass clippings in the bin is decreasing at a rate of 0.164 (or 0.163) lbs/day at time $t = 15$ days.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for $A'(15)$

2-2-2b

1 point is earned for the interpretation

$$A'(15) = -0.164 \text{ (or } -0.163)$$

The amount of grass clippings in the bin is decreasing at a rate of 0.164 (or 0.163) lbs/day at time $t = 15$ days.

Part C

1 point is earned for $\frac{1}{30} \int_0^{30} A(t) dt$

1 point is earned for the answer

$$A(t) = \frac{1}{30} \int_0^{30} A(t) dt \Rightarrow t = 12.415 \text{ (or } 12.414)$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for $\frac{1}{30} \int_0^{30} A(t) dt$

1 point is earned for the answer

$$A(t) = \frac{1}{30} \int_0^{30} A(t) dt \Rightarrow t = 12.415 \text{ (or } 12.414)$$

Part D

2 points are earned for expression for $L(t)$

1 point is earned for $L(t) = 0.5$

1 point is earned for the answer

$$L(t) = A(30) + A'(30) \cdot (t - 30)$$

$$A'(30) = -0.055976$$

2-2-2b

$$A(30) = 0.782928$$

$$L(t) = 0.5 \div t = 35.054$$



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

2 points are earned for expression for $L(t)$

1 point is earned for $L(t) = 0.5$

1 point is earned for the answer

$$L(t) = A(30) + A'(30) \cdot (t - 30)$$

$$A'(30) = -0.055976$$

$$A(30) = 0.782928$$

$$L(t) = 0.5 \div t = 35.054$$

2-2-2b

23. Grass clippings are placed in a bin, where they decompose. For $0 \leq t \leq 30$, the amount of grass clippings remaining in the bin is modeled by $A(t) = 6.687(0.931)^t$, where $A(t)$ is measured in pounds and t is measured in days.



For $t > 30$, $L(t)$, the linear approximation to A at $t = 30$, is a better model for the amount of grass clippings remaining in the bin. Use $L(t)$ to predict the time at which there will be 0.5 pound of grass clippings remaining in the bin. Show the work that leads to your answer.

Part D

2 point is earned for expression for $L(t)$

1 point is earned for $L(t) = 0.5$

1 point is earned for the answer

$$L(t) = A(30) + A'(30) \cdot (t - 30)$$

$$A'(30) = -0.055976$$

$$A(30) = 0.782928$$

$$L(t) = 0.5 \Rightarrow t = 35.054$$



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

2 point is earned for expression for $L(t)$

1 point is earned for $L(t) = 0.5$

1 point is earned for the answer

$$L(t) = A(30) + A'(30) \cdot (t - 30)$$

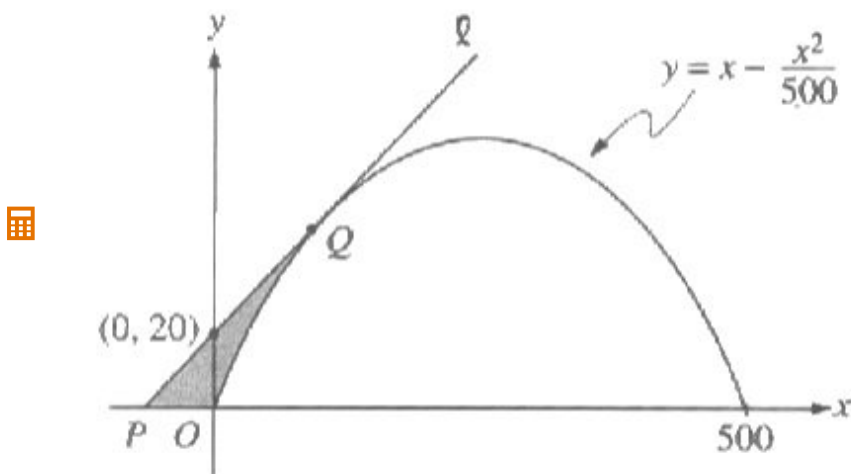
$$A'(30) = -0.055976$$

$$A(30) = 0.782928$$

$$L(t) = 0.5 \Rightarrow t = 35.054$$

2-2-2b

24.



Line ℓ is tangent to the graph of $y = x - \frac{x^2}{500}$ at the point Q , as shown in the figure above.

(a) Find the x -coordinate of point Q .

(b) Write an equation for line ℓ .

(c) Suppose the graph $y = x - \frac{x^2}{500}$ shown in the figure, where x and y are measured in feet, represents a hill. There is a 50-foot tree growing vertically at the top of the hill. Does a spotlight at point P directed along line ℓ shine on any part of the tree? Show the work that leads to your conclusion.

Part A

1 point is earned for the correct slope of tangent line from parabola

Let Q be $\left(a, a - \frac{a^2}{500}\right)$

$$\frac{dy}{dx} = 1 - \frac{x}{250}$$

1 point is earned for correctly using the condition that $\left(a, a - \frac{a^2}{500}\right)$ is on line ℓ

Setting slopes equal:

$$1 - \frac{a}{250} = \frac{\left(a - \frac{a^2}{500}\right) - 20}{a}$$

$$a = 100$$

2-2-2b

OR

Setting y -values equal:

$$\left(1 - \frac{a}{250}\right)a + 20 = a - \frac{a^2}{500}$$

1 point is earned for correctly using the condition that variable slopes are equal at Q

$$1 - \frac{a}{250}$$

OR

Equation for ℓ :

$$y = \left(1 - \frac{a}{250}\right)x + 20$$

1 point is earned for the correct answer (bald ok)

0/1 if student is solving an irrelevant equation

$$a = 100$$

max 2/4 if only uses calculator (and demonstrates)



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

1 point is earned for the correct slope of tangent line from parabola

$$\text{Let } Q \text{ be } \left(a, a - \frac{a^2}{500}\right)$$

$$\frac{dy}{dx} = 1 - \frac{x}{250}$$

2-2-2b

1 point is earned for correctly using the condition that $\left(a, a - \frac{a^2}{500}\right)$ is on line ℓ

Setting slopes equal:

$$1 - \frac{a}{250} = \frac{\left(a - \frac{a^2}{500}\right) - 20}{a}$$

$$a = 100$$

OR

Setting y-values equal:

$$\left(1 - \frac{a}{250}\right)a + 20 = a - \frac{a^2}{500}$$

1 point is earned for correctly using the condition that variable slopes are equal at Q

$$1 - \frac{a}{250}$$

OR

Equation for ℓ :

$$y = \left(1 - \frac{a}{250}\right)x + 20$$

1 point is earned for the correct answer (bald ok)

0/1 if student is solving an irrelevant equation

$$a = 100$$

max 2/4 if only uses calculator (and demonstrates)

Part B

1 point is earned for the correct slope

0/1 if $m \leq 0$

2-2-2b

Note: first type in (a) ok if works in either equation of slope or must commit to value in (a) and use it second type must use value from (a)

1 point is earned for the correct equation and commitment to slope

0/1 bald - no a value in (a)

$$y = \frac{3}{5}x + 20$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the correct slope

0/1 if $m \leq 0$

Note: first type in (a) ok if works in either equation of slope or must commit to value in (a) and use it second type must use value from (a)

1 point is earned for the correct equation and commitment to slope

0/1 bald - no a value in (a)

$$y = \frac{3}{5}x + 20$$

Part C

1 point is earned for the correct height of hill or height of hill and tree = 175 any place on page

$$\text{Height of hill at } x = 250; y = 250 - \frac{250^2}{500} = 125 \text{ feet}$$

1 point is earned for the correct height of line

0/1 if height < 0

$$\text{Height of line at } x = 250; y = \frac{3}{5}(250) + 20 = 170 \text{ feet}$$

2-2-2b

1 point is earned for the correct answer with analysis discussing the tree

0/1 if height < 0

Yes, the spotlight hits the tree since the height of the line is less than the height of the hill + tree which is 175 feet.

max 1 - 1 - 0 if height of line < height of hill



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the correct height of hill or height of hill and tree = 175 any place on page

Height of hill at $x = 250$; $y = 250 - \frac{250^2}{500} = 125$ feet

1 point is earned for the correct height of line

0/1 if height < 0

Height of line at $x = 250$; $y = \frac{3}{5}(250) + 20 = 170$ feet

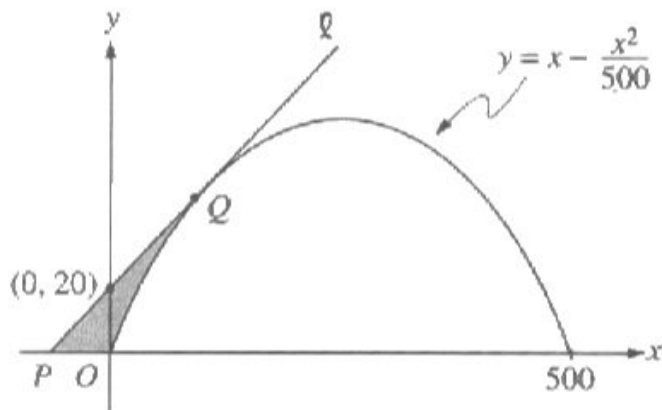
1 point is earned for the correct answer with analysis discussing the tree

0/1 if height < 0

Yes, the spotlight hits the tree since the height of the line is less than the height of the hill + tree which is 175 feet.

max 1 - 1 - 0 if height of line < height of hill

2-2-2b



Line l is tangent to the graph of $y = x - \frac{x^2}{500}$ at the point Q , as shown in the figure above.

25. Find the x -coordinate of point Q .

Part A

1 point is earned for the correct slope of tangent line from parabola

Let Q be $(a, a - \frac{a^2}{500})$

$$\frac{dy}{dx} = 1 - \frac{x}{250}$$

1 point is earned for correctly using the condition that $(a, a - \frac{a^2}{500})$ is on line l

Setting slopes equal:

$$1 - \frac{a}{250} = \frac{(a - \frac{a^2}{500}) - 20}{a}$$

$$a = 100$$

OR

Setting y -values equal:

$$(1 - \frac{a}{250})a + 20 = a - \frac{a^2}{500}$$

1 point is earned for correctly using the condition that variable slopes are equal at Q

$$1 - \frac{a}{250}$$

OR

Equation for l :

2-2-2b

$$y = \left(1 - \frac{a}{250}\right)x + 20$$

1 point is earned for the correct answer (bald ok)

0/1 if student is solving an irrelevant equation

$$a = 100$$

max 2/4 if only uses calculator (and demonstrates)



0	1	2	3	4
---	---	---	---	---

The student response earns four of the following points:

1 point is earned for the correct slope of tangent line from parabola

Let Q be $\left(a, a - \frac{a^2}{500}\right)$

$$\frac{dy}{dx} = 1 - \frac{x}{250}$$

1 point is earned for correctly using the condition that $\left(a, a - \frac{a^2}{500}\right)$ is on line l

Setting slopes equal:

$$1 - \frac{a}{250} = \frac{\left(a - \frac{a^2}{500}\right) - 20}{a}$$

$$a = 100$$

OR

Setting y -values equal:

$$\left(1 - \frac{a}{250}\right)a + 20 = a - \frac{a^2}{500}$$

1 point is earned for correctly using the condition that variable slopes are equal at Q

$$1 - \frac{a}{250}$$

OR

Equation for l :

$$y = \left(1 - \frac{a}{250}\right)x + 20$$

1 point is earned for the correct answer (bald ok)

2-2-2b

0/1 if student is solving an irrelevant equation

$$a = 100$$

max 2/4 if only uses calculator (and demonstrates)

26. Write an equation for line l .

Part B

1 point is earned for the correct slope

0/1 if $m \leq 0$

Note: first type in (a) ok if works in either equation of slope or must commit to value in (a) and use it second type must use value from (a)

1 point is earned for the correct equation and commitment to slope

0/1 bald - no a value in (a)

$$y = \frac{3}{5}x + 20$$



0	1	2
---	---	---

The student response earns two of the following points:

1 point is earned for the correct slope

0/1 if $m \leq 0$

Note: first type in (a) ok if works in either equation of slope or must commit to value in (a) and use it second type must use value from (a)

1 point is earned for the correct equation and commitment to slope

0/1 bald - no a value in (a)

$$y = \frac{3}{5}x + 20$$

2-2-2b

27. In the xy -plane, the line $x + y = k$, where k is a constant, is tangent to the graph of $y = x^2 + 3x + 1$. What is the value of k ?

(A) -3



(B) -2

(C) -1

(D) 0

(E) 1

Consider the closed curve in the xy -plane given by $x^2 + 2x + y^4 + 4y = 5$.

28. Write an equation for the line tangent to the curve at the point $(-2, 1)$.

Part B

1 point is earned for the slope

1 point is earned for the tangent line equation

$$\left. \frac{dy}{dx} \right|_{(-2,1)} = \frac{-(-2+1)}{2(1+1)} = \frac{1}{4}$$

$$\text{Tangent line: } y = 1 + \frac{1}{4}(x + 2)$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the slope

1 point is earned for the tangent line equation

$$\left. \frac{dy}{dx} \right|_{(-2,1)} = \frac{-(-2+1)}{2(1+1)} = \frac{1}{4}$$

$$\text{Tangent line: } y = 1 + \frac{1}{4}(x + 2)$$

2-2-2b

29. Is it possible for this curve to have a horizontal tangent at points where it intersects the x -axis? Explain your reasoning.

Part D

1 point is earned if works with $x = -1$ or $y = 0$

1 point is earned for answer with reason

Horizontal tangents occur at points on the curve where $x = -1$ and $y \neq -1$.

The curve crosses the x -axis where $y = 0$.

$$(-1)^2 + 2(-1) + 0^4 + 4 \cdot 0 \neq 5$$

No, the curve cannot have a horizontal tangent where it crosses the x -axis.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned if works with $x = -1$ or $y = 0$

1 point is earned for answer with reason

Horizontal tangents occur at points on the curve where $x = -1$ and $y \neq -1$.

The curve crosses the x -axis where $y = 0$.

$$(-1)^2 + 2(-1) + 0^4 + 4 \cdot 0 \neq 5$$

No, the curve cannot have a horizontal tangent where it crosses the x -axis.

2-2-2b

30. Find the coordinates of the two points on the curve where the line tangent to the curve is vertical.
-

Part C

1 point is earned for $y = -1$

1 point is earned if substitutes $y = -1$ into the equation of the curve

1 point is earned for the answer

Vertical tangent lines occur at points on the curve where $y^3 + 1 = 0$ (or $y = -1$) and $x \neq -1$.

On the curve, $y = -1$ implies that $x^2 + 2x + 1 - 4 = 5$, so $x = -4$ or $x = 2$. Vertical tangent lines occur at the points $(-4, -1)$ and $(2, -1)$.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for $y = -1$

1 point is earned if substitutes $y = -1$ into the equation of the curve

1 point is earned for the answer

Vertical tangent lines occur at points on the curve where $y^3 + 1 = 0$ (or $y = -1$) and $x \neq -1$.

On the curve, $y = -1$ implies that $x^2 + 2x + 1 - 4 = 5$, so $x = -4$ or $x = 2$. Vertical tangent lines occur at the points $(-4, -1)$ and $(2, -1)$.

2-2-2b

31. Consider the closed curve in the xy -plane given by

$$x^2 + 2x + y^4 + 4y = 5.$$

- (a) Show that $dy/dx = -(x + 1) / 2(y^3 + 1)$.
 (b) Write an equation for the line tangent to the curve at the point $(-2, 1)$.
 (c) Find the coordinates of the two points on the curve where the line tangent to the curve is vertical.
 (d) Is it possible for this curve to have a horizontal tangent at points where it intersects the x -axis? Explain your reasoning.

Part A

1 point is earned for implicit differentiation

1 point is earned for verification

$$2x + 2 + 4y^3 \frac{dy}{dx} + 4 \frac{dy}{dx} = 0$$

$$(4y^3 + 4) \frac{dy}{dx} = -2x - 2$$

$$\frac{dy}{dx} = \frac{-2(x+1)}{4(y^3+1)} = \frac{-(x+1)}{2(y^3+1)}$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for implicit differentiation

1 point is earned for verification

$$2x + 2 + 4y^3 \frac{dy}{dx} + 4 \frac{dy}{dx} = 0$$

$$(4y^3 + 4) \frac{dy}{dx} = -2x - 2$$

$$\frac{dy}{dx} = \frac{-2(x+1)}{4(y^3+1)} = \frac{-(x+1)}{2(y^3+1)}$$

Part B

1 point is earned for the slope

1 point is earned for the tangent line equation

2-2-2b

$$\left. \frac{dy}{dx} \right|_{(-2,1)} = \frac{-(-2+1)}{2(1+1)} = \frac{1}{4}$$

Tangent line: $y = 1 + \frac{1}{4}(x + 2)$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the slope

1 point is earned for the tangent line equation

$$\left. \frac{dy}{dx} \right|_{(-2,1)} = \frac{-(-2+1)}{2(1+1)} = \frac{1}{4}$$

Tangent line: $y = 1 + \frac{1}{4}(x + 2)$

Part C

1 point is earned for $y = -1$

1 point is earned if substitutes $y = -1$ into the equation of the curve

1 point is earned for the answer

Vertical tangent lines occur at points on the curve where $y^3 + 1 = 0$ (or $y = -1$) and $x \neq -1$.

On the curve, $y = -1$ implies that $x^2 + 2x + 1 - 4 = 5$, so $x = -4$ or $x = 2$.

Vertical tangent lines occur at the points $(-4, -1)$ and $(2, -1)$.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for $y = -1$

1 point is earned if substitutes $y = -1$ into the equation of the curve

1 point is earned for the answer

2-2-2b

Vertical tangent lines occur at points on the curve where $y^3 + 1 = 0$ (or $y = -1$) and $x \neq -1$.

On the curve, $y = -1$ implies that $x^2 + 2x + 1 - 4 = 5$, so $x = -4$ or $x = 2$.

Vertical tangent lines occur at the points $(-4, -1)$ and $(2, -1)$.

Part D

1 point is earned if works with $x = -1$ or $y = 0$

1 point is earned for answer with reason

Horizontal tangents occur at points on the curve where $x = -1$ and $y \neq -1$.

The curve crosses the x -axis where $y = 0$.

$$(-1)^2 + 2(-1) + 0^4 + 4 \cdot 0 \neq 5$$

No, the curve cannot have a horizontal tangent where it crosses the x -axis.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned if works with $x = -1$ or $y = 0$

1 point is earned for answer with reason

Horizontal tangents occur at points on the curve where $x = -1$ and $y \neq -1$.

The curve crosses the x -axis where $y = 0$.

$$(-1)^2 + 2(-1) + 0^4 + 4 \cdot 0 \neq 5$$

No, the curve cannot have a horizontal tangent where it crosses the x -axis.

2-2-2b

Consider the curve given by $y^2 = 2 + xy$.

32. Show that there are no points (x,y) on the curve where the line tangent to the curve is horizontal.

Part C

1 point is earned for $y=0$

1 point is earned for the explanation

$$\frac{y}{2y-x} = 0$$

$$y=0$$

The curve has no horizontal tangent since

$$0^2 \neq 2 + x \cdot 0 \text{ for any } x.$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for $y=0$

1 point is earned for the explanation

$$\frac{y}{2y-x} = 0$$

$$y=0$$

The curve has no horizontal tangent since

$$0^2 \neq 2 + x \cdot 0 \text{ for any } x.$$

2-2-2b

33. Find all points (x, y) on the curve where the line tangent to the curve has slope $1/2$.
-

Part B

1 point is earned for $\frac{y}{2y-x} = \frac{1}{2}$

1 point is earned for the answer

$$\frac{y}{2y-x} = \frac{1}{2}$$

$$2y = 2y - x$$

$$x = 0$$

$$y = \pm\sqrt{2}$$

$$(0, \sqrt{2}), (0, -\sqrt{2})$$



0	1	2
---	---	---

The student response earns two of the following points:

1 point is earned for $\frac{y}{2y-x} = \frac{1}{2}$

1 point is earned for the answer

$$\frac{y}{2y-x} = \frac{1}{2}$$

$$2y = 2y - x$$

$$x = 0$$

$$y = \pm\sqrt{2}$$

$$(0, \sqrt{2}), (0, -\sqrt{2})$$

2-2-2b

34. The continuous, piecewise linear function f consists of four line segments. The table shown gives values of f at the endpoints of the line segments.

x	0	8	10	13	15
$f(x)$	2	6	8	9	1

Which of the following values is greatest?

(A) $f'(7)$

(B) $f'(9)$

(C) $f'(11)$

(D) $f'(14)$

**Answer B**

Correct. Since f is piecewise linear, the values of $f'(x)$ are as follows.

$$f'(7) = \frac{f(8) - f(0)}{8 - 0} = \frac{6 - 2}{8} = \frac{1}{2}$$

$$f'(9) = \frac{f(10) - f(8)}{10 - 8} = \frac{8 - 6}{2} = 1$$

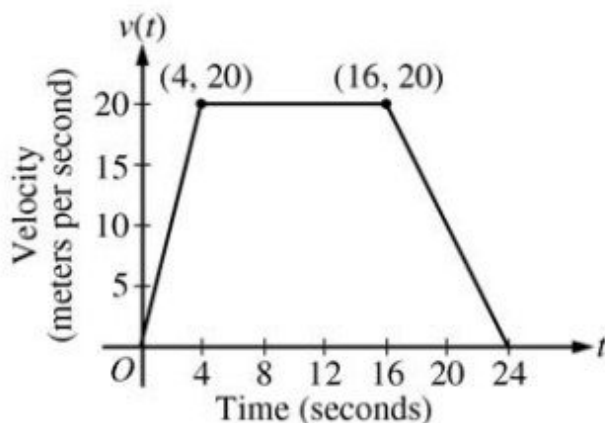
$$f'(11) = \frac{f(13) - f(10)}{13 - 10} = \frac{9 - 8}{3} = \frac{1}{3}$$

$$f'(14) = \frac{f(15) - f(13)}{15 - 13} = \frac{1 - 9}{2} = -4$$

The greatest value is $f'(9)$.

2-2-2b

35.



A car is traveling on a straight road. For $0 \leq t \leq 24$ seconds, the car's velocity $v(t)$, in meters per second, is modeled by the piecewise-linear function defined by the graph above.

For each of $v'(4)$ and $v'(20)$, find the value or explain why it does not exist. Indicate units of measure.

Part B

1 point is earned for: $v'(4)$ does not exist, with explanation

1 point is earned for: $v'(20)$

1 point is earned for the units

$v'(4)$ does not exist because

$$\lim_{t \rightarrow 4^-} \left(\frac{v(t) - v(4)}{t - 4} \right) = 5 \neq 0 = \lim_{t \rightarrow 4^+} \left(\frac{v(t) - v(4)}{t - 4} \right)$$

$$v'(20) = \frac{20 - 0}{16 - 24} = -\frac{5}{2} \text{ m/sec}^2$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for: $v'(4)$ does not exist, with explanation

1 point is earned for: $v'(20)$

1 point is earned for the units

$v'(4)$ does not exist because

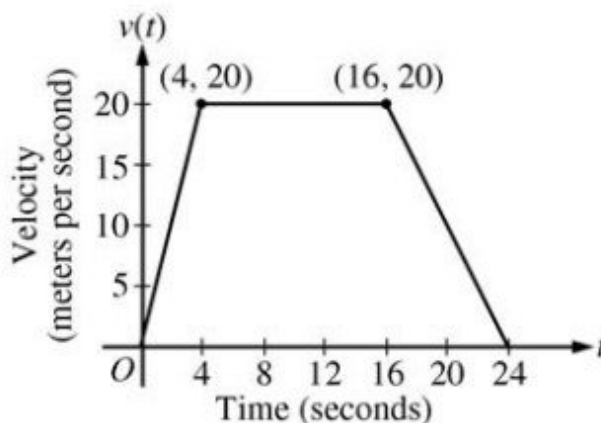
2-2-2b

$$\lim_{t \rightarrow 4^-} \left(\frac{v(t) - v(4)}{t - 4} \right) = 5 \neq 0 = \lim_{t \rightarrow 4^+} \left(\frac{v(t) - v(4)}{t - 4} \right)$$

$$v'(20) = \frac{20 - 0}{16 - 24} = -\frac{5}{2} m / \text{sec}^2$$

2-2-2b

36.



A car is traveling on a straight road. For $0 \leq t \leq 24$ seconds, the car's velocity $v(t)$, in meters per second, is modeled by the piecewise-linear function defined by the graph above.

- (a) Find $\int_0^{24} v(t) dt$. Using correct units, explain the meaning of $\int_0^{24} v(t) dt$.
- (b) For each of $v'(4)$ and $v'(20)$, find the value or explain why it does not exist. Indicate units of measure.
- (c) Let $a(t)$ be the car's acceleration at time t , in meters per second per second. For $0 < t < 24$, write a piecewise-defined function for $a(t)$.
- (d) Find the average rate of change of v over the interval $8 \leq t \leq 20$. Does the Mean Value Theorem guarantee a value of c , for $8 < c < 20$, such that $v'(c)$ is equal to this average rate of change? Why or why not?

Part A

1 point is earned for the value

1 point is earned for the meaning with units

$$\int_0^{24} v(t) dt = \frac{1}{2}(4)(20) + (12)(20) + \frac{1}{2}(8)(20) = 360$$

The car travels 360 meters in these 24 seconds.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the value

1 point is earned for the meaning with units

2-2-2b

$$\int_0^{24} v(t) dt = \frac{1}{2}(4)(20) + (12)(20) + \frac{1}{2}(8)(20) = 360$$

The car travels 360 meters in these 24 seconds.

Part B

1 point is earned for: $v'(4)$ does not exist, with explanation

1 point is earned for: $v'(20)$

1 point is earned for the units

$v'(4)$ does not exist because

$$\lim_{t \rightarrow 4^-} \left(\frac{v(t) - v(4)}{t - 4} \right) = 5 \neq 0 = \lim_{t \rightarrow 4^+} \left(\frac{v(t) - v(4)}{t - 4} \right)$$

$$v'(20) = \frac{20 - 0}{16 - 24} = -\frac{5}{2} \text{ m/sec}^2$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for: $v'(4)$ does not exist, with explanation

1 point is earned for: $v'(20)$

1 point is earned for the units

$v'(4)$ does not exist because

$$\lim_{t \rightarrow 4^-} \left(\frac{v(t) - v(4)}{t - 4} \right) = 5 \neq 0 = \lim_{t \rightarrow 4^+} \left(\frac{v(t) - v(4)}{t - 4} \right)$$

$$v'(20) = \frac{20 - 0}{16 - 24} = -\frac{5}{2} \text{ m/sec}^2$$

Part C

1 point is earned for finds the values 5, 0, $-\frac{5}{2}$

1 point is earned for identifies constants with correct intervals

2-2-2b

$$a(t) = \begin{cases} 5 & \text{if } 0 < t < 4 \\ 0 & \text{if } 4 < t < 16 \\ -\frac{5}{2} & \text{if } 16 < t < 24 \end{cases}$$

$a(t)$ does not exist at $t = 4$ and $t = 16$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for finds the values 5 , 0 , $-\frac{5}{2}$

1 point is earned for identifies constants with correct intervals

$$a(t) = \begin{cases} 5 & \text{if } 0 < t < 4 \\ 0 & \text{if } 4 < t < 16 \\ -\frac{5}{2} & \text{if } 16 < t < 24 \end{cases}$$

$a(t)$ does not exist at $t = 4$ and $t = 16$.

Part D

1 point is earned for the average rate of change of v on $[8, 20]$

1 point is earned for the answer with explanation

The average rate of change of v on $[8, 20]$ is

$$\frac{v(20) - v(8)}{20 - 8} = -\frac{5}{6} \text{ m/sec}^2.$$

No, the Mean Value Theorem does not apply to v on $[8, 20]$ because v is not differentiable at

$t = 16$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the average rate of change of v on $[8, 20]$

1 point is earned for the answer with explanation

2-2-2b

The average rate of change of v on $[8, 20]$ is

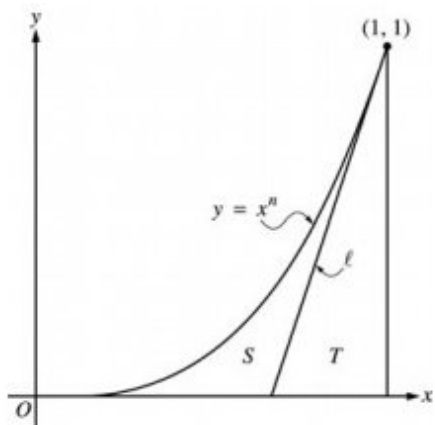
$$\frac{v(20) - v(8)}{20 - 8} = -\frac{5}{6} \text{ m/sec}^2.$$

No, the Mean Value Theorem does not apply to v on $[8, 20]$ because v is not differentiable at

$t = 16$.

2-2-2b

37.



Let l be the line tangent to the graph of $y = x^n$ at the point $(1, 1)$, where $n > 1$, as shown above.

Let T be the triangular region bounded by l , the x -axis, and the line $x=1$. Show that the area of T is $1/2n$.

Part B

1 point is earned if slope of line l is n

1 point is earned if base of T is $\frac{1}{n}$

1 point is earned if shows area is $\frac{1}{2n}$

Let b be the length of the base of triangle T .

$\frac{1}{b}$ is the slope of line l , which is n

$$\text{Area}(T) = \frac{1}{2}b(1) = \frac{1}{2n}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned if slope of line l is n

1 point is earned if base of T is $\frac{1}{n}$

1 point is earned if shows area is $\frac{1}{2n}$

Let b be the length of the base of triangle T .

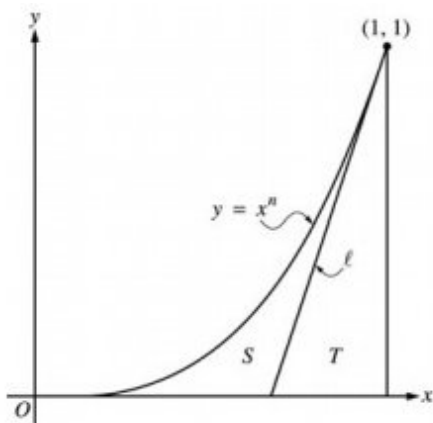
$\frac{1}{b}$ is the slope of line l , which is n

$$\text{Area}(T) = \frac{1}{2}b(1) = \frac{1}{2n}$$

2-2-2b

2-2-2b

38.



Let l be the line tangent to the graph of $y = x^n$ at the point $(1, 1)$, where $n > 1$, as shown above.

(a) Find $\int_0^1 x^n dx$ in terms of n .

(b) Let T be the triangular region bounded by l , the x -axis, and the line $x = 1$. Show that the area of T is $1/2n$.

(c) Let S be the region bounded by the graph of $y = x^n$, the line l , and the x -axis. Express the area of S in terms of n and determine the value of n that maximizes the area of S .

Part A

1 point is earned for the antiderivative of x^n

1 point is earned for the answer

$$\int_0^1 x^n dx = \frac{x^{n+1}}{n+1} \Big|_0^1 = \frac{1}{n+1}$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the antiderivative of x^n

1 point is earned for the answer

$$\int_0^1 x^n dx = \frac{x^{n+1}}{n+1} \Big|_0^1 = \frac{1}{n+1}$$

2-2-2b**Part B**

1 point is earned if slope of line l is n

1 point is earned if base of T is $\frac{1}{n}$

1 point is earned if shows area is $\frac{1}{2n}$

Let b be the length of the base of triangle T .

$\frac{1}{b}$ is the slope of line l , which is n

$$\text{Area}(T) = \frac{1}{2}b(1) = \frac{1}{2n}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned if slope of line l is n

1 point is earned if base of T is $\frac{1}{n}$

1 point is earned if shows area is $\frac{1}{2n}$

Let b be the length of the base of triangle T .

$\frac{1}{b}$ is the slope of line l , which is n

$$\text{Area}(T) = \frac{1}{2}b(1) = \frac{1}{2n}$$

Part C

2-2-2b

1 point is earned for the area of S in terms of n

1 point is earned for the derivative

1 point is earned if sets derivative equal to 0

1 point is earned if solves for n

$$\begin{aligned}\text{Area}(S) &= \int_0^1 x^n dx - \text{Area}(T) \\ &= \frac{1}{n+1} - \frac{1}{2n} \\ \frac{d}{dn}\text{Area}(S) &= -\frac{1}{(n+1)^2} + \frac{1}{2n^2} = 0 \\ 2n^2 &= (n+1)^2 \\ \sqrt{2}n &= (n+1) \\ n &= \frac{1}{\sqrt{2}-1} = 1 + \sqrt{2}\end{aligned}$$



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

1 point is earned for the area of S in terms of n

1 point is earned for the derivative

1 point is earned if sets derivative equal to 0

1 point is earned if solves for n

$$\begin{aligned}\text{Area}(S) &= \int_0^1 x^n dx - \text{Area}(T) \\ &= \frac{1}{n+1} - \frac{1}{2n} \\ \frac{d}{dn}\text{Area}(S) &= -\frac{1}{(n+1)^2} + \frac{1}{2n^2} = 0 \\ 2n^2 &= (n+1)^2 \\ \sqrt{2}n &= (n+1) \\ n &= \frac{1}{\sqrt{2}-1} = 1 + \sqrt{2}\end{aligned}$$

2-2-2b

39. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

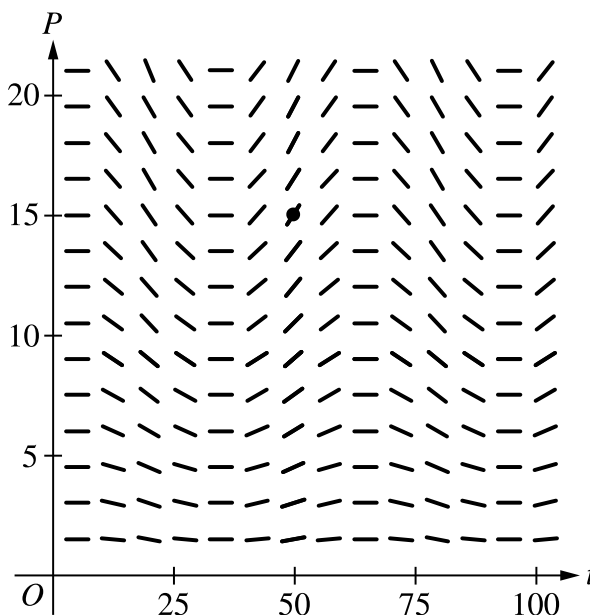
Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Scientists study a population that experiences variations in its rate of growth. The rate of growth of the population is modeled by the differential equation $\frac{dP}{dt} = P \cdot 0.02 \cos\left(\frac{\pi}{30}(t + 10)\right)$, where t is measured in days since the observation of the population began. Let $P = f(t)$ be the particular solution to the given differential equation satisfying $f(50) = 15$.

(a) A slope field for the given differential equation is shown. Sketch the solution curve that passes through the point $(50, 15)$.



(b) Write an equation for the line tangent to the graph of $P = f(t)$ at time $t = 50$.

(c) It is known that $f'(t) > 0$ and $f''(t) > 0$ on the interval $47 < t < 51$. If the tangent line found in part (b) is used to approximate $f(48)$, is the approximation an underestimate or an overestimate for the actual value of $f(48)$? Give a reason for your answer.

(d) Use separation of variables to find the particular solution $P = f(t)$ to the differential equation $\frac{dP}{dt} = P \cdot 0.02 \cos\left(\frac{\pi}{30}(t + 10)\right)$ that satisfies $f(50) = 15$.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

2-2-2b

The solution curve must pass through the point (50, 15), extend reasonably close to the left and right edges of the slope field, and have no obvious conflicts with the given line segments.

Only portions of the solution curve within the given slope field are considered.

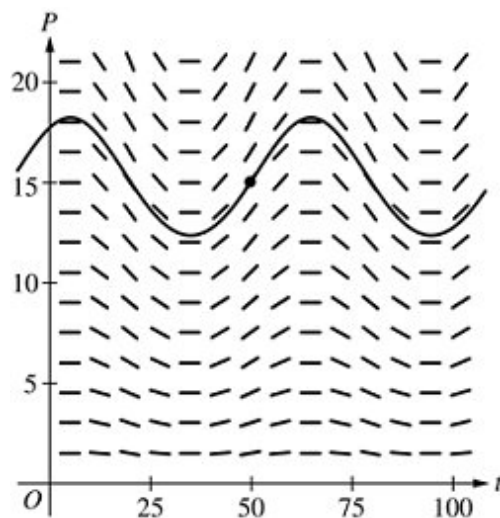
All local maximum/minimum points on the solution curve must occur at horizontal line segments in the slope field.



0	1
---	---

The student response accurately includes the criteria below.

- ☐ Solution curve through (50, 15),

Solution:**Part B**

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The tangent line equation can be presented in any equivalent form and can be written in terms of any variables.

The second point can be earned with a tangent line through the point (50, 15) with a slope equal to the declared value of $\left. \frac{dP}{dt} \right|_{(50, 15)}$.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

2-2-2b

- ☐ Slope at (50, 15)
- ☐ Tangent line equation

Solution:

$$f(50) = 15$$

$$f'(50) = \left. \frac{dP}{dt} \right|_{(50,15)} = 15 \cdot 0.02 \cos\left(\frac{\pi}{30} \cdot 60\right) = 0.3$$

An equation for the line tangent to the graph of $P = f(t)$ at time $t = 50$ is $y = 15 + 0.3(t - 50)$.

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The reason must include one of $f''(t) > 0$, $f'(t)$ is increasing, or $f(t)$ is concave up.

The point is not earned if the reason includes statements about the behavior of $f'(t)$ or $f''(t)$ outside the interval $47 < t < 51$.



0	1
---	---

The student response accurately includes the criteria below.

- ☐ Underestimate with reason

Solution:

The tangent line approximation of $f(48)$ is an underestimate because $f''(t) > 0$ on an interval that contains $t = 48$ and $t = 50$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response with no separation of variables earns 0 out of 5 points.

A response that presents an antiderivative of without absolute value symbols is sufficient for the second point.

A response with no constant of integration can earn at most the first 3 points.

A response is eligible for the fourth point only if it has earned the first point and at least 1 of the 2 antiderivative points.

2-2-2b

An eligible response earns the fourth point by correctly including the constant of integration in an equation and substituting 50 for t and P . A response is eligible for the fifth point only if it has earned the first 4 points.



0	1	2	3	4	5
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The student response accurately includes **all five** of the criteria below.

- ☐ Separates variables
- ☐ Finds one antiderivative
- ☐ Finds the other antiderivative
- ☐ Constant of integration and uses initial condition
- ☐ Solves for P

Solution:

$$\int \frac{dP}{P} = \int 0.02 \cos\left(\frac{\pi}{30}(t + 10)\right) dt$$

$$\ln|P| = \frac{0.6}{\pi} \sin\left(\frac{\pi}{30}(t + 10)\right) + C$$

$$\ln|15| = \frac{0.6}{\pi} \sin\left(\frac{\pi}{30}(50 + 10)\right) + C \Rightarrow C = \ln 15$$

Because $P(50) > 0$, $\ln|P| = \ln P$.

$$\ln P = \frac{0.6}{\pi} \sin\left(\frac{\pi}{30}(t + 10)\right) + \ln 15$$

$$P = 15e^{\frac{0.6}{\pi} \sin\left(\frac{\pi}{30}(t+10)\right)}$$

2-2-2b

40. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

$$f(t) = \begin{cases} 48t + t^2 - \frac{t^3}{12} & \text{for } 0 \leq t < 6 \\ g(t) & \text{for } 6 \leq t \leq 12 \end{cases}$$

t (hours)	6	8	10	12
$g(t)$ (cubic meters)	306	376	428	474

At an excavation site, the amount of dirt that has been removed, in cubic meters, is modeled by the function f defined above, where g is a differentiable function and t is measured in hours. Values of $g(t)$ at selected values of t are given in the table above.

- According to the model f , what is the average rate of change of the amount of dirt removed over the time interval $6 \leq t \leq 12$ hours?
- Use the data in the table to approximate $f'(9)$, the instantaneous rate of change in the amount of dirt removed, in cubic meters per hour, at time $t = 9$ hours. Show the computations that lead to your answer.
- Is f continuous for $0 \leq t \leq 12$? Justify your answer.
- Find $f'(2)$, the instantaneous rate of change in the amount of dirt removed, in cubic meters per hour, at time $t = 2$ hours.

Part A

The point is earned for a correct expression that substitutes both correct values from the table, with or without simplification.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The student response accurately includes a correct average rate of change.

Solution:

2-2-2b

$$\frac{f(12)-f(6)}{12-6} = \frac{g(12)-g(6)}{12-6} = \frac{474-306}{12-6} = \frac{168}{6} = 28$$

The average rate of change of the amount of dirt removed over the time interval $6 \leq t \leq 12$ hours is 28 cubic meters per hour.

Part B

The point is earned for a correct expression that substitutes both correct values from the table, with or without simplification.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The student response accurately includes a correct approximation of $f'(9)$.

Solution:

$$f'(9) \approx \frac{g(10)-g(8)}{10-8} = \frac{428-376}{10-8} = \frac{52}{2} = 26$$

Part C

The response must explicitly connect differentiability and continuity to earn the first point and use the fact that f is a polynomial for $0 \leq t < 6$ to explain why f is continuous on that interval to earn the second point. The third point is for continuity at $t = 6$.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ f is continuous for $0 \leq t < 6$
- ☐ differentiability of g implies continuity
- ☐ $\lim_{t \rightarrow 6^-} f(t) = \lim_{t \rightarrow 6^+} f(t) = f(6)$

Solution:

Because f is a polynomial function for $0 \leq t < 6$, f is continuous for $0 \leq t < 6$.

2-2-2b

Because g is differentiable, g is continuous on $6 \leq t \leq 12$. $f(t) = g(t)$ for $6 \leq t \leq 12$.

$$\lim_{t \rightarrow 6^-} f(t) = \lim_{t \rightarrow 6^-} \left(48t + t^2 - \frac{t^3}{12} \right) = 48(6) + 6^2 - \frac{6^3}{12} = 288 + 36 - 18 = 306$$

$$\lim_{t \rightarrow 6^+} f(t) = f(6) = g(6) = 306$$

Because $\lim_{t \rightarrow 6^-} f(t) = \lim_{t \rightarrow 6^+} f(t) = f(6)$, f is continuous at $t = 6$.

Thus, f is continuous for $0 \leq t \leq 12$.

Part D

The first point is earned for demonstration of a correct derivative for f and does not require a separate statement of $f'(t)$, so $f'(2) = 48 + 2 \cdot 2 - \frac{2^2}{4}$ earns both points. The derivative and evaluation of the derivative can be expressed in one line to earn both points.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

☐ $f'(t)$

☐ $f'(2)$

Solution:

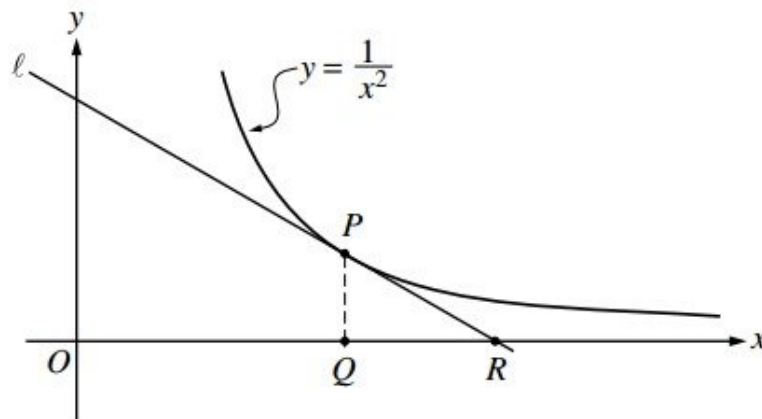
$$f(t) = 48t + t^2 - \frac{t^3}{12} \text{ for } 0 \leq t < 6$$

$$f'(t) = 48 + 2t - \frac{t^2}{4} \text{ for } 0 < t < 6$$

$$f'(2) = 48 + 2 \cdot 2 - \frac{2^2}{4} = 51$$

The instantaneous rate of change in the amount of dirt removed at $t = 2$ hours is 51 cubic meters per hour.

2-2-2b



In the figure above, line L is tangent to the graph of $y = \frac{1}{x^2}$ at point P , with coordinates $(w, \frac{1}{w^2})$, where $w > 0$. Point Q has coordinates $(w, 0)$. Line L crosses the x -axis at point R , with coordinates $(k, 0)$.

41. Find the value of k when $w = 3$.

Part A

1 point is earned for: $\frac{dy}{dx} \Big|_{x=3}$

1 point is earned for the answer

$$\frac{dy}{dx} = -\frac{2}{x^3}; \frac{dy}{dx} \Big|_{x=3} = -\frac{3}{27}$$

Line L through $(3, \frac{1}{9})$ and $(k, 0)$ has slope $-\frac{3}{27}$.

$$\text{Therefore, } \frac{0 - \frac{1}{9}}{k - 3} = -\frac{3}{27} \text{ or } 0 - \frac{1}{9} = -\frac{3}{27}(k - 3)$$

$$k = \frac{9}{2}$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for: $\frac{dy}{dx} \Big|_{x=3}$

2-2-2b

1 point is earned for the answer

$$\frac{dy}{dx} = -\frac{2}{x^3}; \left. \frac{dy}{dx} \right|_{x=3} = -\frac{3}{27}$$

Line L through $\left(3, \frac{1}{9}\right)$ and $(k, 0)$ has slope $-\frac{3}{27}$.

$$\text{Therefore, } \frac{0 - \frac{1}{9}}{k - 3} = -\frac{2}{27} \text{ or } 0 - \frac{1}{9} = -\frac{2}{27}(k - 3)$$
$$k = \frac{9}{2}$$

2-2-2b

42. For all $w > 0$, find k in terms of w .

Part B

1 point is earned for equation relating w and k , using slopes

1 point is earned for the answer

Line L through $(w, \frac{1}{w^2})$ and $(k, 0)$ has slope $-\frac{2}{w^3}$.

$$\text{Therefore, } \frac{0 - \frac{1}{w^2}}{k - w} = -\frac{2}{w^3} \text{ or } 0 - \frac{1}{w^2} = -\frac{2}{w^3}(k - w)$$

$$k = \frac{3}{2}w$$



0	1	2
---	---	---

The student response earns none of the following points:

1 point is earned for equation relating w and k , using slopes

1 point is earned for the answer

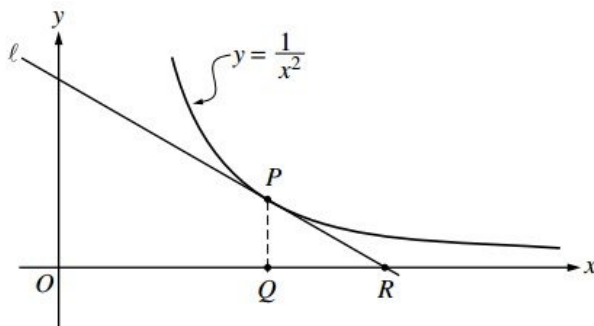
Line L through $(w, \frac{1}{w^2})$ and $(k, 0)$ has slope $-\frac{2}{w^3}$.

$$\text{Therefore, } \frac{0 - \frac{1}{w^2}}{k - w} = -\frac{2}{w^3} \text{ or } 0 - \frac{1}{w^2} = -\frac{2}{w^3}(k - w)$$

$$k = \frac{3}{2}w$$

2-2-2b

43.



In the figure above, line ℓ is tangent to the graph of $y = \frac{1}{x^2}$ at point P , with coordinates $(w, \frac{1}{w^2})$, where $w > 0$. Point Q has coordinates $(w, 0)$. Line ℓ crosses the x -axis at point R , with coordinates $(k, 0)$.

(a) Find the value of k when $w = 3$.

(b) For all $w > 0$, find k in terms of w .

(c) Suppose that w is increasing at the constant rate of 7 units per second. When $w = 5$, what is the rate of change of k with respect to time?

(d) Suppose that w is increasing at the constant rate of 7 units per second. When $w = 5$, what is the rate of change of the area of $\triangle PQR$ with respect to time? Determine whether the area is increasing or decreasing at this instant.

Part A

1 point is earned for: $\left. \frac{dy}{dx} \right|_{x=3}$

1 point is earned for the answer

$$\frac{dy}{dx} = -\frac{2}{x^3}; \left. \frac{dy}{dx} \right|_{x=3} = -\frac{2}{27}$$

Line ℓ through $(3, \frac{1}{9})$ and $(k, 0)$ has slope $-\frac{2}{27}$.

Therefore, $\frac{0 - \frac{1}{9}}{k - 3} = -\frac{2}{27}$ or, $0 - \frac{1}{9} = -\frac{2}{27}(k - 3)$

$$k = \frac{9}{2}$$

2-2-2b



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for: $\frac{dy}{dx} \Big|_{x=3}$

1 point is earned for the answer

$$\frac{dy}{dx} = -\frac{2}{x^3}; \frac{dy}{dx} \Big|_{x=3} = -\frac{2}{27}$$

Line ℓ through $(3, \frac{1}{9})$ and $(k, 0)$ has slope $-\frac{2}{27}$.

Therefore, $\frac{0 - \frac{1}{9}}{k - 3} = -\frac{2}{27}$ or, $0 - \frac{1}{9} = -\frac{2}{27}(k - 3)$

$$k = \frac{9}{2}$$

Part B

1 point is earned for equation relating w and k , using slopes

1 point is earned for the answer

Line ℓ through $(w, \frac{1}{w^2})$ and $(k, 0)$ has slope $-\frac{2}{w^3}$.

Therefore, $\frac{0 - \frac{1}{w^2}}{k - w} = -\frac{2}{w^3}$ or $0 - \frac{1}{w^2} = -\frac{2}{w^3}(k - w)$

$$k = \frac{3}{2}w$$



0	1	2
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2-2-2b

The student response earns all of the following points:

1 point is earned for equation relating w and k , using slopes

1 point is earned for the answer

Line ℓ through $(w, \frac{1}{w^2})$ and $(k, 0)$ has slope $-\frac{2}{w^3}$.

Therefore, $\frac{0 - \frac{1}{w^2}}{k - w} = -\frac{2}{w^3}$ or $0 - \frac{1}{w^2} = -\frac{2}{w^3}(k - w)$
 $k = \frac{3}{2}w$

Part C

1 point is earned for the answer using: $\frac{dw}{dt} = 7$

$$\frac{dk}{dt} = \frac{3}{2} \frac{dw}{dt} = \frac{3}{2} \cdot 7 = \frac{21}{2}; \left. \frac{dk}{dt} \right|_{w=5} = \frac{21}{2}$$



0	1
---	---

The student response earns all of the following points:

1 point is earned for the answer using: $\frac{dw}{dt} = 7$

$$\frac{dk}{dt} = \frac{3}{2} \frac{dw}{dt} = \frac{3}{2} \cdot 7 = \frac{21}{2}; \left. \frac{dk}{dt} \right|_{w=5} = \frac{21}{2}$$

Part D

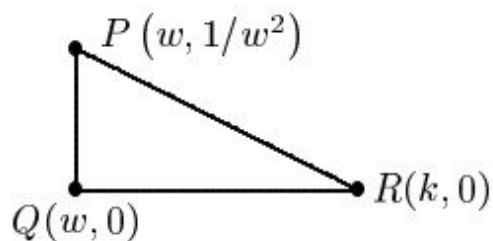
1 point is earned for the area in term of w and/or k

1 point is earned for: $\frac{dA}{dt}$ implicitly

2-2-2b

1 point is earned for: $\frac{dA}{dt} \Big|_{w=5}$ using $\frac{dw}{dt} = 7$

1 point is earned for the conclusion



$$A = \frac{1}{2}(k - w)\frac{1}{w^2} = \frac{1}{2}\left(\frac{3}{2}w - w\right)\frac{1}{w^2} = \frac{1}{4w}$$

$$\frac{dA}{dt} = -\frac{1}{4w^2} \frac{dw}{dt}$$

$$\frac{dA}{dt} \Big|_{w=5} = -\frac{1}{100} \cdot 7 = -0.07$$

Therefore, area is decreasing.



0	1	2	3	4
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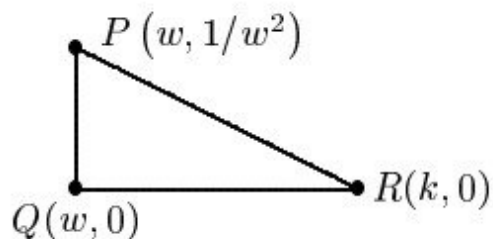
The student response earns all of the following points:

1 point is earned for the area in term or w and/or k

1 point is earned for: $\frac{dA}{dt}$ implicitly

1 point is earned for: $\frac{dA}{dt} \Big|_{w=5}$ using $\frac{dw}{dt} = 7$

1 point is earned for the conclusion



2-2-2b

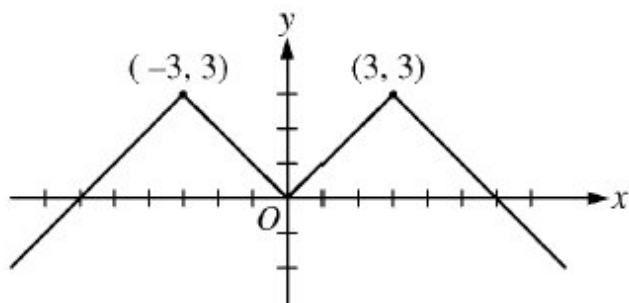
$$A = \frac{1}{2}(k - w)\frac{1}{w^2} = \frac{1}{2}\left(\frac{3}{2}w - w\right)\frac{1}{w^2} = \frac{1}{4w}$$

$$\frac{dA}{dt} = -\frac{1}{4w^2} \frac{dw}{dt}$$

$$\left. \frac{dA}{dt} \right|_{w=5} = -\frac{1}{100} \cdot 7 = -0.07$$

Therefore, area is decreasing.

44.



The graph of the even function $y = f(x)$ consists of 4 line segments, as shown above. Which of the following statements about f is false?

(A) $\lim_{x \rightarrow 0} (f(x) - f(0)) = 0$

(B) $\lim_{x \rightarrow 0} \frac{f(x) - f(0)}{x} = 0$ ✓

(C) $\lim_{x \rightarrow 0} \frac{f(x) - f(-x)}{2x} = 0$

(D) $\lim_{x \rightarrow 2} \frac{f(x) - f(2)}{x - 2} = 1$

(E) $\lim_{x \rightarrow 3} \frac{f(x) - f(3)}{x - 3}$ does not exist.

2-2-2b

45. At time $t = 0$, a storage tank is empty and begins filling with water. For $t > 0$ hours, the depth of the water in the tank is increasing at a rate of $W(t)$ feet per hour.

Which of the following is the best interpretation of the statement $W'(2) > 3$?

- (A) Two hours after the tank begins filling with water, the depth of the water is increasing at a rate greater than 3 feet per hour.
- (B) Over the first two hours after the tank begins filling with water, the depth of the water is always increasing at a rate greater than 3 feet per hour.
- (C) Two hours after the tank begins filling with water, the rate at which the depth of the water is rising is increasing at a rate greater than 3 feet per hour per hour. ✓
- (D) Over the first two hours after the tank begins filling with water, the rate at which the depth of the water is rising is always increasing at a rate greater than 3 feet per hour per hour.

Answer C

Correct. In the expression $W'(2)$, the 2 represents the value of the independent variable and is therefore the number of hours since the tank began filling with water. $W'(2)$, being the value of a derivative, is the rate of change of W , that is, the rate of change of the rate at which the depth of the water is rising; in this case, 2 hours after the tank begins filling with water. The units for the derivative would be the units of W per unit of time; thus, feet per hour per hour. The statement says that at time 2 hours after the tank begins filling with water, the rate at which the depth of the water is rising, $W(t)$, is increasing at a rate that is greater than 3 feet per hour per hour.

2-2-2b

46. Let f be the function given by $f(x) = x^3 - 7x + 6$.

Write an equation of the line tangent to the graph of f at $x = -1$.

Part B

1 point is earned for $f'(x)$

1 point is earned for evaluating $f'(-1)$ and $f(-1)$

1 point is earned for equation of line

$$f'(x) = 3x^2 - 7$$

$$f'(-1) = -4, f(-1) = 12$$

$$\begin{array}{l} y - 12 = -4(x + 1) \\ \text{or} \\ 4x + y = 8 \\ \text{or} \\ y = -4x + 8 \end{array}$$



0	1	2	3
---	---	---	---

The student response earns three of the following points:

1 point is earned for $f'(x)$

1 point is earned for evaluating $f'(-1)$ and $f(-1)$

1 point is earned for equation of line

$$f'(x) = 3x^2 - 7$$

$$f'(-1) = -4, f(-1) = 12$$

$$\begin{array}{l} y - 12 = -4(x + 1) \\ \text{or} \\ 4x + y = 8 \\ \text{or} \\ y = -4x + 8 \end{array}$$

2-2-2b

47. Let f be the function given by $f(x) = x^3 - 7x + 6$.

(a) Find the zeros of f .

(b) Write an equation of the line tangent to the graph of f at $x = -1$.

(c) Find the number c that satisfies the conclusion of the Mean Value Theorem for f on the closed interval $[1, 3]$.

Part A

1 point is earned for first correct zero

2 points are earned for zeros of f

1 point is **deducted** for incorrect zero

$$\begin{aligned} f(x) &= x^3 - 7x + 6 \\ &= (x - 1)(x - 2)(x + 3) \end{aligned}$$

$$x = 1, x = 2, x = -3$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for first correct zero

2 points are earned for zeros of f

1 point is **deducted** for incorrect zero

$$\begin{aligned} f(x) &= x^3 - 7x + 6 \\ &= (x - 1)(x - 2)(x + 3) \end{aligned}$$

$$x = 1, x = 2, x = -3$$

Part B

1 point is earned for $f'(x)$

1 point is earned for evaluating $f'(-1)$ and $f(-1)$

1 point is earned for equation of line

2-2-2b

$$f'(x) = 3x^2 - 7$$

$$f'(-1) = -4, f(-1) = 12$$

$$y - 12 = -4(x + 1)$$

or

$$4x + y = 8$$

or

$$y = -4x + 8$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for $f'(x)$

1 point is earned for evaluating $f'(-1)$ and $f(-1)$

1 point is earned for equation of line

$$f'(x) = 3x^2 - 7$$

$$f'(-1) = -4, f(-1) = 12$$

$$y - 12 = -4(x + 1)$$

or

$$4x + y = 8$$

or

$$y = -4x + 8$$

Part C

1 point is earned for the answer

1 point is earned for solving $f(c)$

1 point is earned for solving c

$$\frac{f(3) - f(1)}{3 - 1} = \frac{12 - 0}{2} = 6$$

$$3c^2 - 7 = f'(c) = 6$$

$$c = \sqrt{\frac{13}{3}}$$

2-2-2b

0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the answer

1 point is earned for solving $f(c)$

1 point is earned for solving c

$$\frac{f(3)-f(1)}{3-1} = \frac{12-0}{2} = 6$$

$$3c^2 - 7 = f'(c) = 6$$

$$c = \sqrt{\frac{13}{3}}$$

2-2-2b

48. Let f be the function defined by $f(x) = 3x^5 - 5x^3 + 2$.

Write the equation of each horizontal tangent line to the graph of f .

Part C

1 point is earned for solving student's $f'(x) = 0$

1 point is earned for answer, $y =$ number

1 point is earned for consistent answer

$f'(x) = 0$ when $x = -1, 0, 1$

$x = -1 \Rightarrow f(x) = 4 ; y = 4$

$f(0) = 2 ; y = 2$

$f(1) = 0 ; y = 0$



0	1	2	3
---	---	---	---

The student response earns three of the following points:

1 point is earned for solving student's $f'(x) = 0$

1 point is earned for answer, $y =$ number

1 point is earned for consistent answer

$f'(x) = 0$ when $x = -1, 0, 1$

$x = -1 \Rightarrow f(x) = 4 ; y = 4$

$f(0) = 2 ; y = 2$

$f(1) = 0 ; y = 0$

2-2-2b

49. Let f be the function defined by $f(x) = 3x^5 - 5x^3 + 2$.
- On what intervals is f increasing?
 - On what intervals is the graph of f concave upward?
 - Write the equation of each horizontal tangent line to the graph of f .

Part A

1 point is earned for $f'(x)$

1 point is earned for analyze sign of $f'(x)$ or explicitly setting students $f'(x) > 0$

1 point is earned for answer

$$f'(x) = 15x^4 - 15x^2 = 15x^2(x^2 - 1)$$



Answer: f is increasing on the intervals $(-\infty, -1]$ and $[1, \infty)$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for $f'(x)$

1 point is earned for analyze sign of $f'(x)$ or explicitly setting students $f'(x) > 0$

1 point is earned for answer

$$f'(x) = 15x^4 - 15x^2 = 15x^2(x^2 - 1)$$



Answer: f is increasing on the intervals $(-\infty, -1]$ and $[1, \infty)$

2-2-2b

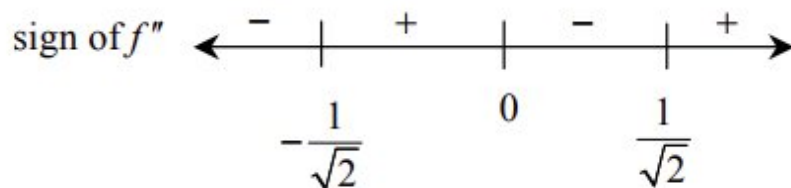
Part B

1 point is earned for $f''(x)$

1 point is earned for analyze sign of $f''(x)$ or explicitly setting students $f''(x) > 0$

1 point is earned for answer

$$f''(x) = 60x^3 - 30x = 30x(2x^2 - 1)$$



Answer: f is concave upward on $\left(-\frac{1}{\sqrt{2}}, 0\right)$ and on $\left(\frac{1}{\sqrt{2}}, \infty\right)$



0	1	2	3
---	---	---	---

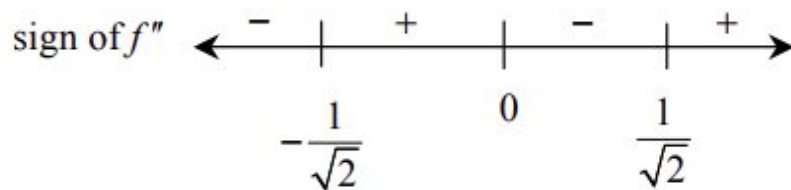
The student response earns all of the following points:

1 point is earned for $f''(x)$

1 point is earned for analyze sign of $f''(x)$ or explicitly setting students $f''(x) > 0$

1 point is earned for answer

$$f''(x) = 60x^3 - 30x = 30x(2x^2 - 1)$$



Answer: f is concave upward on $\left(-\frac{1}{\sqrt{2}}, 0\right)$ and on $\left(\frac{1}{\sqrt{2}}, \infty\right)$

Part C

2-2-2b

1 point is earned for solving student's $f'(x) = 0$

1 point is earned for answer, $y = \text{number}$

1 point is earned for consistent answer

$$f'(x) = 0 \text{ when } x = -1, 0, 1$$

$$x = -1 \Rightarrow f(x) = 4; y = 4$$

$$f(0) = 2; y = 2$$

$$f(1) = 0; y = 0$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for solving student's $f'(x) = 0$

1 point is earned for answer, $y = \text{number}$

1 point is earned for consistent answer

$$f'(x) = 0 \text{ when } x = -1, 0, 1$$

$$x = -1 \Rightarrow f(x) = 4; y = 4$$

$$f(0) = 2; y = 2$$

$$f(1) = 0; y = 0$$

50. An equation of the line tangent to $y = x^3 + 3x^2 + 2$ at its point of inflection is

(A) $y = -6x - 6$

(B) $y = -3x + 1$

(C) $y = 2x + 10$

(D) $y = 3x - 1$

(E) $y = 4x + 1$



2-2-2b

51. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Consider the curve given by the equation $2(x - y) = 3 + \cos y$. For all points on the curve, $\frac{2}{3} \leq \frac{dy}{dx} \leq 2$.

(a) Show that $\frac{dy}{dx} = \frac{2}{2 - \sin y}$.

(b) For $-\frac{\pi}{2} < y < \frac{\pi}{2}$, there is a point P on the curve through which the line tangent to the curve has slope 1. Find the coordinates of the point P .

(c) Determine the concavity of the curve at points for which $-\frac{\pi}{2} < y < \frac{\pi}{2}$. Give a reason for your answer.

(d) Let $y = f(x)$ be a function, defined implicitly by $2(x - y) = 3 + \cos y$, that is continuous on the closed interval $[2, 2.1]$ and differentiable on the open interval $(2, 2.1)$. Use the Mean Value Theorem on the interval $[2, 2.1]$ to show that $\frac{1}{15} \leq f(2.1) - f(2) \leq \frac{1}{5}$.

Part A

1 point is earned for: implicit differentiation

1 point is earned for: verification

$$\frac{d}{dx}(2(x - y)) = \frac{d}{dx}(3 + \cos y)$$

$$2 - 2\frac{dy}{dx} = (-\sin y)\frac{dy}{dx}$$

$$2 = (2 - \sin y)\frac{dy}{dx}$$

$$\frac{dy}{dx} = \frac{2}{2 - \sin y}$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for: implicit differentiation

1 point is earned for: verification

$$\frac{d}{dx}(2(x - y)) = \frac{d}{dx}(3 + \cos y)$$

$$2 - 2\frac{dy}{dx} = (-\sin y)\frac{dy}{dx}$$

$$2 = (2 - \sin y)\frac{dy}{dx}$$

$$\frac{dy}{dx} = \frac{2}{2 - \sin y}$$

Part B

2-2-2b

1 point is earned for: $\frac{dy}{dx} = 1$

1 point is earned for: answer

$$\frac{dy}{dx} = \frac{2}{2-\sin y} = 1 \Rightarrow \sin y = 0 \Rightarrow y = 0$$

$$2(x - 0) = 3 + \cos 0 \Rightarrow 2x = 4 \Rightarrow x = 2$$

Point P has coordinates $(2, 0)$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for: $\frac{dy}{dx} = 1$

1 point is earned for: answer

$$\frac{dy}{dx} = \frac{2}{2-\sin y} = 1 \Rightarrow \sin y = 0 \Rightarrow y = 0$$

$$2(x - 0) = 3 + \cos 0 \Rightarrow 2x = 4 \Rightarrow x = 2$$

Point P has coordinates $(2, 0)$.

Part C

2 points are earned for: $\frac{d^2y}{dx^2}$

1 point is earned for: answer with reason

$$\frac{d^2y}{dx^2} = \frac{-2}{(2-\sin y)^2}(-\cos y)\frac{dy}{dx} = \frac{4\cos y}{(2-\sin y)^3}$$

$$\frac{d^2y}{dx^2} > 0 \text{ for all } y \text{ in the interval } -\frac{\pi}{2} < y < \frac{\pi}{2}.$$

Therefore, the curve is concave up for $-\frac{\pi}{2} < y < \frac{\pi}{2}$.



0	1	2	3
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The student response earns all of the following points:

2-2-2b

2 points are earned for: $\frac{d^2y}{dx^2}$

1 point is earned for: answer with reason

$$\frac{d^2y}{dx^2} = \frac{-2}{(2-\sin y)^2}(-\cos y)\frac{dy}{dx} = \frac{4\cos y}{(2-\sin y)^3}$$

$$\frac{d^2y}{dx^2} > 0 \text{ for all } y \text{ in the interval } -\frac{\pi}{2} < y < \frac{\pi}{2}.$$

Therefore, the curve is concave up for $-\frac{\pi}{2} < y < \frac{\pi}{2}$.

Part D

1 point is earned for: applies Mean Value Theorem

1 point is earned for: verification

By the Mean Value Theorem, for some value c in the interval $(2, 2.1)$, $f'(c) = \frac{f(2.1)-f(2)}{0.1}$.

For all points on the curve, $\frac{2}{3} \leq f'(x) \leq 2$.

$$\text{Thus, } \frac{2}{3} \leq \frac{f(2.1)-f(2)}{0.1} \leq 2 \Rightarrow \frac{1}{15} \leq f(2.1) - f(2) \leq \frac{1}{5}.$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for: applies Mean Value Theorem

1 point is earned for: verification

By the Mean Value Theorem, for some value c in the interval $(2, 2.1)$, $f'(c) = \frac{f(2.1)-f(2)}{0.1}$.

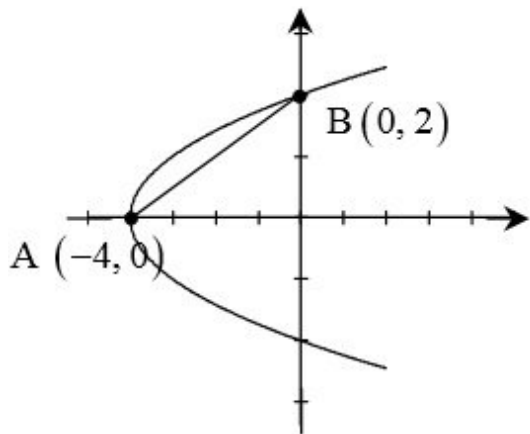
For all points on the curve, $\frac{2}{3} \leq f'(x) \leq 2$.

$$\text{Thus, } \frac{2}{3} \leq \frac{f(2.1)-f(2)}{0.1} \leq 2 \Rightarrow \frac{1}{15} \leq f(2.1) - f(2) \leq \frac{1}{5}.$$

2-2-2b

52. Consider the curve $y^2=4+x$ and chord AB joining the points $A(-4,0)$ and $B(0,2)$ on the curve. Find the x - and y -coordinates of the point on the curve where the tangent line is parallel to chord AB .

Part A



$$\text{slope } AB = \frac{2}{4} = \frac{1}{2}$$

Method 1:

1 point is earned for the correct differentiation

1 point is earned for correctly setting student's derivative = student's slope of AB

$$y = \sqrt{4+x}; \frac{dy}{dx} = \frac{1}{2\sqrt{4+x}}; \frac{1}{2\sqrt{4+x}} = \frac{1}{2}$$

1 point is earned for the correct answer

$$\text{so } x = -3, y = 1$$

Method 2:

1 point is earned for the correct differentiation

1 point is earned for correctly setting student's derivative = student's slope of AB

$$2y \frac{dy}{dx} = 1; \text{ so } 2y \left(\frac{1}{2} \right) = 1$$

1 point is earned for the correct answer

$$\text{and } y = 1, x = -3$$

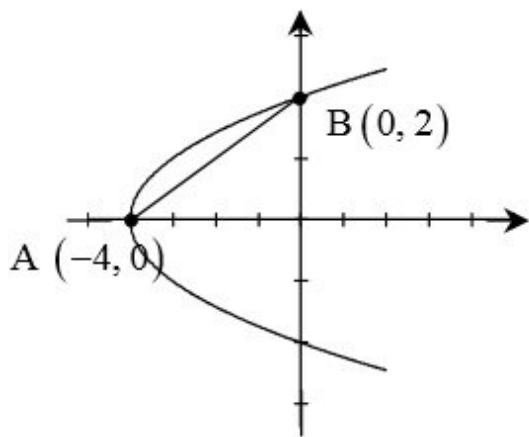
Note: In (a), (b), and (c) any arithmetic error results in loss of last point.

2-2-2b

✓

0	1	2	3
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The student response earns three of the following points:



$$\text{slope } AB = \frac{2}{4} = \frac{1}{2}$$

Method 1:

1 point is earned for the correct differentiation

1 point is earned for correctly setting student's derivative = student's slope of AB

$$y = \sqrt{4+x}; \frac{dy}{dx} = \frac{1}{2\sqrt{4+x}}; \frac{1}{2\sqrt{4+x}} = \frac{1}{2}$$

1 point is earned for the correct answer

$$\text{so } x = -3, y = 1$$

Method 2:

1 point is earned for the correct differentiation

1 point is earned for correctly setting student's derivative = student's slope of AB

$$2y \frac{dy}{dx} = 1; \text{ so } 2y \left(\frac{1}{2} \right) = 1$$

1 point is earned for the correct answer

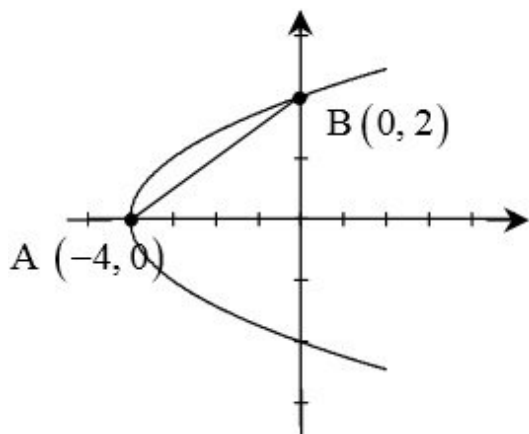
$$\text{and } y = 1, x = -3$$

Note: In (a), (b), and (c) any arithmetic error results in loss of last point.

2-2-2b

53. Consider the curve $y^2 = 4 + x$ and chord AB joining the points $A(-4, 0)$ and $B(0, 2)$ on the curve.
- Find the x - and y -coordinates of the point on the curve where the tangent line is parallel to chord AB .
 - Find the area of the region R enclosed by the curve and the chord AB .
 - Find the volume of the solid generated when the region R , defined in part (b), is revolved about the x -axis.

Part A



$$\text{slope of } AB = \frac{2}{4} = \frac{1}{2}$$

Method 1:

1 point is earned for the correct differentiation

1 point is earned for correctly setting student's derivative = student's slope of AB

$$y = \sqrt{4 + x}; \frac{dy}{dx} = \frac{1}{2\sqrt{4 + x}}; \frac{1}{2\sqrt{4 + x}} = \frac{1}{2}$$

1 point is earned for the correct answer

$$\text{so } x = -3, y = 1$$

Method 2:

1 point is earned for the correct differentiation

1 point is earned for correctly setting student's derivative = student's slope of AB

2-2-2b

$$2y \frac{dy}{dx} = 1, \text{ so } 2y \left(\frac{1}{2} \right) = 1$$

1 point is earned for the correct answer

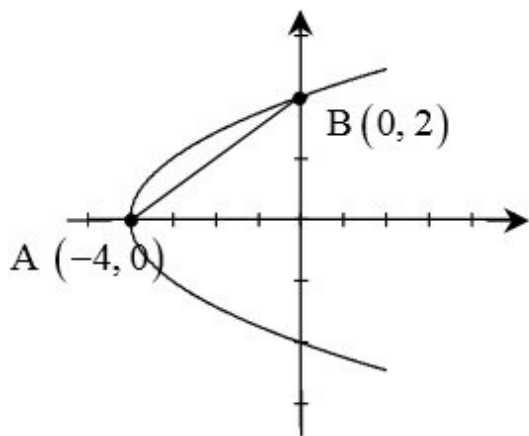
and $y = 1, x = -3$

Note: In (a), (b), and (c) any arithmetic error results in loss of last point.



0	1	2	3
---	---	---	---

The student response earns all of the following points:



$$\text{slope of } AB = \frac{2}{4} = \frac{1}{2}$$

Method 1:

1 point is earned for the correct differentiation

1 point is earned for correctly setting student's derivative = student's slope of AB

$$y = \sqrt{4+x}; \frac{dy}{dx} = \frac{1}{2\sqrt{4+x}}; \frac{1}{2\sqrt{4+x}} = \frac{1}{2}$$

1 point is earned for the correct answer

so $x = -3, y = 1$

Method 2:

2-2-2b

1 point is earned for the correct differentiation

1 point is earned for correctly setting student's derivative = student's slope of AB

$$2y \frac{dy}{dx} = 1, \text{ so } 2y \left(\frac{1}{2} \right) = 1$$

1 point is earned for the correct answer

and $y = 1, x = -3$

Note: In (a), (b), and (c) any arithmetic error results in loss of last point.

Part BMethod 1:

1 point is earned for the correct bounds and constant

1 point is earned for the correct integrand

1 point is earned for the correct antidifferentiation and evaluation

OR

1 point is earned for the correct bounds and constant

1 point is earned for the correct subtraction of areas

1 point is earned for the correct answer

$$\begin{aligned} \int_{-4}^0 \left[(\sqrt{4+x} - \left(\frac{1}{2}x + 2 \right)) \right] dx &= \frac{2}{3}(4+x)^{3/2} - \frac{1}{4}x^2 - 2x \Big|_{-4}^0 \\ &= \frac{2}{3}(4)^{3/2} - (-4 + 8) = \frac{16}{3} - 4 = \frac{4}{3} \end{aligned}$$

Method 2:

1 point is earned for the correct bounds and constant

1 point is earned for the correct integrand

1 point is earned for the correct antidifferentiation and evaluation

OR

1 point is earned for the correct bounds and constant

2-2-2b

1 point is earned for the correct subtraction of areas

1 point is earned for the correct answer

$$\int_0^2 [(2y - 4) - (y^2 - 4)] dy = y^2 - \frac{y^3}{3} \Big|_0^2$$

$$= 4 - \frac{8}{3} = \frac{4}{3}$$

Method 3:

1 point is earned for the correct bounds and constant

1 point is earned for the correct integrand

1 point is earned for the correct antidifferentiation and evaluation

OR

1 point is earned for the correct bounds and constant

1 point is earned for the correct subtraction of areas

1 point is earned for the correct answer

$$\int_{-4}^0 \sqrt{4+x} dx = \frac{16}{3}; \text{ Area of triangle} = 4$$

$$\text{Area of region} = \frac{16}{3} - 4 = \frac{4}{3}$$

Note: In (a), (b), and (c) any arithmetic error results in loss of last point.



0	1	2	3
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The student response earns all of the following points:

Method 1:

1 point is earned for the correct bounds and constant

1 point is earned for the correct integrand

1 point is earned for the correct antidifferentiation and evaluation

OR

2-2-2b

1 point is earned for the correct bounds and constant

1 point is earned for the correct subtraction of areas

1 point is earned for the correct answer

$$\int_{-4}^0 \left[(\sqrt{4+x} - \left(\frac{1}{2}x + 2\right)) \right] dx = \frac{2}{3}(4+x)^{3/2} - \frac{1}{4}x^2 - 2x \Big|_{-4}^0$$
$$= \frac{2}{3}(4)^{3/2} - (-4 + 8) = \frac{16}{3} - 4 = \frac{4}{3}$$

Method 2:

1 point is earned for the correct bounds and constant

1 point is earned for the correct integrand

1 point is earned for the correct antidifferentiation and evaluation

OR

1 point is earned for the correct bounds and constant

1 point is earned for the correct subtraction of areas

1 point is earned for the correct answer

$$\int_0^2 [(2y - 4) - (y^2 - 4)] dy = y^2 - \frac{y^3}{3} \Big|_0^2$$
$$= 4 - \frac{8}{3} = \frac{4}{3}$$

Method 3:

1 point is earned for the correct bounds and constant

1 point is earned for the correct integrand

1 point is earned for the correct antidifferentiation and evaluation

OR

1 point is earned for the correct bounds and constant

1 point is earned for the correct subtraction of areas

1 point is earned for the correct answer

2-2-2b

$$\int_{-4}^0 \sqrt{4+x} dx = \frac{16}{3}; \text{ Area of triangle} = 4$$

$$\text{Area of region} = \frac{16}{3} - 4 = \frac{4}{3}$$

Note: In (a), (b), and (c) any arithmetic error results in loss of last point.

Part CMethod 1:

1 point is earned for the correct bounds and constant

1 point is earned for the correct integrand

1 point is earned for the correct antidifferentiation and evaluation

Note: 0 of 1 if integral not of the form $\int_a^b ([f(x)]^2 - [g(x)]^2) dx$ or $\int_c^d y[f(y) - g(y)] dy$

OR

1 point is earned for the correct bounds and constant

1 point is earned for the correct subtraction of volumes

1 point is earned for the correct answer

$$\begin{aligned} & \pi \int_{-4}^0 \left((\sqrt{4+x})^2 - \left(\frac{1}{2}x + 2 \right)^2 \right) dx \\ &= \pi \int_{-4}^0 \left(4 + x - \left(\frac{1}{4}x^2 + 2x + 4 \right) \right) dx \end{aligned}$$

$$= \pi(8 - 16/3) = 8\pi/3 \approx 8.378$$

Method 2:

1 point is earned for the correct bounds and constant

1 point is earned for the correct integrand

1 point is earned for the correct antidifferentiation and evaluation

2-2-2b

Note: 0 of 1 if integral not of the form $\int_a^b \left([f(x)]^2 - [g(x)]^2 dx \right)$ or $\int_c^d y [f(y) - g(y)] dy$

OR

1 point is earned for the correct bounds and constant

1 point is earned for the correct subtraction of volumes

1 point is earned for the correct answer

$$\int_0^2 2\pi y [(2y - 4) - (y^2 - 4)] dy = \frac{8\pi}{3}$$

Method 3:

1 point is earned for the correct bounds and constant

1 point is earned for the correct integrand

1 point is earned for the correct antidifferentiation and evaluation

Note: 0 of 1 if integral not of the form $\int_a^b \left([f(x)]^2 - [g(x)]^2 dx \right)$ or $\int_c^d y [f(y) - g(y)] dy$

OR

1 point is earned for the correct bounds and constant

1 point is earned for the correct subtraction of volumes

1 point is earned for the correct answer

$$\pi \int_{-4}^0 \left(\sqrt{4+x} \right)^2 dx = \pi \left(4x + \frac{x^2}{2} \right) \bigg|_{-4}^0$$

$$= 0 - \pi(-16 + 8) = 8\pi$$

$$\text{Volume of cone} = 1/3 \pi (2)^2 (4) = 16\pi/3$$

$$\text{Volume} = 8\pi - 16\pi/3 = 8\pi/3$$

2-2-2b

Note: In (a), (b), and (c) any arithmetic error results in loss of last point.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

Method 1:

1 point is earned for the correct bounds and constant

1 point is earned for the correct integrand

1 point is earned for the correct antidifferentiation and evaluation

Note: 0 of 1 if integral not of the form $\int_a^b ([f(x)]^2 - [g(x)]^2) dx$ or $\int_c^d y[f(y) - g(y)] dy$

OR

1 point is earned for the correct bounds and constant

1 point is earned for the correct subtraction of volumes

1 point is earned for the correct answer

$$\begin{aligned} & \pi \int_{-4}^0 \left((\sqrt{4+x})^2 - \left(\frac{1}{2}x + 2 \right)^2 \right) dx \\ &= \pi \int_{-4}^0 \left(4 + x - \left(\frac{1}{4}x^2 + 2x + 4 \right) \right) dx \end{aligned}$$

$$= \pi(8 - 16/3) = 8\pi/3 \approx 8.378$$

Method 2:

1 point is earned for the correct bounds and constant

1 point is earned for the correct integrand

1 point is earned for the correct antidifferentiation and evaluation

2-2-2b

Note: 0 of 1 if integral not of the form $\int_a^b \left([f(x)]^2 - [g(x)]^2 dx \right)$ or $\int_c^d y [f(y) - g(y)] dy$

OR

1 point is earned for the correct bounds and constant

1 point is earned for the correct subtraction of volumes

1 point is earned for the correct answer

$$\int_0^2 2\pi y [(2y - 4) - (y^2 - 4)] dy = \frac{8\pi}{3}$$

Method 3:

1 point is earned for the correct bounds and constant

1 point is earned for the correct integrand

1 point is earned for the correct antidifferentiation and evaluation

Note: 0 of 1 if integral not of the form $\int_a^b \left([f(x)]^2 - [g(x)]^2 dx \right)$ or $\int_c^d y [f(y) - g(y)] dy$

OR

1 point is earned for the correct bounds and constant

1 point is earned for the correct subtraction of volumes

1 point is earned for the correct answer

$$\pi \int_{-4}^0 \left(\sqrt{4+x} \right)^2 dx = \pi \left(4x + \frac{x^2}{2} \right) \bigg|_{-4}^0$$

$$= 0 - \pi(-16 + 8) = 8\pi$$

$$\text{Volume of cone} = 1/3 \pi (2)^2 (4) = 16\pi/3$$

$$\text{Volume} = 8\pi - 16\pi/3 = 8\pi/3$$

2-2-2b

Note: In (a), (b), and (c) any arithmetic error results in loss of last point.

2-2-2b

54. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Consider the curve defined by $y^2 - x^2y = 6$ for $y > 0$.

- Show that $\frac{dy}{dx} = \frac{2xy}{2y-x^2}$.
- Write an equation for the line tangent to the curve at the point $(1, 3)$.
- Show that there is a point P with x -coordinate 0 at which the line tangent to the curve at P is horizontal. Find the y -coordinate of P .
- Find the value of $\frac{d^2y}{dx^2}$ at the point P found in part (c).

Part A

The first point is earned for $2y\frac{dy}{dx} - 2xy - x^2\frac{dy}{dx} = 0$ or equivalent.

The second point is earned for a response that arrives at the given expression rather than an algebraically equivalent expression.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ implicit differentiation
- ☐ verification

Solution:

2-2-2b

$$\frac{d}{dx}(y^2 - x^2y) = \frac{d}{dx}(6)$$

$$\Rightarrow 2y\frac{dy}{dx} - 2xy - x^2\frac{dy}{dx} = 0$$

$$\Rightarrow (2y - x^2)\frac{dy}{dx} = 2xy$$

$$\Rightarrow \frac{dy}{dx} = \frac{2xy}{2y - x^2}$$

Part B

The second point may be earned if response contains an incorrect value for slope based on one computational error.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ slope
- ☐ equation for tangent line

Solution:

$$\left. \frac{dy}{dx} \right|_{(x,y)=(1,3)} = \frac{2 \cdot 1 \cdot 3}{2 \cdot 3 - 1^2} = \frac{6}{5}$$

An equation for the tangent line is $y = 3 + \frac{6}{5}(x - 1)$.

Part C

The second point may be earned if response contains an incorrect value for the y -coordinate based on one computational error.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ y -coordinate
- ☐ verification

2-2-2b**Solution:**

At the point P with $x = 0$, it follows that $y^2 - 0 \cdot y = 6 \Rightarrow y = \sqrt{6}$.

$$\left. \frac{dy}{dx} \right|_{(x,y)=(0,\sqrt{6})} = \frac{2 \cdot 0 \cdot \sqrt{6}}{2 \cdot \sqrt{6} - 0^2} = 0$$

Therefore, the line tangent to the curve at P is horizontal.

Part D

At most 1 out of 2 implicit differentiation points is earned for correct implicit differentiation with a correct application of the quotient rule and chain rule with a maximum of one computational error.

The third point requires substitution of values with a consistent use of the y -coordinate from part (c). A simplified answer is not required although the expression should simplify to 1 regardless of the y -coordinate value.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3
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The student response accurately includes all three of the criteria below.

- ☐ implicit differentiation (2 points for full credit)
- ☐ answer

Solution:

$$\frac{d^2y}{dx^2} = \frac{d}{dx} \left(\frac{2xy}{2y-x^2} \right) = \frac{\left(2y + 2x \frac{dy}{dx} \right) (2y-x^2) - 2xy \left(2 \frac{dy}{dx} - 2x \right)}{(2y-x^2)^2}$$

$$\left. \frac{d^2y}{dx^2} \right|_{(x,y)=(0,\sqrt{6})} = \frac{(2\sqrt{6}+0)(2\sqrt{6}-0)-0}{(2\sqrt{6}-0)^2} = \frac{(2\sqrt{6})^2}{(2\sqrt{6})^2} = 1$$

2-2-2b

55. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Consider the curve defined by $xy^2 - 2x^3 = 2$ for $y \geq 0$.

- Show that $\frac{dy}{dx} = \frac{6x^2 - y^2}{2xy}$.
- Write an equation for the line tangent to the curve at the point $(1, 2)$.
- Find the x -coordinate of the point P at which the line tangent to the curve at P is horizontal.
- Find the value of $\frac{d^2y}{dx^2}$ at the point $(1, 2)$.

Part A

The first point is earned for $x \cdot 2y \frac{dy}{dx} + 1 \cdot y^2 - 6x^2 = 0$ or equivalent. The second point is earned for a response that arrives at the given expression rather than an algebraically equivalent expression. Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ implicit differentiation
- ☐ verification

Solution:

$$\frac{d}{dx}(xy^2 - 2x^3) = \frac{d}{dx}(2)$$

$$\Rightarrow x \cdot 2y \frac{dy}{dx} + 1 \cdot y^2 - 6x^2 = 0$$

$$\Rightarrow 2xy \frac{dy}{dx} = 6x^2 - y^2$$

$$\Rightarrow \frac{dy}{dx} = \frac{6x^2 - y^2}{2xy}$$

2-2-2b

Part B

The second point may be earned if response contains an incorrect value for slope based on one computational error. Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ slope
- ☐ equation for tangent line

Solution:

$$\left. \frac{dy}{dx} \right|_{(x,y)=(1,2)} = \frac{6(1)^2 - (2)^2}{2(1)(2)} = \frac{2}{4} = \frac{1}{2}$$

An equation for the tangent line is $y = 2 + \frac{1}{2}(x - 1)$.

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2
---	---	---

The student response does not accurately include any of the criteria below.

- ☐ $6x^2 - y^2 = 0$
- ☐ x -coordinate

Solution:

If the line tangent to the curve at P is horizontal, $\frac{dy}{dx} = 0$.

Therefore, $6x^2 - y^2 = 0$, $x \neq 0$, $y \neq 0$.

Because $xy^2 - 2x^3 = 2$, $6x^2 - \left(\frac{2x^3+2}{x}\right) = 0$.

$$6x^2 - \left(\frac{2x^3+2}{x}\right) = 0 \Rightarrow 6x^3 - 2x^3 = 2 \Rightarrow 4x^3 = 2 \Rightarrow x = \frac{1}{\sqrt[3]{2}}$$

2-2-2b

Part D

At most 1 out of 2 implicit differentiation points is earned for correct implicit differentiation with a correct application of the quotient rule and chain rule with a maximum of one computational error.

The third point requires substitution of values with correct or consistent use of the $\frac{dy}{dx}$ value from part (b). A simplified answer is not required.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ implicit differentiation (2 points for full credit)
- ☐ answer

Solution:

$$\left. \frac{dy}{dx} \right|_{(x,y)=(1,2)} = \frac{1}{2}$$

$$\frac{d^2y}{dx^2} = \frac{d}{dx} \left(\frac{6x^2 - y^2}{2xy} \right) = \frac{2xy \left(12x - 2y \frac{dy}{dx} \right) - (6x^2 - y^2) \left(2x \frac{dy}{dx} + 2y \right)}{(2xy)^2}$$

$$\left. \frac{d^2y}{dx^2} \right|_{(x,y)=(1,2)} = \frac{4 \left(12 - 4 \cdot \frac{1}{2} \right) - (6 - 4) \left(2 \cdot \frac{1}{2} + 4 \right)}{4^2} = \frac{40 - 10}{16} = \frac{30}{16} = \frac{15}{8}$$

2-2-2b

56. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Consider the curve given by the equation $(2y + 1)^3 - 24x = -3$.

- (a) Show that $\frac{dy}{dx} = \frac{4}{(2y+1)^2}$.
- (b) Write an equation for the line tangent to the curve at the point $(-1, -2)$.
- (c) Evaluate $\frac{d^2y}{dx^2}$ at the point $(-1, -2)$.
- (d) The point $(\frac{1}{6}, 0)$ is on the curve. Find the value of $(y^{-1})'(0)$.

Part A

The first point is earned for $3(2y + 1)^2 \cdot 2\frac{dy}{dx} - 24 = 0$ or equivalent.

The second point cannot be earned without the first point. The second point is earned for a response that arrives at the given expression rather than an algebraically equivalent expression.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ implicit differentiation
- ☐ verification

Solution:

2-2-2b

$$\frac{d}{dx}((2y+1)^3 - 24x) = \frac{d}{dx}(-3)$$

$$\Rightarrow 3(2y+1)^2 \cdot 2 \frac{dy}{dx} - 24 = 0$$

$$\Rightarrow \frac{dy}{dx} = \frac{24}{6(2y+1)^2} = \frac{4}{(2y+1)^2}$$

Part B

The second point may be earned if response contains an incorrect value for slope based on one computational error.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ slope
- ☐ equation for tangent line

Solution:

$$\frac{dy}{dx} \Big|_{(x,y)=(-1,-2)} = \frac{4}{(2(-2)+1)^2} = \frac{4}{9}$$

An equation for the tangent line is $y = -2 + \frac{4}{9}(x+1)$.

Part C

The first point is earned with $4(-2)(2y+1)^{-3} \cdot 2 \frac{dy}{dx}$ or equivalent.

The second point is earned with an algebraic or numerical substitution for $\frac{dy}{dx}$. The response does not require an explicit expression for $\frac{d^2y}{dx^2}$. The third point does not require a simplified answer.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2	3
---	---	---	---

The student responses accurately includes all three of the criteria below.

2-2-2b

- ☐ implicit differentiation (2 points for full credit)
- ☐ answer

Solution:

$$\begin{aligned}\frac{d^2y}{dx^2} &= \frac{d}{dx} \left(4(2y+1)^{-2} \right) \\ &= 4(-2)(2y+1)^{-3} \cdot 2 \frac{dy}{dx} \\ &= \frac{-16}{(2y+1)^3} \cdot \frac{4}{(2y+1)^2} = \frac{-64}{(2y+1)^5}\end{aligned}$$

$$\left. \frac{d^2y}{dx^2} \right|_{(x,y)=(-1,-2)} = \frac{-64}{(2(-2)+1)^5} = \frac{-64}{-243} = \frac{64}{243}$$

Part D

The second point may be earned if response contains an incorrect value based on one computational error. Substitution of function values is required.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ $\frac{dy}{dx}$ at $(\frac{1}{6}, 0)$
- ☐ answer

Solution:

$$\left. \frac{dy}{dx} \right|_{(x,y)=(\frac{1}{6}, 0)} = \frac{4}{(2 \cdot 0 + 1)^2} = 4$$

Because $(\frac{1}{6}, 0)$ is on the curve, $(0, \frac{1}{6})$ is on the inverse.

$$(y^{-1})'(0) = \frac{1}{y'(y^{-1}(0))} = \frac{1}{y'(\frac{1}{6})} = \frac{1}{4}$$

2-2-2b

57. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Consider the curve given by the equation $2xy + y^2 = 8$ for $y > 0$.

- Show that $\frac{dy}{dx} = \frac{-y}{x+y}$.
- Write an equation for the line tangent to the curve at the point $(1, 2)$.
- Evaluate $\frac{d^2y}{dx^2}$ at the point $(1, 2)$.
- The points $(1, 2)$ and $(\frac{7}{2}, 1)$ are on the curve. Find the value of $(y^{-1})'(1)$.

Part A

The first point is earned for $2x\frac{dy}{dx} + 2y + 2y\frac{dy}{dx} = 0$ or equivalent

The second point cannot be earned without the first point. The second point is earned for a response that arrives at the given expression rather than an algebraically equivalent expression.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ implicit differentiation
- ☐ verification

Solution:

2-2-2b

$$\begin{aligned}\frac{d}{dx}(2xy + y^2) &= \frac{d}{dx}(8) \\ \Rightarrow 2x \frac{dy}{dx} + 2y + 2y \frac{dy}{dx} &= 0 \\ \Rightarrow x \frac{dy}{dx} + y \frac{dy}{dx} &= -y \\ \Rightarrow \frac{dy}{dx} &= \frac{-y}{x+y}\end{aligned}$$

Part B

The second point may be earned if response contains an incorrect value for slope based on one computational error.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below

- ☐ slope
- ☐ equation for tangent line

Solution:

$$\left. \frac{dy}{dx} \right|_{(x,y)=(1,2)} = \frac{-2}{1+2} = -\frac{2}{3}$$

An equation for the tangent line is $y = 2 - \frac{2}{3}(x - 1)$.

Part C

The first point is earned with

$$\frac{(x+y) \cdot \left(-1 \frac{dy}{dx}\right) - (-y) \left(1 + 1 \frac{dy}{dx}\right)}{(x+y)^2} \text{ or equivalent.}$$

The second point is earned with an algebraic or numerical substitution for $\frac{dy}{dx}$.

The response does not require an explicit expression for $\frac{d^2y}{dx^2}$.

The third point does not require a simplified answer.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

2-2-2b



0	1	2	3
---	---	---	---

The student responses accurately includes all three of the criteria below.

- ☐ implicit differentiation (2 points for full credit)
- ☐ answer

Solution:

$$\begin{aligned}\frac{d^2y}{dx^2} &= \frac{d}{dx} \left(\frac{-y}{x+y} \right) \\ &= \frac{(x+y) \cdot \left(-1 \frac{dy}{dx} \right) - (-y) \left(1 + 1 \frac{dy}{dx} \right)}{(x+y)^2} \\ &= \frac{(x+y) \cdot \left(\frac{y}{x+y} \right) + y \left(1 + \left(\frac{-y}{x+y} \right) \right)}{(x+y)^2} = \frac{y^2 + 2xy}{(x+y)^3}\end{aligned}$$

$$\left. \frac{d^2y}{dx^2} \right|_{(x,y)=(1,2)} = \frac{2^2 + 2(1)(2)}{(1+2)^3} = \frac{8}{27}$$

Part D

The second point may be earned if response contains an incorrect value based on one computational error. Substitution of function values is required.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The student response does not accurately include any of the criteria below.

- ☐ $\frac{dy}{dx}$ at $\left(\frac{7}{2}, 1\right)$
- ☐ answer

Solution:

$$\left. \frac{dy}{dx} \right|_{(x,y)=\left(\frac{7}{2}, 1\right)} = \frac{-1}{\left(\frac{7}{2}+1\right)} = -\frac{2}{9}$$

2-2-2b

Because $(\frac{7}{2}, 1)$ is on the curve, $(1, \frac{7}{2})$ is on the inverse.

$$(y^{-1})'(1) = \frac{1}{y'(y^{-1}(1))} = \frac{1}{y'(\frac{7}{2})} = -\frac{9}{2}$$

$$(y^{-1})'(1) = \frac{1}{y'(y^{-1}(1))} = \frac{1}{y'(\frac{7}{2})} = -\frac{9}{2}$$

2-2-2b

58. Consider the curve given by $y^2 = 2 + xy$.

(a) Show that $\frac{dy}{dx} = \frac{y}{2y-x}$.

(b) Find all points (x, y) on the curve where the line tangent to the curve has slope $1/2$.

(c) Show that there are no points (x, y) on the curve where the line tangent to the curve is horizontal.

(d) Let x and y be functions of time t that are related by the equation $y^2 = 2 + xy$. At time $t = 5$, the value of y is 3 and $dy/dt = 6$. Find the value of dx/dt at time $t = 5$.

Part A

1 point is earned for the implicit differentiation

1 point is earned if solved for y'

$$2yy' = y + xy'$$

$$(2y - x)y' = y$$

$$y' = \frac{y}{2y - x}$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the implicit differentiation

1 point is earned if solved for y'

$$2yy' = y + xy'$$

$$(2y - x)y' = y$$

$$y' = \frac{y}{2y - x}$$

Part B

1 point is earned for $\frac{y}{2y-x} = \frac{1}{2}$

1 point is earned for the answer

2-2-2b

$$\frac{y}{2y-x} = \frac{1}{2}$$

$$2y = 2y - x$$

$$x = 0$$

$$y = \pm\sqrt{2}$$

$$(0, \sqrt{2}), (0, -\sqrt{2})$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for $\frac{y}{2y-x} = \frac{1}{2}$

1 point is earned for the answer

$$\frac{y}{2y-x} = \frac{1}{2}$$

$$2y = 2y - x$$

$$x = 0$$

$$y = \pm\sqrt{2}$$

$$(0, \sqrt{2}), (0, -\sqrt{2})$$

Part C

1 point is earned for $y = 0$

1 point is earned for the explanation

$$\frac{y}{2y-x} = 0$$

$$y = 0$$

The curve has no horizontal tangent since

2-2-2b

$$0^2 \neq 2 + x \cdot 0 \text{ for any } x.$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for $y = 0$

1 point is earned for the explanation

$$\frac{y}{2y-x} = 0$$

$$y = 0$$

The curve has no horizontal tangent since

$$0^2 \neq 2 + x \cdot 0 \text{ for any } x.$$

Part D

1 point is earned if solves for x

1 point is earned for the chain rule

1 point is earned for the answer

When $y = 3$, $3^2 = 2 + 3x$ so $x = 7/3$.

$$\frac{dy}{dt} = \frac{dy}{dx} \cdot \frac{dx}{dt} = \frac{y}{2y-x} \cdot \frac{dx}{dt}$$

$$\text{At } t = 5, 6 = \frac{3}{6 - \frac{7}{3}} \cdot \frac{dx}{dt} = \frac{9}{11} \cdot \frac{dx}{dt}$$

$$\frac{dx}{dt} \Big|_{t=5} = \frac{22}{3}$$



0	1	2	3
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The student response earns all of the following points:

1 point is earned if solves for x

1 point is earned for the chain rule

2-2-2b

1 point is earned for the answer

When $y = 3$, $3^2 = 2 + 3x$ so $x = 7/3$.

$$\frac{dy}{dt} = \frac{dy}{dx} \cdot \frac{dx}{dt} = \frac{y}{2y-x} \cdot \frac{dx}{dt}$$

$$\text{At } t = 5, 6 = \frac{3}{6 - \frac{7}{3}} \cdot \frac{dx}{dt} = \frac{9}{11} \cdot \frac{dx}{dt}$$

$$\frac{dx}{dt} \Big|_{t=5} = \frac{22}{3}$$

2-2-2b

59. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Consider the function $y = f(x)$ whose curve is given by the equation $2y^2 - 6 = y \sin x$ for $y > 0$.

(a) Show that $\frac{dy}{dx} = \frac{y \cos x}{4y - \sin x}$.

(b) Write an equation for the line tangent to the curve at the point $(0, \sqrt{3})$.

(c) For $0 \leq x \leq \pi$ and $y > 0$, find the coordinates of the point where the line tangent to the curve is horizontal.

(d) Determine whether f has a relative minimum, a relative maximum, or neither at the point found in part (c). Justify your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned only for correctly implicitly differentiating $2y^2 - 6 = y \sin x$. Responses may use alternative notations for $\frac{dy}{dx}$, such as y' .

The second point may not be earned without the first point.

It is sufficient to present $\frac{dy}{dx}(4y - \sin x) = y \cos x$ to earn the second point, provided that there are no subsequent errors.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Implicit differentiation
- ☐ Verification

Solution:

2-2-2b

$$\begin{aligned}\frac{d}{dx}(2y^2 - 6) &= \frac{d}{dx}(y \sin x) \Rightarrow 4y \frac{dy}{dx} = \frac{dy}{dx} \sin x + y \cos x \\ \Rightarrow 4y \frac{dy}{dx} - \frac{dy}{dx} \sin x &= y \cos x \Rightarrow \frac{dy}{dx}(4y - \sin x) = y \cos x \\ \Rightarrow \frac{dy}{dx} &= \frac{y \cos x}{4y - \sin x}\end{aligned}$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Any correct tangent line equation will earn the point. No supporting work is required. Simplification of the slope value is not required.



0	1
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The student response accurately includes the criteria below.

☐ Answer

Solution:

At the point $(0, \sqrt{3})$, $\frac{dy}{dx} = \frac{\sqrt{3} \cos 0}{4\sqrt{3} - \sin 0} = \frac{1}{4}$.

An equation for the tangent line is $y = \sqrt{3} + \frac{1}{4}x$.

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned by any of $\frac{dy}{dx} = 0$, $\frac{y \cos x}{4y - \sin x} = 0$, or $\cos x = 0$.

If additional “correct” x -values are considered outside of the given domain, the response must commit to only $x = \frac{\pi}{2}$ to earn the second point. Any presented y -values, correct or incorrect, are not considered for the second point.

Entering with $x = \frac{\pi}{2}$ does not earn the first point, earns the second point, and is eligible for the third point. The third point is earned for finding $y = 2$. The coordinates do not have to be presented as an ordered pair.

The third point is not earned with additional points present unless the response commits to the correct point.

2-2-2b



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Sets $\frac{dy}{dx} = 0$
- ☐ $x = \frac{\pi}{2}$
- ☐ $y = 2$

Solution:

$$\frac{dy}{dx} = \frac{y \cos x}{4y - \sin x} = 0 \Rightarrow y \cos x = 0 \text{ and } 4y - \sin x \neq 0$$

$$y \cos x = 0 \text{ and } y > 0 \Rightarrow x = \frac{\pi}{2}$$

$$\text{When } x = \frac{\pi}{2}, y \sin x = 2y^2 - 6 \Rightarrow y \sin \frac{\pi}{2} = 2y^2 - 6$$

$$\Rightarrow y = 2y^2 - 6 \Rightarrow 2y^2 - y - 6 = 0$$

$$\Rightarrow (2y + 3)(y - 2) = 0 \Rightarrow y = 2$$

When $x = \frac{\pi}{2}$ and $y = 2$, $4y - \sin x = 8 - 1 \neq 0$. Therefore, the line tangent to the curve is horizontal at the point $(\frac{\pi}{2}, 2)$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned for an attempt to use the quotient rule (or product rule) to find $\frac{d^2y}{dx^2}$.

The second point is earned for correctly finding $\frac{d^2y}{dx^2}$ and evaluating to find that $\frac{d^2y}{dx^2} < 0$ at $(\frac{\pi}{2}, 2)$. The explicit value of $-\frac{2}{7}$ or the equivalent does not need to be reported, but any reported values must be correct in order to earn this point.

The third point can be earned without the second point by reaching a consistent conclusion based on the reported sign of a nonzero value of $\frac{d^2y}{dx^2}$ obtained utilizing $\frac{dy}{dx} = 0$.

Imports: A response is eligible to earn all 3 points in part (d) with a point of the form $(\frac{\pi}{2}, k)$ with $k > 0$, imported from part (c).

2-2-2b



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ Considers $\frac{d^2y}{dx^2}$
- ☐ $\frac{d^2y}{dx^2}$ at $(\frac{\pi}{2}, 2)$
- ☐ Answer with justification

Solution:

$$\frac{d^2y}{dx^2} = \frac{(4y - \sin x) \left(\frac{dy}{dx} \cos x - y \sin x \right) - (y \cos x) \left(4 \frac{dy}{dx} - \cos x \right)}{(4y - \sin x)^2}$$

When $x = \frac{\pi}{2}$ and $y = 2$,

$$\begin{aligned} \frac{d^2y}{dx^2} &= \frac{(4 \cdot 2 - \sin \frac{\pi}{2}) (0 \cdot \cos \frac{\pi}{2} - 2 \cdot \sin \frac{\pi}{2}) - (2 \cos \frac{\pi}{2}) (4 \cdot 0 - \cos \frac{\pi}{2})}{(4 \cdot 2 - \sin \frac{\pi}{2})^2} \\ &= \frac{(7)(-2) - (0)(0)}{(7)^2} = \frac{-2}{7} < 0. \end{aligned}$$

f has a relative maximum at the point $(\frac{\pi}{2}, 2)$ because $\frac{dy}{dx} = 0$ and $\frac{d^2y}{dx^2} < 0$.

2-2-2b

60. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Consider the curve defined by the equation $xy = x + y^2 + 3$.

- (a) Show that $\frac{dy}{dx} = \frac{1-y}{x-2y}$.
- (b) Find the slope of the line tangent to the curve at the point $(7, 2)$.
- (c) Find an equation for each vertical line tangent to the curve.
- (d) Evaluate $\frac{d^2y}{dx^2}$ at the point $(7, 2)$. Show the work that leads to your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned only for correct implicit differentiation of $xy = x + y^2 + 3$. Responses may use alternative notation for $\frac{dy}{dx}$, such as y' .

The second point may not be earned without the first point.

It is sufficient to present $(x - 2y)\frac{dy}{dx} = 1 - y$ to earn the second point, provided that there are no subsequent errors.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Implicit differentiation
- ☐ Verification

Solution:

2-2-2b

$$\frac{d}{dx}(xy) = \frac{d}{dx}(x + y^2 + 3) \Rightarrow y + x \frac{dy}{dx} = 1 + 2y \frac{dy}{dx}$$

$$\Rightarrow (x - 2y) \frac{dy}{dx} = 1 - y \Rightarrow \frac{dy}{dx} = \frac{1-y}{x-2y}$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response of $-\frac{1}{3}$ with no supporting work earns the point.



0	1
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The student response accurately includes the criteria below.

☐ Answer

Solution:

$$\text{Slope} = \frac{dy}{dx} \Big|_{(x,y)=(7,2)} = \frac{1-2}{7-4} = -\frac{1}{3}$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response need not specify that $y - 1 \neq 0$ in order to earn any point in this part.

The third point is earned only for both answers $x = -2$ and $x = 6$, accompanied by supporting work.

Alternate solution:

Substituting $y = \frac{x}{2}$ in $xy = x + y^2 + 3$ yields the equation $\frac{x^2}{2} = x + \frac{x^2}{4} + 3$.

$$\Rightarrow \frac{1}{4}(x^2 - 4x - 12) = \frac{1}{4}(x + 2)(x - 6) = 0$$

$$\Rightarrow x = -2 \text{ or } x = 6,$$

$$\text{If } x = -2 \text{ then } -2y = -2 + y^2 + 3 \Rightarrow y = -1 \neq 1.$$

$$\text{If } x = 6, \text{ then } 6y = 6 + y^2 + 3 \Rightarrow y = 3 \neq 1.$$

The vertical tangent lines to the curve have equations $x = -2$ and $x = 6$.

2-2-2b



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ Sets $x - 2y = 0$
- ☐ Substitutes $x = 2y$ in $xy = x + y^2 + 3$
- ☐ Answers

Solution:

For the line tangent to the curve to be vertical, it is necessary that $x - 2y = 0$ (so $x = 2y$) and $y - 1 \neq 0$ ($y \neq 1$).

Substituting $x = 2y$ in $xy = x + y^2 + 3$ yields the equation $2y^2 = 2y + y^2 + 3$.

$$\Rightarrow y^2 - 2y - 3 = (y + 1)(y - 3) = 0$$

$$\Rightarrow y = -1 \text{ or } y = 3$$

$$\text{If } y = -1, \text{ then } -x = x + 1 + 3 \Rightarrow x = -2.$$

$$\text{If } y = 3, \text{ then } 3x = x + 9 + 3 \Rightarrow x = 6.$$

The vertical tangent lines to the curve have equations $x = -2$ and $x = 6$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first two points can be earned in either order.

A response that presents at most one error (copy, derivative, missing parenthesis) in applying the quotient rule or product rule earns the first point.

- Any error in a presented expression for $\frac{d^2y}{dx^2}$ is not eligible for the third point.

To earn the second point, a response must present both occurrences of $\frac{dy}{dx}$ (or y') in their expression for $\frac{d^2y}{dx^2}$ (or y'').

A response must have earned both of the first 2 points in order to be eligible for the third point.

The third point is earned with a correct expression for $\frac{d^2y}{dx^2} \Big|_{(x,y)=(7,2)}$ consistent with the declared value of $\frac{dy}{dx} \Big|_{(x,y)=(7,2)}$ from part (b).

2-2-2b

The value of $\frac{d^2y}{dx^2} \Big|_{(x,y)=(7,2)}$ does not need to be simplified. Any error in simplification does not earn the third point.

Alternate solution:

$$xy = x + y^2 + 3 \Rightarrow y + x \frac{dy}{dx} = 1 + 2y \frac{dy}{dx}$$

$$\frac{dy}{dx} + x \frac{d^2y}{dx^2} + \frac{dy}{dx} \cdot 1 = 0 + 2y \cdot \frac{d^2y}{dx^2} + \frac{dy}{dx} \cdot 2 \cdot \frac{dy}{dx}$$

$$\text{From part (b)} \quad \frac{dy}{dx} \Big|_{(x,y)=(7,2)} = -\frac{1}{3}$$

$$-\frac{1}{3} + (7) \frac{d^2y}{dx^2} + \left(-\frac{1}{3}\right) = 2(2) \cdot \frac{d^2y}{dx^2} + \left(-\frac{1}{3}\right) \cdot 2 \cdot \left(-\frac{1}{3}\right) \Rightarrow \frac{d^2y}{dx^2} = \frac{8}{27}$$



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ Quotient rule (or product rule)
- ☐ Chain rule
- ☐ Answer

Solution:

$$\frac{dy}{dx} = \frac{1-y}{x-2y}$$

$$\text{Using the quotient rule, } \frac{d^2y}{dx^2} = \frac{-\frac{dy}{dx}(x-2y) - (1-y)\left(1-2\frac{dy}{dx}\right)}{(x-2y)^2}.$$

$$\text{From part (b), at } (x, y) = (7, 2), \frac{dy}{dx} \Big|_{(x,y)=(7,2)} = -\frac{1}{3}.$$

$$\text{Thus, } \frac{d^2y}{dx^2} \Big|_{(x,y)=(7,2)} = \frac{-\left(-\frac{1}{3}\right)(7-4) - (1-2)\left(1-2\left(-\frac{1}{3}\right)\right)}{(7-4)^2} = \frac{1+\frac{5}{3}}{9} = \frac{8}{27}.$$

2-2-2b

61.  Line L is tangent to the graph of $y = e^x$ at the point (k, e^k) . What is the positive value of k for which the y -intercept of L is $\frac{1}{2}$?

(A) 0.405

(B) 0.768

(C) 1.500

(D) 1.560

(E) There is no such value of k .

62. What is the slope of the line tangent to the graph of $y = \frac{e^{-x}}{x+1}$ at $x = 1$?

(A) $-\frac{1}{e}$

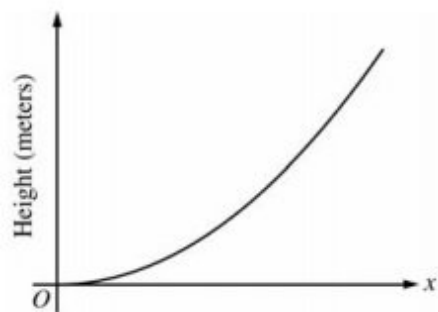
(B) $-\frac{3}{4e}$

(C) $-\frac{1}{4e}$

(D) $\frac{1}{4e}$

(E) $\frac{1}{e}$

2-2-2b



The figure above is the graph of a function of x , which models the height of a skateboard ramp. The function meets the following requirements.

- (i) At $x=0$, the value of the function is 0, and the slope of the graph of the function is 0.
- (ii) At $x=4$, the value of the function is 1, and the slope of the graph of the function is 1.
- (iii) Between $x=0$ and $x=4$, the function is increasing.

63. Let $h(x) = x^n / k$, where k is a nonzero constant and n is a positive integer. Find the values of k and n so that h meets requirement (ii) above. Show that h also meets requirements (i) and (iii) above.

Part D

1 point is earned for $\frac{4^n}{k} = 1$

1 point is earned for $\frac{n4^{n-1}}{k} = 1$

1 point is earned for values for k and n

1 point is earned for verification

$$h(4) = \frac{4^n}{k} = 1 \text{ implies that } 4^n = k.$$

$$h'(4) = \frac{n4^{n-1}}{k} = \frac{n4^{n-1}}{4^n} = \frac{n}{4} = 1 \text{ gives } n=4 \text{ and } k = 4^4 = 256.$$

$$h(x) = \frac{x^4}{256} \Rightarrow h(0) = 0.$$

$$h'(x) = \frac{4x^3}{256} \Rightarrow h'(0) = 0 \text{ and } h'(x) > 0 \text{ for } 0 < x < 4.$$



0	1	2	3	4
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The student response earns all of the following points:

2-2-2b

1 point is earned for $\frac{4^n}{k} = 1$

1 point is earned for $\frac{n4^{n-1}}{k} = 1$

1 point is earned for values for k and n

1 point is earned for verification

$$h(4) = \frac{4^n}{k} = 1 \text{ implies that } 4^n = k.$$

$$h'(4) = \frac{n4^{n-1}}{k} = \frac{n4^{n-1}}{4^n} = \frac{n}{4} = 1 \text{ gives } n=4 \text{ and } k = 4^4 = 256.$$

$$h(x) = \frac{x^4}{256} \Rightarrow h(0) = 0.$$

$$h'(x) = \frac{4x^3}{256} \Rightarrow h'(0) = 0 \text{ and } h'(x) > 0 \text{ for } 0 < x < 4.$$

- 64.** Let $g(x) = cx^3 - \frac{x^2}{16}$ where c is a nonzero constant. Find the value of c so that g meets requirement (ii) above. Show the work that leads to your answer.

Part B

1 point is earned for the value of c

$$g(4) = 64c - 1 = 1 \text{ implies that } c = \frac{1}{32}.$$

$$\text{When } c = \frac{1}{32}, g'(4) = 3c(4)^2 - \frac{2(4)}{16} = 3\left(\frac{1}{32}\right)(16) - \frac{1}{2} = 1$$



0	1
---	---

The student response earns none of the following points:

1 point is earned for the value of c

$$g(4) = 64c - 1 = 1 \text{ implies that } c = \frac{1}{32}.$$

$$\text{When } c = \frac{1}{32}, g'(4) = 3c(4)^2 - \frac{2(4)}{16} = 3\left(\frac{1}{32}\right)(16) - \frac{1}{2} = 1$$

2-2-2b

65. Let $f(x) = ax^2$, where a is a nonzero constant. Show that it is not possible to find a value for a so that f meets requirement (ii) above.
-

Part A

1 point is earned for if $a = \frac{1}{16}$ or $a = \frac{1}{8}$

1 point is earned if shows a does not work

$f(4)=1$ implies that $a = \frac{1}{16}$ and $f(4)=2a(4)=1$ implies that $a = \frac{1}{8}$. Thus, f cannot satisfy (ii).



0	1	2
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The student response earns all of the following points:

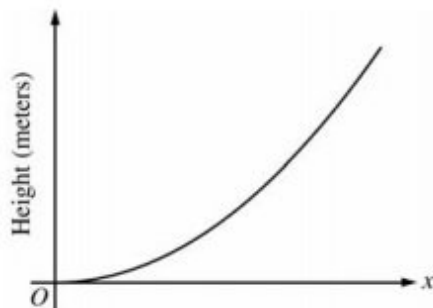
1 point is earned for if $a = \frac{1}{16}$ or $a = \frac{1}{8}$

1 point is earned if shows a does not work

$f(4)=1$ implies that $a = \frac{1}{16}$ and $f(4)=2a(4)=1$ implies that $a = \frac{1}{8}$. Thus, f cannot satisfy (ii).

2-2-2b

66.



The figure above is the graph of a function of x , which models the height of a skateboard ramp. The function meets the following requirements.

(i) At $x = 0$, the value of the function is 0, and the slope of the graph of the function is 0.

(ii) At $x = 4$, the value of the function is 1, and the slope of the graph of the function is 1.

(iii) Between $x = 0$ and $x = 4$, the function is increasing.

(a) Let $f(x) = ax^2$, where a is a nonzero constant. Show that it is not possible to find a value for a so that f meets requirement (ii) above.

(b) Let $g(x) = cx^3 - \frac{x^2}{16}$, where c is a nonzero constant. Find the value of c so that g meets requirement (ii) above. Show the work that leads to your answer.

(c) Using the function g and your value of c from part (b), show that g does not meet requirement (iii) above.

(d) Let $h(x) = x^n/k$, where k is a nonzero constant and n is a positive integer. Find the values of k and n so that h meets requirement (ii) above. Show that h also meets requirements (i) and (iii) above.

Part A

1 point is earned for if $a = 1/16$ or $a = 1/8$

1 point is earned if shows a does not work

$f(4) = 1$ implies that $a = 1/16$ and $f'(4) = 2a(4) = 1$ implies that $a = 1/8$. Thus, f cannot satisfy (ii).



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for if $a = 1/16$ or $a = 1/8$

1 point is earned if shows a does not work

$f(4) = 1$ implies that $a = 1/16$ and $f'(4) = 2a(4) = 1$ implies that $a = 1/8$. Thus, f cannot satisfy (ii).

2-2-2b

Part B

1 point is earned for the value of c

$$g(4) = 64c - 1 = 1 \text{ implies that } c = \frac{1}{32}.$$

$$\text{When } c = \frac{1}{32}, g'(4) = 3c(4)^2 - \frac{2(4)}{16} = 3\left(\frac{1}{32}\right)(16) - \frac{1}{2} = 1$$



0	1
---	---

The student response earns all of the following points:

1 point is earned for the value of c

$$g(4) = 64c - 1 = 1 \text{ implies that } c = \frac{1}{32}.$$

$$\text{When } c = \frac{1}{32}, g'(4) = 3c(4)^2 - \frac{2(4)}{16} = 3\left(\frac{1}{32}\right)(16) - \frac{1}{2} = 1$$

Part C

1 point is earned for $g'(x)$

1 point is earned for the explanation

$$g'(x) = \frac{3}{32}x^2 - \frac{x}{8} = \frac{1}{32}x(3x - 4)$$

$g'(x) < 0$ for $0 < x < 4/3$, so g does not satisfy (iii).



0	1	2
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The student response earns all of the following points:

2-2-2b

1 point is earned for $g'(x)$

1 point is earned for the explanation

$$g'(x) = \frac{3}{32}x^2 - \frac{x}{8} = \frac{1}{32}x(3x - 4)$$

$g'(x) < 0$ for $0 < x < 4/3$, so g does not satisfy (iii).

Part D

1 point is earned for $\frac{4^n}{k} = 1$

1 point is earned for $\frac{n4^{n-1}}{k} = 1$

1 point is earned for values for k and n

1 point is earned for verification

$$h(4) = \frac{4^n}{k} = 1 \text{ implies that } 4^n = k.$$

$$h'(4) = \frac{n4^{n-1}}{k} = \frac{n4^{n-1}}{4^n} = \frac{n}{4} = 1 \text{ gives } n = 4 \text{ and } k = 4^4 = 256.$$

$$h(x) = \frac{x^4}{256} \Rightarrow h(0) = 0.$$

$$h'(x) = \frac{4x^3}{256} \Rightarrow h'(0) = 0 \text{ and } h'(x) > 0 \text{ for } 0 < x < 4.$$



0	1	2	3	4
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The student response earns all of the following points:

1 point is earned for $\frac{4^n}{k} = 1$

1 point is earned for $\frac{n4^{n-1}}{k} = 1$

1 point is earned for values for k and n

2-2-2b

1 point is earned for verification

$$h(4) = \frac{4^n}{k} = 1 \text{ implies that } 4^n = k.$$

$$h'(4) = \frac{n4^{n-1}}{k} = \frac{n4^{n-1}}{4^n} = \frac{n}{4} = 1 \text{ gives } n = 4 \text{ and } k = 4^4 = 256.$$

$$h(x) = \frac{x^4}{256} \Rightarrow h(0) = 0.$$

$$h'(x) = \frac{4x^3}{256} \Rightarrow h'(0) = 0 \text{ and } h'(x) > 0 \text{ for } 0 < x < 4.$$

67. The following are related to this scenario:

Consider the differential equation $\frac{dy}{dx} = \left(1 - \frac{2}{x^2}\right)(y - 1)$, where $x \neq 0$. Let $y = f(x)$ be the particular solution to the differential equation with initial condition $f(1) = 2$.

Find the slope of the line tangent to the graph of f at the point $(1, 2)$.

Part A

The response can earn up to 1 point:

1 point: For the correct answer

$$\left. \frac{dy}{dx} \right|_{(x,y)=(1,2)} = \left(1 - \frac{2}{1}\right)(2 - 1) = -1$$



0	1
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The response earns the following point:

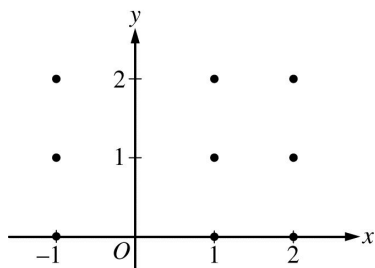
1 point: For the correct answer

$$\left. \frac{dy}{dx} \right|_{(x,y)=(1,2)} = \left(1 - \frac{2}{1}\right)(2 - 1) = -1$$

2-2-2b

68. Consider the differential equation $\frac{dy}{dx} = \left(1 - \frac{2}{x^2}\right)(y - 1)$, where $x \neq 0$. Let $y = f(x)$ be the particular solution to the differential equation with initial condition $f(1) = 2$.

- (a) Find the slope of the line tangent to the graph of f at the point $(1, 2)$.
- (b) On the axes provided, sketch a slope field for the given differential equation at the nine points indicated.



- (c) Find the particular solution $y = f(x)$ to the differential equation $\frac{dy}{dx} = \left(1 - \frac{2}{x^2}\right)(y - 1)$ with initial condition $f(1) = 2$.

Part A

1 point is earned for the correct answer

$$\left. \frac{dy}{dx} \right|_{(x,y)=(1,2)} = \left(1 - \frac{2}{1}\right)(2 - 1) = -1$$



0

1

The student earns all of the following point:

1 point is earned for the correct answer

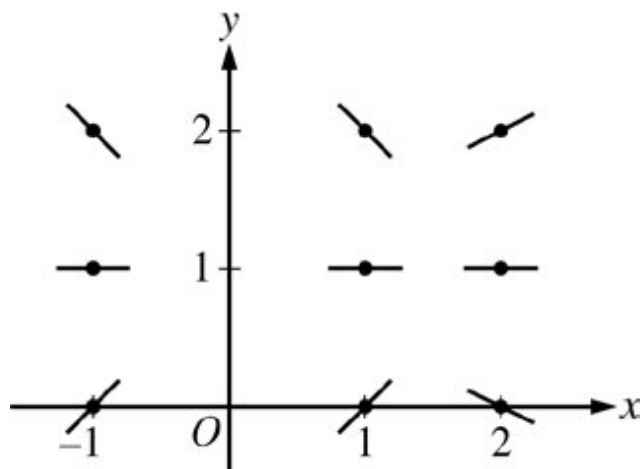
$$\left. \frac{dy}{dx} \right|_{(x,y)=(1,2)} = \left(1 - \frac{2}{1}\right)(2 - 1) = -1$$

Part B

1 point is earned for zero slope at each point (x, y) where $y = 1$

1 point is earned for the remaining slopes

2-2-2b



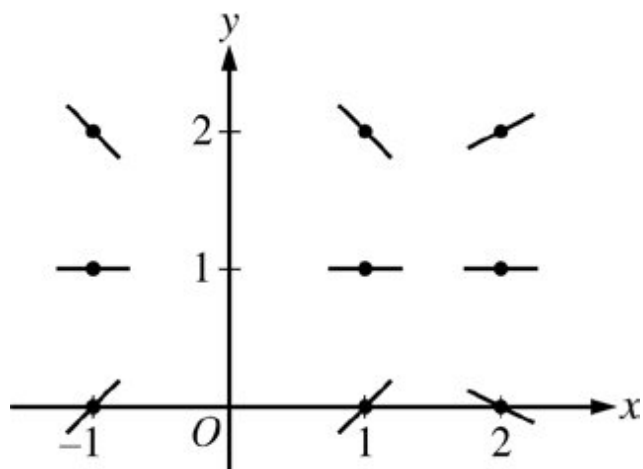
✓

0	1	2
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The student earns all of the following points:

1 point is earned for zero slope at each point (x, y) where $y = 1$

1 point is earned for the remaining slopes



Part C

1 point is earned for the separation of variables

2 points are earned for the antiderivatives

1 point is earned for the constant of integration

1 point is earned for uses of the initial condition

1 point is earned for the solution for y

2-2-2b

$$\frac{dy}{dx} = \left(1 - \frac{2}{x^2}\right) (y - 1)$$

$$\int \frac{dy}{y - 1} = \int \left(1 - \frac{2}{x^2}\right) dx$$

$$\ln |y - 1| = x + \frac{2}{x} + C$$

$$\ln |2 - 1| = 1 + \frac{2}{1} + C \Rightarrow C = -3$$

$$\ln |y - 1| = x + \frac{2}{x} - 3$$

Note that $y - 1 > 0$ since the solution curve includes the point $(1, 2)$.

$$\ln(y - 1) = x + \frac{2}{x} - 3$$

$$y = f(x) = e^{(x + \frac{2}{x} - 3)} + 1$$

Note: This solution is valid for $x > 0$.

Note: max 3/6 [1-2-0-0-0] if no constant of integration.

Note: 0/6 if no separation of variables



0	1	2	3	4	5	6
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The student earns all of the following points:

1 point is earned for the separation of variables

2 points are earned for the antiderivatives

1 point is earned for the constant of integration

1 point is earned for uses of the initial condition

1 point is earned for the solution for y

$$\frac{dy}{dx} = \left(1 - \frac{2}{x^2}\right) (y - 1)$$

$$\int \frac{dy}{y - 1} = \int \left(1 - \frac{2}{x^2}\right) dx$$

$$\ln |y - 1| = x + \frac{2}{x} + C$$

$$\ln |2 - 1| = 1 + \frac{2}{1} + C \Rightarrow C = -3$$

$$\ln |y - 1| = x + \frac{2}{x} - 3$$

Note that $y - 1 > 0$ since the solution curve includes the point $(1, 2)$.

2-2-2b

$$\ln(y - 1) = x + \frac{2}{x} - 3$$

$$y = f(x) = e^{(x + \frac{2}{x} - 3)} + 1$$

Note: This solution is valid for $x > 0$.

Note: max 3/6 [1-2-0-0-0] if no constant of integration.

Note: 0/6 if no separation of variables

2-2-2b

69. Let f be the function given by $f(x) = e^{-x}$, and let g be the function given by $g(x) = kx$, where k is the nonzero constant such that the graph of f is tangent to the graph of g .

Find the x -coordinate of the point of tangency and the value of k .

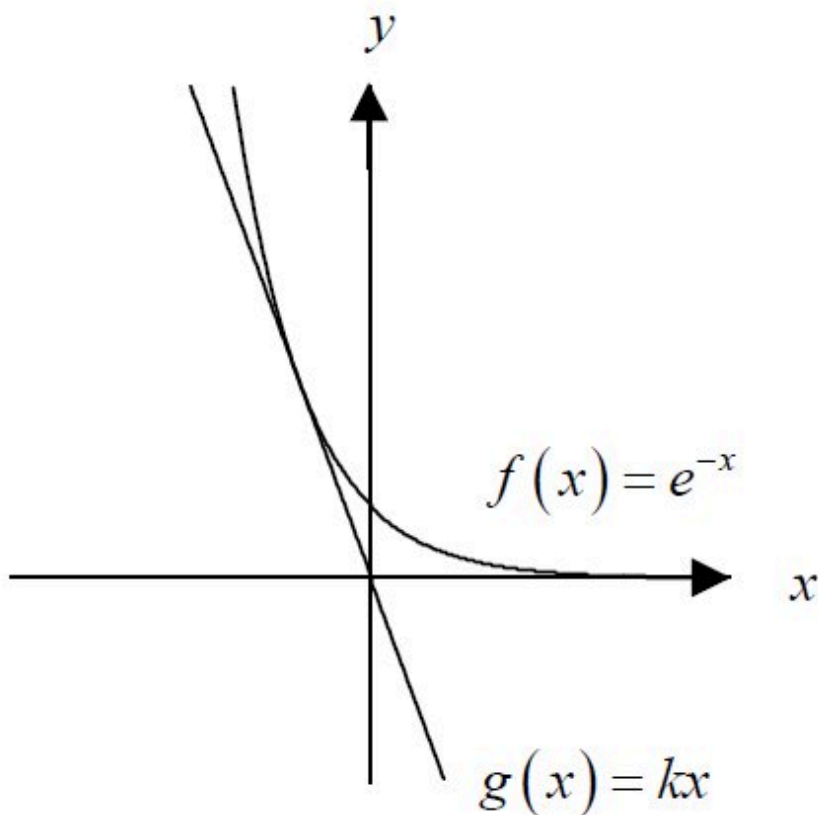
Part A

1 point is earned if finds $f(x)$ and $g'(x)$ explicitly

1 point is earned for setting student's $f(x)$ equal to student's $g'(x)$ explicitly

1 point is earned for setting $f(x) = g(x)$

1 point is earned for solving for x and k



$$f'(x) = -e^{-x}; g'(x) = K - e^{-x} = k \leftarrow +2$$

$$e^{-x} = kx$$

$$x = 1 \text{ and } k = (-e)$$



0	1	2	3	4
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2-2-2b

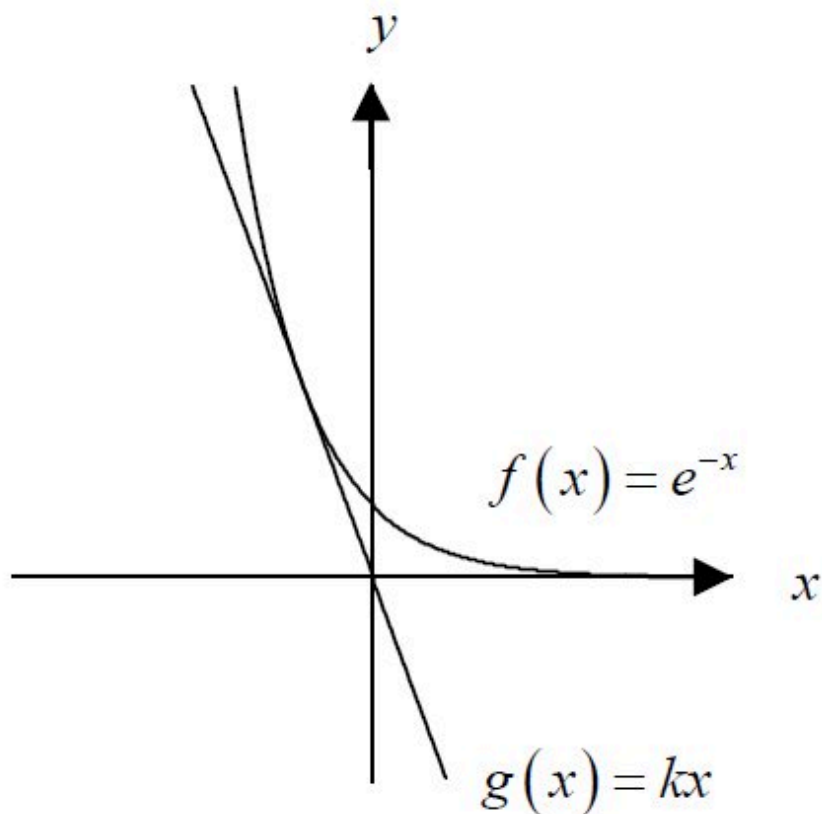
The student response earns four of the following points:

1 point is earned if finds $f'(x)$ and $g'(x)$ explicitly

1 point is earned for setting student's $f'(x)$ equal to student's $g'(x)$ explicitly

1 point is earned for setting $f(x)=g(x)$

1 point is earned for solving for x and k



$$f'(x) = -e^{-x}; g'(x) = K - e^{-x} = k \leftarrow +2$$

$$e^{-x} = kx$$

$$x = 1 \text{ and } k = (-e)$$

2-2-2b

70. Let f be the function given by $f(x) = e^{-x}$, and let g be the function given by $g(x) = kx$, where k is the nonzero constant such that the graph of f is tangent to the graph of g .
- (a) Find the x -coordinate of the point of tangency and the value of k .
- (b) Let R be the region enclosed by the y -axis and the graphs of f and g . Using the results found in part (a), determine the area of R .
- (c) Set up, but do not integrate, an integral expression in terms of a single variable for the volume of the solid generated by revolving the region R , given in part (b), about the x -axis.

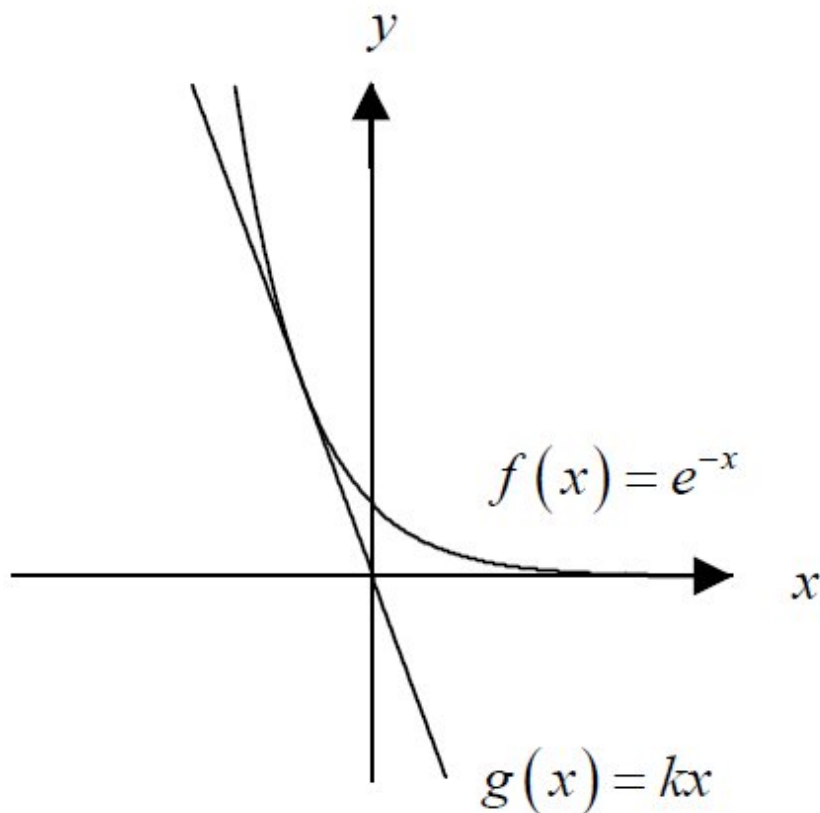
Part A

1 point is earned if finds $f'(x)$ and $g'(x)$ explicitly

1 point is earned for setting student's $f'(x)$ equal to student's $g'(x)$ explicitly

1 point is earned for setting $f(x) = g(x)$

1 point is earned for solving for x and k



$$f'(x) = -e^{-x}; \quad g'(x) = k$$

$$-e^{-x} = k \leftarrow +2$$

$$e^{-x} = kx$$

$$x = -1 \text{ and } k = -e$$

2-2-2b



0	1	2	3	4
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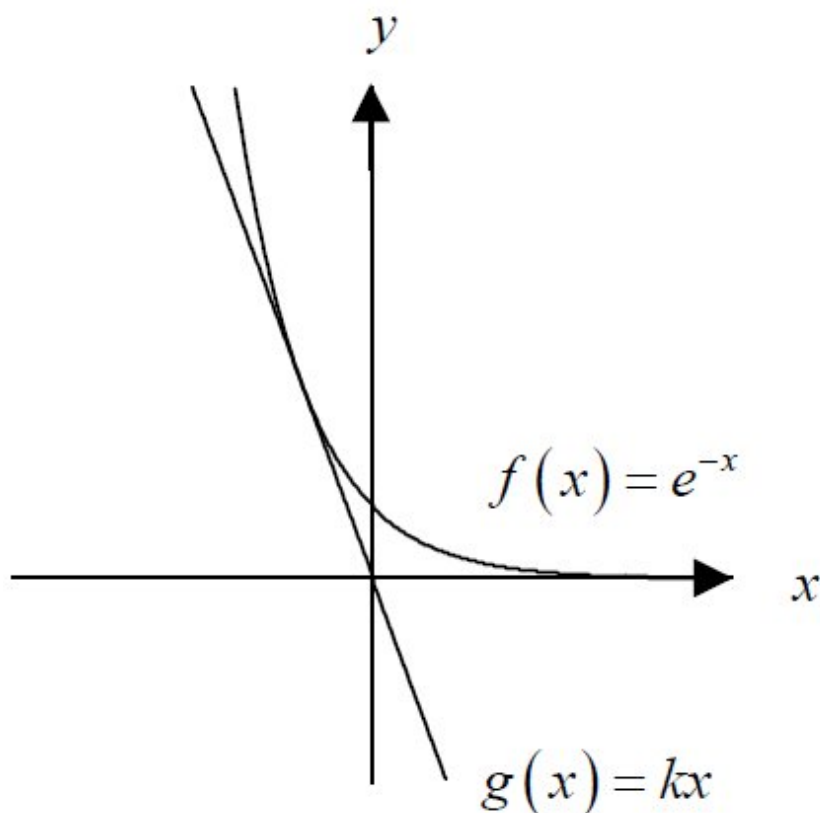
The student response earns all of the following points:

1 point is earned if finds $f'(x)$ and $g'(x)$ explicitly

1 point is earned for setting student's $f'(x)$ equal to student's $g'(x)$ explicitly

1 point is earned for setting $f(x) = g(x)$

1 point is earned for solving for x and k



$$f'(x) = -e^{-x}; g'(x) = k$$

$$-e^{-x} = k \leftarrow +2$$

$$e^{-x} = kx$$

$$x = -1 \text{ and } k = -e$$

Part B

2 points are earned for integral

2-2-2b

1 point is earned if anti-differentiates

$$\begin{aligned}\int_{-1}^0 (e^{-x} - (-ex)) dx &= \int_{-1}^0 (e^{-x} + ex) dx \\ &= \left(-e^{-x} + \frac{ex^2}{2} \right) \Big|_{-1}^0 \\ &= (-1 + 0) - \left(-e + \frac{e}{2} \right) \\ &= \frac{e}{2} - 1\end{aligned}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

2 points are earned for integral

1 point is earned if anti-differentiates

$$\begin{aligned}\int_{-1}^0 (e^{-x} - (-ex)) dx &= \int_{-1}^0 (e^{-x} + ex) dx \\ &= \left(-e^{-x} + \frac{ex^2}{2} \right) \Big|_{-1}^0 \\ &= (-1 + 0) - \left(-e + \frac{e}{2} \right) \\ &= \frac{e}{2} - 1\end{aligned}$$

Part C

2 points are earned for setting up integral with correct integrand

1 point **deducted** for incorrect limits/error involving π

$$\pi \int_{-1}^0 \left((e^{-x})^2 - (-ex)^2 \right) dx$$



0	1	2
---	---	---

The student response earns all of the following points:

2-2-2b

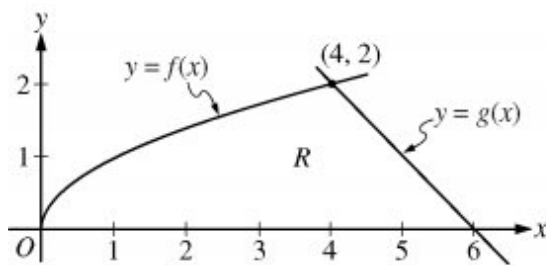
2 points are earned for setting up integral with correct integrand

1 point **deducted** for incorrect limits/error involving π

$$\pi \int_{-1}^0 \left((e^{-x})^2 - (-ex)^2 \right) dx$$

2-2-2b

71.



The functions f and g are given by $f(x) = \sqrt{x}$ and $g(x) = 6 - x$. Let R be the region bounded by the x -axis and the graphs of f and g , as shown in the figure above.

There is a point P on the graph of f at which the line tangent to the graph of f is perpendicular to the graph of g . Find the coordinates of point P .

Part C

1 point is earned for $f'(x)$

1 point is earned for the equation

1 point is earned for the answer

$$g'(x) = -1$$

Thus a line perpendicular to the graph of g has slope 1.

$$f'(x) = \frac{1}{2\sqrt{x}}$$

$$\frac{1}{2\sqrt{x}} = 1 \Rightarrow x = \frac{1}{4}$$

The point P has coordinates $\left(\frac{1}{4}, \frac{1}{2}\right)$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for $f'(x)$

1 point is earned for the equation

1 point is earned for the answer

$$g'(x) = -1$$

Thus a line perpendicular to the graph of g has slope 1.

2-2-2b

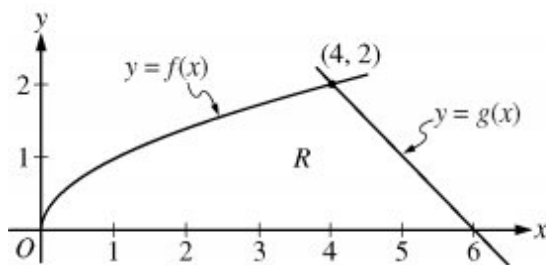
$$f'(x) = \frac{1}{2\sqrt{x}}$$

$$\frac{1}{2\sqrt{x}} = 1 \Rightarrow x = \frac{1}{4}$$

The point P has coordinates $(\frac{1}{4}, \frac{1}{2})$

2-2-2b

72.



The functions f and g are given by $f(x) = \sqrt{x}$ and $g(x) = 6 - x$. Let R be the region bounded by the x -axis and the graphs of f and g , as shown in the figure above.

(a) Find the area of R .

(b) The region R is the base of a solid. For each y , where $0 \leq y \leq 2$, the cross section of the solid taken perpendicular to the y -axis is a rectangle whose base lies in R and whose height is $2y$. Write, but do not evaluate, an integral expression that gives the volume of the solid.

(c) There is a point P on the graph of f at which the line tangent to the graph of f is perpendicular to the graph of g . Find the coordinates of point P .

Part A

1 point is earned for the integral

1 point is earned for the antiderivative

1 point is earned for the answer

$$\text{Area} = \int_0^4 \sqrt{x} \, dx + \frac{1}{2} \cdot 2 \cdot 2 = \frac{2}{3} x^{3/2} \Big|_{x=0}^{x=4} + 2 = \frac{22}{3}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the integral

1 point is earned for the antiderivative

1 point is earned for the answer

$$\text{Area} = \int_0^4 \sqrt{x} \, dx + \frac{1}{2} \cdot 2 \cdot 2 = \frac{2}{3} x^{3/2} \Big|_{x=0}^{x=4} + 2 = \frac{22}{3}$$

Part B

2-2-2b

2 points are earned for integrand

1 point is earned for the answer

$$y = \sqrt{x} \Rightarrow x = y^2$$

$$y = 6 - x \Rightarrow x = 6 - y$$

$$\text{Width} = (6 - y) - y^2$$

$$\text{Volume} = \int_0^2 2y(6 - y - y^2) dy$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

2 points are earned for integrand

1 point is earned for the answer

$$y = \sqrt{x} \Rightarrow x = y^2$$

$$y = 6 - x \Rightarrow x = 6 - y$$

$$\text{Width} = (6 - y) - y^2$$

$$\text{Volume} = \int_0^2 2y(6 - y - y^2) dy$$

Part C

1 point is earned for $f'(x)$

1 point is earned for the equation

1 point is earned for the answer

$$g'(x) = -1$$

Thus a line perpendicular to the graph of g has slope 1.

$$f'(x) = \frac{1}{2\sqrt{x}}$$

$$\frac{1}{2\sqrt{x}} = 1 \Rightarrow x = \frac{1}{4}$$

2-2-2b

The point P has coordinates $\left(\frac{1}{4}, \frac{1}{2}\right)$.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for $f'(x)$

1 point is earned for the equation

1 point is earned for the answer

$$g'(x) = -1$$

Thus a line perpendicular to the graph of g has slope 1.

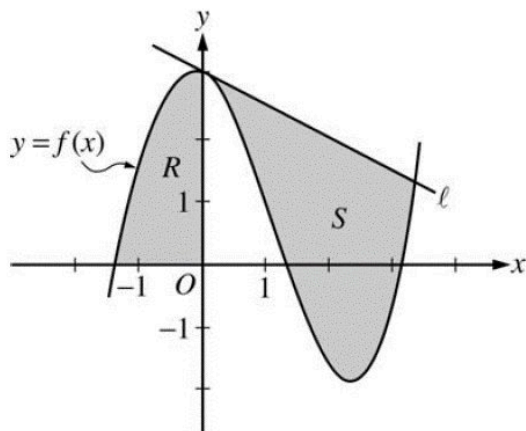
$$f'(x) = \frac{1}{2\sqrt{x}}$$

$$\frac{1}{2\sqrt{x}} = 1 \Rightarrow x = \frac{1}{4}$$

The point P has coordinates $\left(\frac{1}{4}, \frac{1}{2}\right)$.

2-2-2b

73.



Let f be the function given by $f(x) = \frac{x^3}{4} - \frac{x^2}{3} - \frac{x}{2} + 3 \cos x$. Let R be the shaded region in the second quadrant bounded by the graph of f , and let S be the shaded region bounded by the graph of f and line ℓ , the line tangent to the graph of f at $x = 0$, as shown above.



Write, but do not evaluate, an integral expression that can be used to find the area of S .

Part C

1 point is earned for the tangent line

1 point is earned for the integrand

1 point is earned for the limits

The equation of the tangent line ℓ is $y = 3 - \frac{1}{2}x$.

The graph of f and line ℓ intersect at $A=3.38987$.

$$\text{Area of } S = \int_0^A \left(\left(3 - \frac{1}{2}x \right) - f(x) \right) dx$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the tangent line

1 point is earned for the integrand

1 point is earned for the limits

The equation of the tangent line ℓ is $y = 3 - \frac{1}{2}x$.

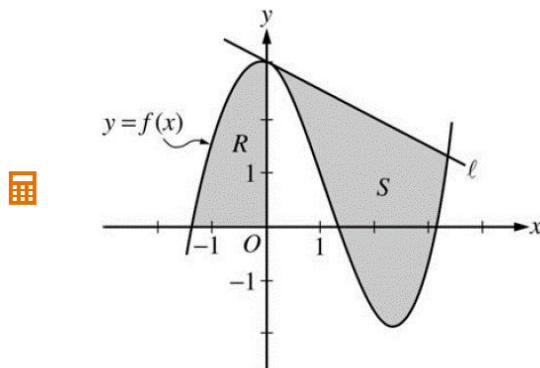
2-2-2b

The graph of f and line l intersect at $A=3.38987$.

$$\text{Area of } S = \int_0^A \left(\left(3 - \frac{1}{2}x \right) - f(x) \right) dx$$

2-2-2b

74.



Let f be the function given by $f(x) = \frac{x^3}{4} - \frac{x^2}{3} - \frac{x}{2} + 3\cos x$. Let R be the shaded region in the second quadrant bounded by the graph of f , and let S be the shaded region bounded by the graph of f and line ℓ , the line tangent to the graph of f at $x = 0$, as shown above.

- Find the area of R .
- Find the volume of the solid generated when R is rotated about the horizontal line $y = -2$.
- Write, but do not evaluate, an integral expression that can be used to find the area of S .

Part A

1 point is earned for integral

1 point is earned for the answer

For $x < 0$, $f(x) = 0$ when $x = -1.37312$

Let $P = -1.37312$.

$$\text{Area of } R = \int_P^0 f(x) dx = 2.903$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for integral

1 point is earned for the answer

For $x < 0$, $f(x) = 0$ when $x = -1.37312$

Let $P = -1.37312$.

2-2-2b

$$\text{Area of } R = \int_P^0 f(x)dx = 2.903$$

Part B

1 point is earned for limits and constant

2 points are earned for the integrand

1 point is earned for the answer

$$\text{Volume} = \pi \int_P^0 \left((f(x) + 2)^2 - 4 \right) dx = 59.361$$



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

1 point is earned for limits and constant

2 points are earned for the integrand

1 point is earned for the answer

$$\text{Volume} = \pi \int_P^0 \left((f(x) + 2)^2 - 4 \right) dx = 59.361$$

Part C

1 point is earned for the tangent line

1 point is earned for the integrand

1 point is earned for the limits

The equation of the tangent line ℓ is $y = 3 - \frac{1}{2}x$.

The graph of f and line ℓ intersect at $A = 3.38987$.

2-2-2b

$$\text{Area of } S = \int_0^A \left(\left(3 - \frac{1}{2}x \right) - f(x) \right) dx$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the tangent line

1 point is earned for the integrand

1 point is earned for the limits

The equation of the tangent line ℓ is $y = 3 - \frac{1}{2}x$.

The graph of f and line ℓ intersect at $A = 3.38987$.

$$\text{Area of } S = \int_0^A \left(\left(3 - \frac{1}{2}x \right) - f(x) \right) dx$$

75. Which of the following is an equation of the line tangent to the graph of $y = \cos x$ at $x = \frac{\pi}{2}$?

(A) $y = x + \frac{\pi}{2}$

(B) $y = x - \frac{\pi}{2}$

(C) $y = -x + \frac{\pi}{2}$

(D) $y = -x - \frac{\pi}{2}$



Answer C

Correct. The slope of the tangent line is the value of the derivative at $x = \frac{\pi}{2}$.

$$\frac{dy}{dx} = -\sin x \Rightarrow \frac{dy}{dx} \Big|_{x=\frac{\pi}{2}} = -\sin\left(\frac{\pi}{2}\right) = -1$$

$$\text{At } x = \frac{\pi}{2}, y = \cos\left(\frac{\pi}{2}\right) = 0.$$

$$\text{An equation of the tangent line at the point } \left(\frac{\pi}{2}, 0\right) \text{ is therefore } y = -1\left(x - \frac{\pi}{2}\right) + 0 = -x + \frac{\pi}{2}.$$

2-2-2b

76.  The derivative of the function f is given by $f'(x) = 1 - 5 \cos(3x^2 - 2)$. For how many values of x in the interval $0 \leq x \leq 2$ is the line tangent to the graph of $y = f(x)$ at the point $(x, f(x))$ parallel to the line $y = 1 + 4x$?
- (A) One
(B) Two
(C) Three ✓
(D) Four

Answer C

Correct. If the line tangent to the graph of $y = f(x)$ at a point on the graph is parallel to the line $y = 1 + 4x$, then $f'(x) = 4$ at that point, because the line $y = 1 + 4x$ has slope 4. The graph of f' intersects the horizontal line $y = 4$ in three places on the interval $0 \leq x \leq 2$.

77.  The derivative of a function f is given by $f'(x) = 0.1x + e^{0.25x}$. At what value of x for $x > 0$ does the line tangent to the graph of f at x have slope 2?
- (A) 0.512
(B) 1.849
(C) 2.287 ✓
(D) 8.113

Answer C

Correct. Since the derivative at a point is the slope of the line tangent to the graph at that point, the calculator is used to solve $f'(x) = 0.1x + e^{0.25x} = 2$.

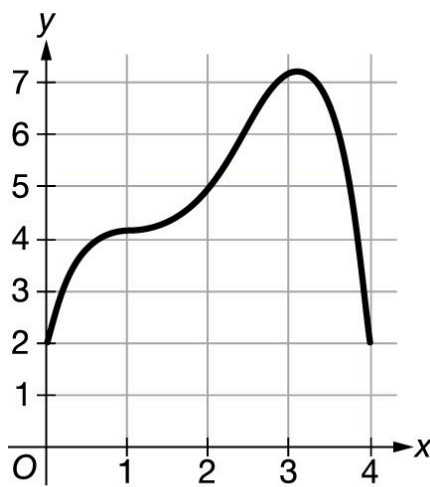
2-2-2b

78. The derivative of the function f is given by $f'(x) = -3x + 4$ for all x , and $f(-1) = 6$. Which of the following is an equation of the line tangent to the graph of f at $x = -1$?
- (A) $y = -3x + 3$
- (B) $y = -3x + 4$
- (C) $y = 7x + 6$
- (D) $y = 7x + 13$ ✓

Answer D

Correct. The slope of the line tangent to the graph of f at $x = -1$ is $f'(-1) = -3(-1) + 4 = 7$. An equation of the line containing the point $(-1, 6)$ with slope 7 is $y = 7(x + 1) + 6 = 7x + 13$.

79.

Graph of f'

The graph of f' , the derivative of a function f , is shown above. The points $(2, 6)$ and $(4, 18)$ are on the graph of f . Which of the following is an equation of the line tangent to the graph of f at $x = 2$?

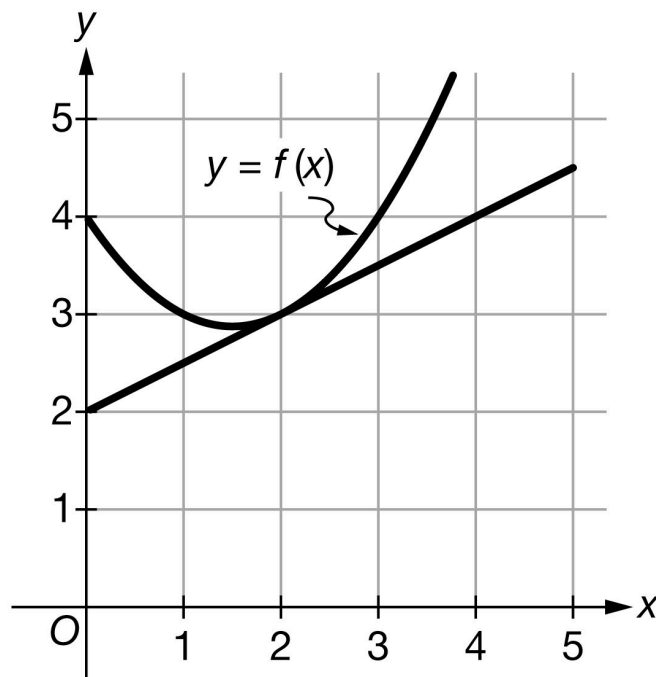
- (A) $y = 2x + 1$
- (B) $y = 5x - 4$ ✓
- (C) $y = 5x - 10$
- (D) $y = 6x - 6$

Answer B

Correct. The slope of the line tangent to the graph of f at $x = 2$ is $f'(2)$. This value can be read from the graph of f' and is equal to 5. An equation of the line with slope 5 that contains the point $(2, 6)$ is $y = 5(x - 2) + 6 = 5x - 4$.

2-2-2b

80.



Shown above is the graph of the differentiable function f , along with the line tangent to the graph of f at $x = 2$. What is the value of $f'(2)$?

- (A) $\frac{1}{2}$
(B) 2
(C) 3
(D) 4

**Answer A**

Correct. The derivative of f at $x = 2$ is the slope of the line tangent to the graph of f at $x = 2$. The slope of the line is

$$\frac{\Delta y}{\Delta x} = \frac{4-3}{4-2} = \frac{1}{2}.$$

2-2-2b

81. An equation for the line tangent to the graph of the differentiable function f at $x = 2$ is $y = 9x - 12$. Which of the following statements must be true?

- I. $f(0) = -12$
- II. $f(2) = 6$
- III. $f'(2) = 9$

- (A) None
- (B) I and II only
- (C) II and III only
- (D) I, II, and III

**Answer C**

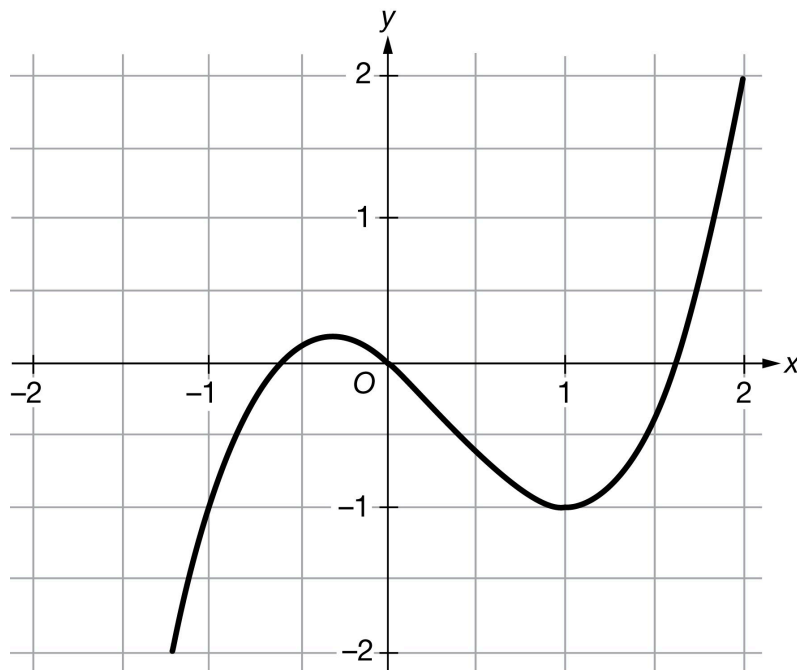
Correct. Statement II must be true. Since the line $y = 9x - 12$ is tangent to the curve at $x = 2$, the value of the tangent line at $x = 2$ must equal the value of the function f at $x = 2$. Therefore $f(2) = 9 \cdot 2 - 12 = 6$.

Statement III must be true. Since the value of the derivative f' at $x = 2$ is the slope of the line tangent to the graph of f at $x = 2$, it follows that $f'(2) = 9$.

Statement I could be false. Although the tangent line does pass through the point $(0, -12)$, this does not imply that the function f equals -12 at $x = 0$.

2-2-2b

82.

Graph of f'

Let f be a differentiable function with $f(1) = -2$. The graph of f' , the derivative of f , is shown above. Which of the following statements is true about the line tangent to the graph of f at $x = 1$?

- (A) The tangent line has slope -1 and passes through the point $(1, -2)$. ✓
- (B) The tangent line has slope -1 and passes through the point $(1, -1)$.
- (C) The tangent line has slope 0 and passes through the point $(1, -2)$.
- (D) The tangent line has slope 0 and passes through the point $(1, -1)$.

Answer A

Correct. The slope of the line tangent to the graph of f at $x = 1$ is given by $f'(1)$. From the graph, $f'(1) = -1$. Because $f(1) = -2$, the tangent line passes through the point $(1, -2)$.

2-2-2b

83. The derivative of a function f is given by $f'(x) = 0.2x + e^{0.15x}$. Which of the following procedures can be used to determine the value of x at which the line tangent to the graph of f has slope 2?
- (A) Evaluate $0.2x + e^{0.15x}$ at $x = 2$.
- (B) Evaluate $\frac{d}{dx}(0.2x + e^{0.15x})$ at $x = 2$.
- (C) Solve $0.2x + e^{0.15x} = 2$ for x . ✓
- (D) Solve $\frac{d}{dx}(0.2x + e^{0.15x}) = 2$ for x .

Answer C

Correct. Since the derivative at a point is the slope of the line tangent to the graph at that point, solving $f'(x) = 0.2x + e^{0.15x} = 2$ will yield the value of x at which the line tangent to the graph of f has slope 2.

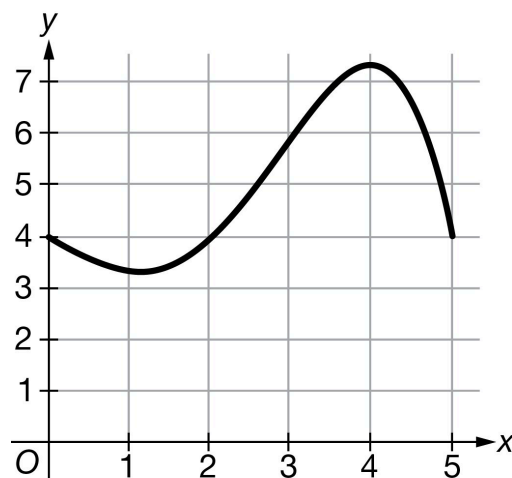
84. The derivative of the function f is given by $f'(x) = -2x + 4$ for all x , and $f(-1) = 5$. Which of the following is an equation for the line tangent to the graph of f at $x = -1$?
- (A) $y = -2x + 3$
- (B) $y = -2x + 4$
- (C) $y = 6x + 5$
- (D) $y = 6x + 11$ ✓

Answer D

Correct. The slope of the line tangent to the graph of f at $x = -1$ is $f'(-1) = -2(-1) + 4 = 2 + 4 = 6$. An equation for the line containing the point $(-1, 5)$ with slope 6 is $y = 6(x + 1) + 5 = 6x + 11$.

2-2-2b

85.

Graph of f'

The graph of f' , the derivative of a function f , is shown above. The points $(2, 7)$ and $(4, 18.8)$ are on the graph of f . Which of the following is an equation for the line tangent to the graph of f at $x = 2$?

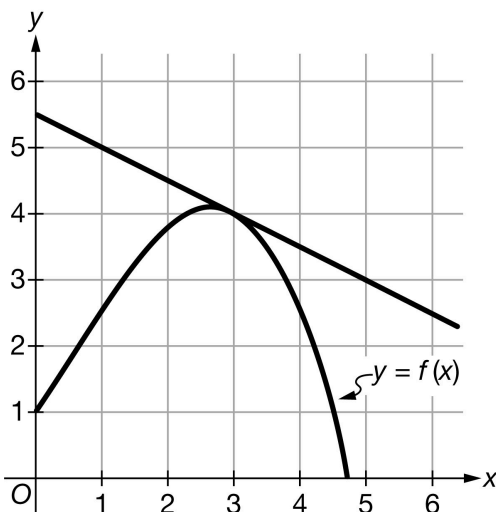
- (A) $y = 2x - 1$
- (B) $y = 4x - 1$
- (C) $y = 4x - 8$
- (D) $y = 5.9x - 4.8$

**Answer B**

Correct. The slope of the line tangent to the graph of f at $x = 2$ is $f'(2)$. This value can be read from the graph of f' and is equal to 4. An equation for the line with slope 4 that contains the point $(2, 7)$ is $y = 4(x - 2) + 7 = 4x - 1$.

2-2-2b

86.



Shown above is the graph of the differentiable function f along with the line tangent to the graph of f at $x = 3$. What is the value of $f'(3)$?

- (A) $-\frac{1}{2}$
- (B) -2
- (C) 4
- (D) $\frac{11}{2}$

**Answer A**

Correct. The derivative of f at $x = 3$ is the slope of the line tangent to the graph of f at $x = 3$. The slope of the line is $\frac{\Delta y}{\Delta x} = \frac{4-3}{3-5} = -\frac{1}{2}$.

2-2-2b

87. An equation for the line tangent to the graph of the differentiable function f at $x = 3$ is $y = 4x + 6$. Which of the following statements must be true?

- I. $f(0) = 6$
- II. $f(3) = 18$
- III. $f'(3) = 4$

- (A) None
- (B) I and II only
- (C) II and III only
- (D) I, II, and III

**Answer C**

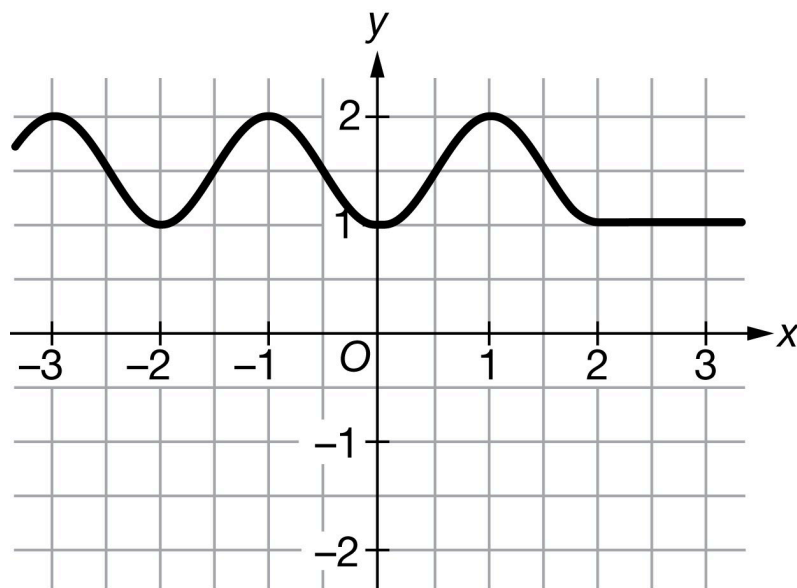
Correct. Statement II must be true. Since the line $y = 4x + 6$ is tangent to the curve at $x = 3$, the value of the tangent line at $x = 3$ must equal the value of the function f at $x = 3$. Therefore, $f(3) = 4 \cdot 3 + 6 = 18$.

Statement III must be true. Since the value of the derivative f' at $x = 3$ is the slope of the line tangent to the graph of f at $x = 3$, it follows that $f'(3) = 4$.

Statement I could be false. Although the tangent line does pass through the point $(0, 6)$, this does not imply that the function f equals 6 at $x = 0$.

2-2-2b

88.

Graph of f'

Let f be a differentiable function with $f(1) = 3$. The graph of f' , the derivative of f , is shown above. Which of the following statements is true about the line tangent to the graph of f at $x = 1$?

- (A) The tangent line has slope 2 and passes through the point $(1, 3)$. ✓
- (B) The tangent line has slope 2 and passes through the point $(1, 2)$.
- (C) The tangent line has slope 0 and passes through the point $(1, 3)$.
- (D) The tangent line has slope 0 and passes through the point $(1, 2)$.

Answer A

Correct. The slope of the line tangent to the graph of f at $x = 1$ is given by $f'(1)$. From the graph, $f'(1) = 2$. Because $f(1) = 3$, the tangent line passes through the point $(1, 3)$.

2-2-2b

89. Let ℓ be the line tangent to the graph of $y = \frac{1}{x}$ at $x = 3$. What is the x -coordinate of the point of intersection of ℓ and the x -axis?

(A) 6

(B) 4

(C) $3 - \frac{1}{3 \ln 3}$

(D) 0

**Answer A**

Correct. The slope of the tangent line is the value of $\frac{dy}{dx}$ at $x = 3$.

$$\frac{dy}{dx} = -\frac{1}{x^2} \Rightarrow \left. \frac{dy}{dx} \right|_{x=3} = -\frac{1}{9}$$

The equation of the tangent line is therefore $y = -\frac{1}{9}(x - 3) + \frac{1}{3}$. The point of intersection of the line and the x -axis occurs when $y = 0$.

$$0 = -\frac{1}{9}(x - 3) + \frac{1}{3} \Rightarrow x - 3 = (-9) \left(-\frac{1}{3}\right) = 3 \Rightarrow x = 6$$